

Caffevolve™

CAFFEINE MANAGEMENT & DELIVERY PLATFORM

Platform Reference

THE PLATFORM	CAFFEINE MANAGEMENT	ADAPTIVE FEEDBACK	THE USER PROFILE	THE DEVICE	CAFFEVOLVEGO	USER EXPERIENCE

Confidential

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1. The Caffevolve™ Platform 

Caffevolve™ is a comprehensive caffeine management platform built to provide coffee consumers with timely, precise guidance on their caffeine intake from the moment they wake to the moment they retire to bed. The primary objective of this guidance is to keep the consumer within a caffeine range that keeps them alert and focused when they need to be, without crossing over that fine line that typically results in agitation, nervousness, and ultimately, a night of poor sleep. There is a science to this objective and the Caffevolve™ Platform has captured it precisely.

Platform Intelligence

At the heart of the Caffevolve™ Platform lies Platform Intelligence — a layer that stretches across your entire day, from the device at home, to your drinks at work, and even to the moment you're standing in a café line. It is built on a simple but powerful idea: not everyone responds to caffeine the same way, and the same person responds differently depending on how much caffeine is already in their system. Caffevolve™ accounts for both. Every brew recommendation begins with the user's personal caffeine sensitivity — their individual tolerance level — and combines it with a real-time picture of how much caffeine is still active in their body from prior consumption. The goal is to keep the user in their personal **Caffeine Comfort Zone**: alert, focused, and performing well — above the level where caffeine has a muted impact, and below the level where too much of it starts working against them. Get it right, and caffeine becomes a precision tool rather than a daily gamble.

The Caffevolve™ Platform is always thinking ahead. It maintains a running caffeine record, and every brew you make on a Caffevolve device is added to it automatically. From that record, it continuously tracks how the caffeine in your system now will "decay" in the hours ahead. And it projects forward to your declared bedtime — so it knows whether a brew right now will clear your system in time to let you sleep well. If it won't, the caffeine recommendation is adjusted to keep that from happening. The result is a system that manages caffeine as a full-day variable, not a cup-by-cup decision — one that works for you around the clock, not just in the moment you press brew.

Wherever the next cup comes from, Platform Intelligence maintains a real-time picture of how much caffeine is active in your system so that well-reasoned drink recommendations can always be made.

The Caffevolve™ Device

The Caffevolve™ device is where Platform Intelligence is put into practice at home — and what makes it unlike anything else is that once the caffeine blend is confirmed, it brews it with precision. Using a dual-hopper system that blends caffeinated and decaffeinated beans in the exact ratio the caffeine blend recommendation calls for, the device closes the loop between calculation and cup.

The device's reach has a limit, though. The moment you step away and have caffeine somewhere away from home — the office, a coffee shop, a convenience store — the picture goes incomplete. What the device doesn't see, it can't account for, and the recommendations that follow are built on a partial view. This is where the companion app comes in.

CaffevolveGo

The CaffevolveGo app is where Platform Intelligence is applied when you're away from home. It keeps the caffeine picture complete, and it does this in two ways. First, it lets the user add any caffeine consumed away from the device — a quick entry that immediately updates the running record, as if the device had brewed it. Second, it advises. Before the user orders, the app looks at their current caffeine level and the time remaining before bedtime, and recommends what still fits. When the user walks out the door, the app keeps the day whole — nothing goes unaccounted for.

Together, the device and the **CaffevolveGo** app form a closed loop — one that follows the user through their entire day, from the first cup to the last. The Caffevolve™ Platform doesn't manage coffee. It manages caffeine — intelligently, continuously, and always in service of the user's performance and rest. And because the platform works entirely from the user's own behavioral signals and declared bedtime, it is fully schedule-agnostic: **it works equally well for a nurse on rotating shifts, a third-shift parts assembler, a student, or a traditional nine-to-five professional.**

To do this well, the platform is built on a working model of how caffeine behaves in the human body — how it accumulates, how it clears as the hours pass, and how individual caffeine sensitivity shapes the experience. The next section covers exactly that.

2. The Caffevolve™ Approach to Caffeine Management 

At the center of the Caffevolve™ approach to caffeine management is the development of a user's personal **Caffeine Comfort Zone** — the caffeine band where they are alert, focused, and performing well, above the level where caffeine has a muted impact and below the level where it starts working against them. Three boundaries define this band: the **Peak Zone Floor**, the Wake Hours Ceiling, and the Bedtime Ceiling — and in Caffevolve™, all three are derived from the user's own reported behavior.

2.1 Building the Caffeine Comfort Zone

Caffeine Comfort Zone — Defined

> Throughout this document we refer to the user's "Caffeine Comfort Zone" — the band where caffeine is actively working for them.
>
> Caffevolve™ also calls this the Peak Zone. The Peak Zone sits between three boundaries: the Peak Zone Floor at the bottom, the Wake Hours Ceiling at the top, and the Bedtime Ceiling between them. The two ceilings play different roles: the **Wake Hours Ceiling** governs the active hours of the day, capping how much caffeine the user can carry at any moment before the focus and productivity benefit reverses. The **Bedtime Ceiling** governs the night, setting the maximum residual caffeine targeted in the user's system at the moment they go to sleep — and shaping every recommendation made during the hours that lead there. Within the Peak Zone, the caffeine present in the system delivers its intended benefit: alertness, focus, and sustained cognitive performance. Below it, caffeine in the system isn't enough to produce a meaningful effect. Above it, the benefit reverses — alertness gives way to agitation, focus to restlessness, and performance suffers, as does the ability to sleep.

How Caffevolve™ Learns You — The Caffeine Behavior Instrument (CBI)

The **Caffeine Behavior Instrument** is the platform's onboarding tool for establishing the initial boundaries of the user's Caffeine Comfort Zone and their personal **Caffeine Decay Rate** — the two inputs the platform needs before it can make a meaningful recommendation. Both are derived from the same place: the user's own reported behavior. How do they respond to caffeine over the course of the day? How quickly does it clear their system? The conventional approach to answering these questions is to ask the user to classify themselves: "Are you sensitive to caffeine?" The problem is that this question is nearly unanswerable.

The problem is that sensitivity, clearance, and tolerance are biologically independent traits that feel identical from the inside. A heavy drinker who reports "caffeine doesn't affect me" may be genuinely insensitive — or simply tolerant to a dose that would still wreck their sleep. Self-assessment can't tell the difference.

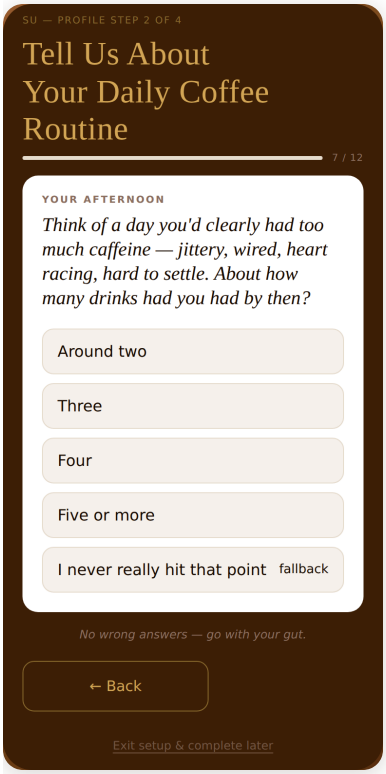
Caffevolve™ sidesteps this dilemma entirely. Instead of asking users to rate themselves on a trait they cannot accurately perceive, the platform asks about their own familiar behavior against known quantities — things a user can report reliably because they live them virtually every day. Some of these questions include:

- What kind of caffeinated drink do you enjoy first thing in the morning?
- How many cups does it take before you begin to feel alert, focused, and ready for the day?
- How many cups do you usually have before you feel wired, jittery, or overstimulated?
- What is the latest time of day you can have caffeine without it having a negative impact on your sleep?

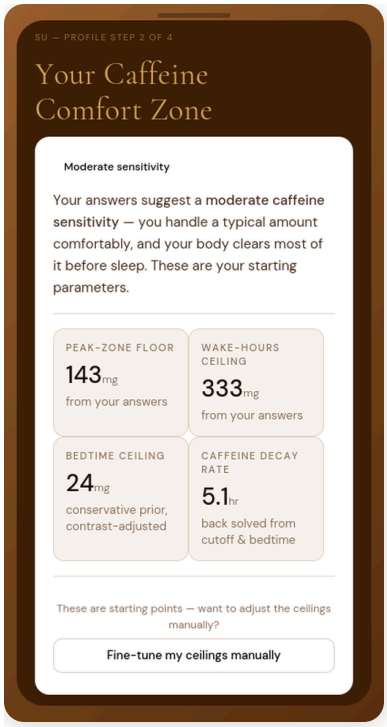
From this initial round of questions, the platform begins to compute the user's three Caffeine Comfort Zone boundaries and the user's personal Caffeine Decay Rate (Section 3).

No sensitivity tier is assigned. No category is selected. The boundaries themselves are the quantified expression of the user's Caffeine Sensitivity Level — derived from actual user behavior rather than self-assessed. Descriptive labels used elsewhere in this document — such as "moderate sensitivity" — characterize a computed result for the reader; they are not tiers, and no such category is selected or stored for **Resident Users**. (Guest profiles are the sole exception, where Higher / Moderate / Lower is a deliberately selected conservatism shortcut.)

These boundaries are starting points — designed to be right enough to guide the first brew and conservative enough to do no harm. From there, the platform's **Adaptive Feedback System** (described in Section 3) refines them from real outcomes, so the Caffeine Comfort Zone continues to sharpen as the system gathers real-world user feedback.



The Caffeine Behavior Instrument — question 07/12. The user describes a day they felt overstimulated. The platform derives the Wake Hours Ceiling from the reported cup count.



Moderate Caffeine Sensitivity Level — the initial, computed Caffeine Comfort Zone and the Caffeine Decay Rate derived from the responses to the twelve CBI questions.

Caffeine Comfort Zone — Establishing the Boundaries

Peak Zone Floor

The Peak Zone Floor is the minimum residual caffeine level at which caffeine is actively working for a given user — below it, the caffeine in their system isn't producing a meaningful alertness benefit, and the user may feel sluggish, unfocused, or simply flat. The platform derives this from the user's answer to a straightforward question: how many cups of their typical morning drink does it take before they feel alert and focused? The answer, paired with the known caffeine content of that drink, yields the residual level at which their body crosses from "caffeine present but not yet working" to "caffeine engaged." This is their floor — the lower boundary of their Peak Zone.

Wake Hours Ceiling

The Wake Hours Ceiling is the upper boundary of the Caffeine Comfort Zone during the user's active hours — the line above which caffeine stops being productive and starts being counterproductive. Alertness becomes agitation; focus becomes scattered attention; the user is overcaffeinated and, paradoxically, less productive than they would have been at a lower level. The platform derives this from the same onboarding sequence: how many cups of that same morning drink before the user feels overstimulated? The answer, against the same known caffeine content, yields the personal level at which the benefit reverses. This ceiling holds steady throughout active hours and applies regardless of when those hours fall — making it fully schedule-agnostic.

The Wake Hours Ceiling is constant — it does not change based on time of day. It is the overstimulation threshold at all times. It only comes under meaningful pressure during waking hours, when caffeine is being consumed and accumulating; during sleep, caffeine decays and the ceiling is never challenged.

Bedtime Ceiling

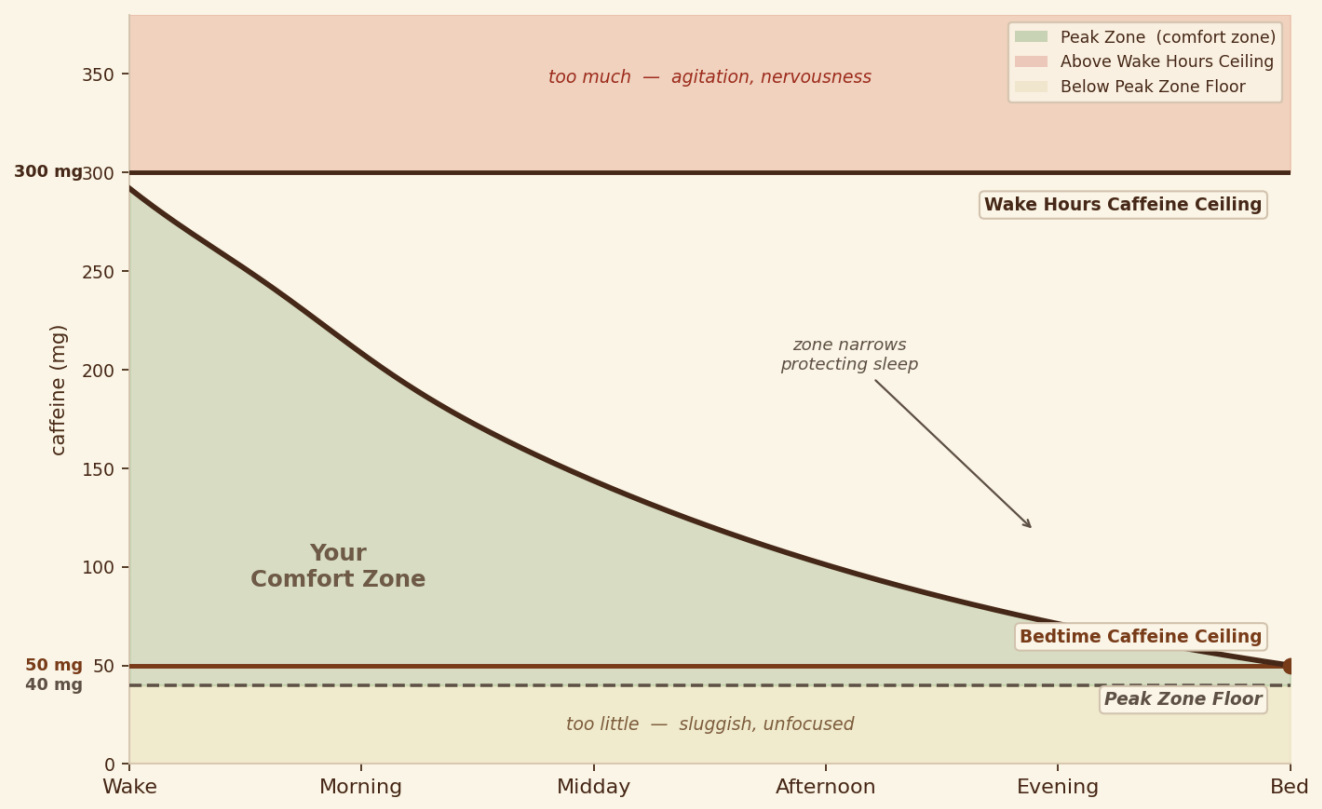
The Bedtime Ceiling is the second of the platform's two ceilings — and the one that increasingly shapes recommendations as the day progresses and bedtime draws closer. Where the Wake Hours Ceiling is an instantaneous check against overstimulation right now, the Bedtime Ceiling is a forward-looking target: the maximum residual caffeine the platform targets at the moment the user goes to sleep. Every caffeine blend the platform recommends, from the first morning cup onward, is calibrated to ensure this target is honored — because a disrupted night of sleep is the single costliest failure the platform can produce.

The platform derives this boundary rather than presetting it: from the user's reported routine it reconstructs a typical day's caffeine, decays each drink forward at the user's established Caffeine Decay Rate to find the residual carried into their declared evening cutoff, then projects that residual forward to their declared bedtime — the result is the Bedtime Ceiling. It is set conservatively at onboarding — erring on the side of protecting sleep — and refined from how the user actually sleeps. Within days, the conservative starting point gives way to a boundary shaped by real outcomes.

Connecting the Dots

Together, these three boundaries define the user's personal Caffeine Comfort Zone — and the Caffeine Comfort Zone is the first component of the equation that drives every caffeine recommendation the platform makes. The other component is based on the user's Caffeine Decay Rate.

Defining Your Caffeine Comfort Zone



2.2 Determining Distance to the Ceilings

The Caffeine Decay Rate (CDR)

The Caffeine Decay Rate is the personalized rate at which a user's body clears caffeine from their system — established by the Caffeine Behavior Instrument as a research-anchored starting estimate, then refined by the Adaptive Feedback System, as described below. It is the engine behind every time-based projection the platform makes: projected residual at the moment, projected residual at any hour, and projected residual at bedtime.

The underlying model is standard pharmacokinetics — exponential decay, in which the body clears a fixed proportion of its active caffeine in each unit of time. Peer-reviewed literature puts the individual half-life range at roughly 2.5 to 10 hours, with a population mean around five to six hours — which is where the platform begins before the user's personal rate is established. For every six hours that pass, approximately half of the user's active caffeine clears. A cup of coffee delivering 95 mg at 7:00 AM leaves roughly 47 mg by 1:00 PM, about 24 mg by

7:00 PM, and approximately 12 mg by 1:00 AM. Each cup follows this curve independently, and the platform sums them to produce the user's total active caffeine at any moment.

Because caffeine clearance is biologically independent of caffeine sensitivity, the decay rate cannot be inferred from the answers that set the Caffeine Comfort Zone boundaries — it must be isolated from separate, clearance-specific signals. The platform therefore begins from a research-anchored 6.0-hour population prior and adjusts it up or down from the two behavioral factors most strongly tied to clearance rather than sensitivity: regular cigarette smoking, which accelerates caffeine metabolism through the CYP1A2 pathway, and how often the user feels overstimulated in the afternoon on their usual intake. The adjusted rate — not the population baseline — is what the platform then applies to every subsequent projection, and what it uses to derive the Bedtime Ceiling. Because the Caffeine Decay Rate underlies every projection the platform makes — residual level, caffeine headroom, bedtime estimate — its accuracy has a compounding effect on the quality of every recommendation. The Adaptive Feedback System treats CDR refinement as its highest early priority, gathering decay-relevant signals as quickly as possible in the first weeks of use so that the platform's projections converge on the user's actual biology without delay.

Residual Caffeine Level

> The Residual Caffeine Level is the total caffeine active in the user's system at any given moment — the computed output of applying the Caffeine Decay Rate to everything the user has consumed.
>
> Accumulated from all brews and any externally logged caffeine entries, and continuously decaying on a 24-hour clock, it operates as a cumulative 24-hour model: not a stored value but a living projection that changes with every passing minute and every new brew.
>
> Think of it as a running tally that moves in both directions. Every new brew or logged drink pushes it up; every hour that passes pulls it back down as the Caffeine Decay Rate works. At any point, the platform computes this number and compares it against the Caffeine Comfort Zone boundaries from Section 2.
>
> With the residual level known, the picture is complete. The boundaries tell the platform where the edges are; the Caffeine Decay Rate tells it how fast caffeine is clearing; and the Residual Caffeine Level tells it where the user stands right now. From here, every recommendation is arithmetic.

The Dual-Ceiling Architecture

> A single ceiling would be simple to manage. Caffevolve™ works with two, and it evaluates both at once.
>
> The Dual-Ceiling Architecture is the platform's core decision model: every brew recommendation must satisfy the Wake Hours Ceiling and the Bedtime Ceiling simultaneously. Neither ceiling operates in isolation, and neither is allowed to override the other.
>
> The two ceilings ask different questions. The Wake Hours Ceiling is an instantaneous check — it asks whether adding this brew's caffeine to the user's current residual would push them above their focused and productive range and into overstimulation. The Bedtime Ceiling is a forward-looking check — it asks whether the user's

> total residual at bedtime, with this brew included, will clear the targeted caffeine ceiling and leave the door open for sleep. One governs the next hour; the other governs the night.

Because the two checks measure different things, they rarely constrain at the same time. Early in the day, with a distant bedtime, the Wake Hours Ceiling usually does the limiting — there are plenty of hours for caffeine to clear, so bedtime is not yet the constraint. Late in the day the situation reverses: the user may sit well within their wake-hours range, but with bedtime only a few hours off, the Bedtime Ceiling becomes the tighter of the two. The platform checks both ceilings, every time. If either would be violated by the brew being considered, the caffeine recommendation is adjusted until the user stays under both — the strongest brew that clears each ceiling simultaneously.

Caffeine Headroom

Caffeine Headroom is the "caffeine distance" between the user's current residual caffeine level and the nearest ceiling — the room available before the next boundary is reached. For every brew, the platform computes headroom independently against each ceiling: how much additional caffeine can be added before the Wake Hours Ceiling is reached, and how much before the Bedtime Ceiling is reached when projected forward to the user's declared bedtime. The two checks are different in nature — the Wake Hours check is instantaneous, the Bedtime check is forward-looking, projecting the brew's residual forward to the declared bedtime using the user's personal CDR. Whichever ceiling leaves the least room becomes the binding headroom for that brew. The recommendation that follows is then calculated to fill that headroom exactly — the highest caffeinated blend the architecture allows at that moment.

The platform computes caffeine headroom with two tests — one for each ceiling — and takes the smaller of the two results.

THE TWO-TEST CAFFEINE HEADROOM CALCULATION

TEST 1 — WAKE HOURS (INSTANTANEOUS)

```
Caffeine Headroom = Wake Hours Ceiling - Current Residual
```

TEST 2 — BEDTIME (DECAY-AWARE)

```
Caffeine Headroom = ( Bedtime Ceiling - Projected Bedtime Residual (prior) ) ÷ Decay Factor
```

```
Decay Factor = 0.5 ^ ( Hours Until Bedtime ÷ Caffeine Decay Rate )
```

Projected Bedtime Residual (prior): the residual projected to bedtime from prior brews and logged drinks only — this brew not yet included.

Binding Caffeine Headroom = `the lower result of Test 1 and Test 2`

The first test checks the Wake Hours Ceiling directly. The second is decay-aware: it projects forward to bedtime using the Caffeine Decay Rate. The further off bedtime is, the more caffeine can be added now and still decay away in time — which is why the same residual level yields generous caffeine headroom at 8 AM and almost none at 9 PM. The platform runs both tests and adopts the smaller answer. That smaller number is the binding caffeine headroom — the true ceiling on what the platform will recommend.

Caffeine Headroom is the platform's internal working figure; the user never sees a milligram-of-headroom number on the device. What they see is the recommendation it produces — which is where the platform turns next.

Projected Residual at Bedtime

Of every number the platform computes, one matters more than the rest: the projected bedtime caffeine residual level. It is the answer to the question that defines the entire platform — if the user brews this now, where will they stand when they try to sleep?

2.3 Building the Caffeine Recommendation

The Candidate Brew

Caffeine Headroom tells the platform how much caffeine it has to work with. The next step is turning that allowance into a Candidate Brew — and the answer is a direct calculation, not a search.

The platform takes the binding caffeine headroom as its target caffeine output and applies the Blend Ratio Formula — factoring in the user's serving size, serving count, and brew strength — to compute the exact caffeinated-to-decaffeinated ratio that delivers that amount. One formula, one answer: the strongest brew that both ceilings allow. The result is the Candidate Brew — the fully-specified brew the platform has determined best serves the user at this moment, expressed as a blend ratio ready to be confirmed and brewed. That ratio becomes the Caffeine Blend Recommendation.

The Caffeine Blend Recommendation

The Caffeine Blend Recommendation is the Candidate Brew presented to the user — the platform's complete answer translated into a form the user can read, understand, and act on. It is expressed as a blend ratio — the proportion of caffeinated to decaffeinated beans the device will grind together for this cup — because the ratio is the lever the user controls. A recommendation might read 70% caffeinated / 30% decaffeinated. That number tells the user exactly where they are, and exactly which direction to move if they want something different.

Alongside the ratio, the platform shows the user the two numbers that explain it: their post-brew caffeine residual level — where they'll stand after this cup — and the projected caffeine residual at bedtime — where this brew will

leave them when it counts. The user sees not only what the platform recommends, but the reasoning behind it.

And then the platform stops recommending.

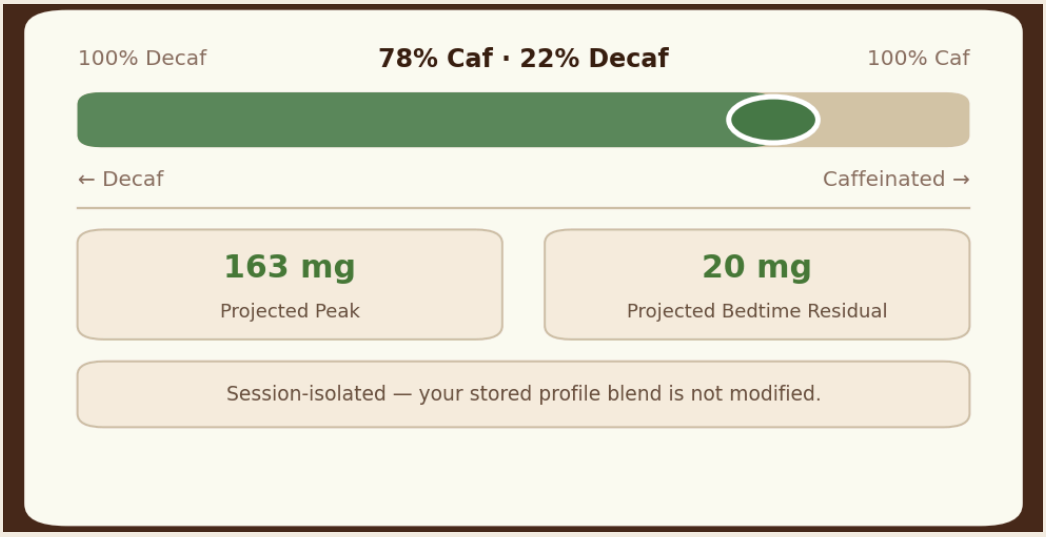
The Caffevolve™ Advisory Philosophy

The system recommends. The user decides. The machine always brews.

The caffeine-blend recommendation is always the starting point — never a restriction. The user can accept it, adjust it, or override it entirely — a change on **Adjust Caffeine Blend** that the platform honors without question. Caffevolve™ is an intelligent advisor, not a gatekeeper. What ends up in your mug is always your call.

If the user wants to adjust the recommendation, they are advanced to **Adjust Caffeine Blend** — a slider that lets them dial the caffeine level up or down from the recommended blend. As the slider moves, the Projected Caffeine Peak and the Projected Bedtime Caffeine Residual update automatically, giving the user a live numeric guide to the consequence of every position. Whatever blend they choose becomes the blend that is brewed — and the platform adds it to the running record, updating the caffeine picture from that point forward.

The Adjust Caffeine Blend



The platform computes the projected bedtime residual (post-brew) — the figure the user sees on the recommendation screen, distinct from the prior figure the headroom test consumes — by taking the user's residual level the moment the contemplated brew lands (current residual plus the caffeine the brew adds) and decaying that total forward to the declared bedtime using the user's personal Caffeine Decay Rate. The result is a single figure in milligrams: the caffeine that will still be active in the user's system at the moment they lie down.

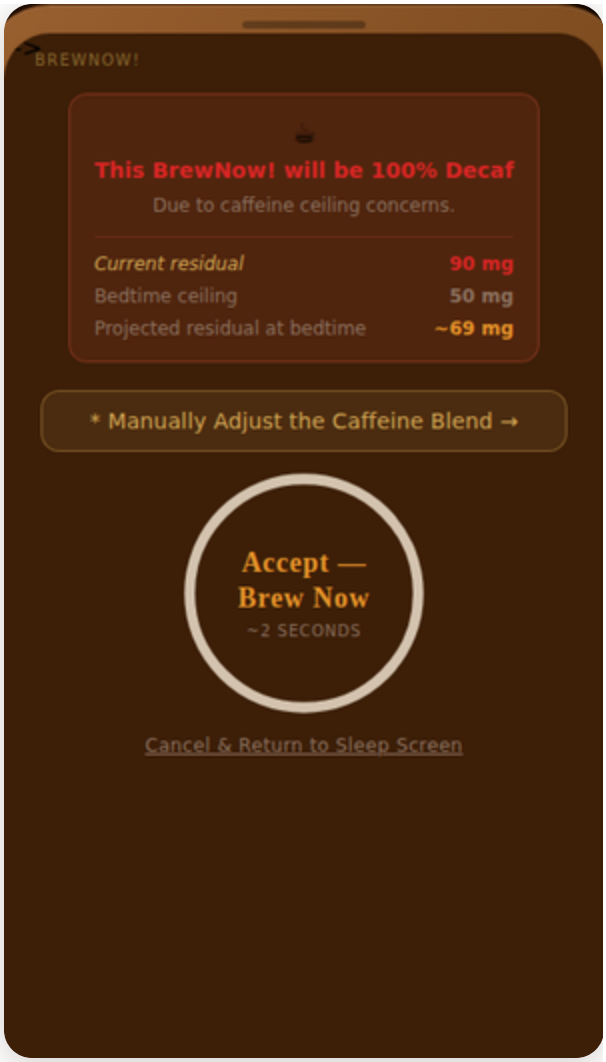
This is the number that turns caffeine management from a daytime concern into a full-day one. A brew can sit comfortably within the Wake Hours Ceiling and still be the wrong brew — if its projected residual at bedtime sits above the Bedtime Ceiling, it will cost the user sleep. By projecting every candidate forward to bedtime before recommending it, the platform protects the night while it serves the day — or vice versa for a shift worker. It is the reason a Caffevolve™ recommendation is never just about the next cup — it is about how that cup lands hours later, when the lights go out.

The Zero-Caffeine-Headroom State

There is one situation the platform must handle with particular care: the moment when no caffeinated brew fits and a new brew is requested. It happens when the user's residual level — from earlier brews, logged drinks, or both — has already climbed so high, or bedtime sits so close, that even the smallest caffeinated candidate would breach the binding ceiling. The caffeine headroom has gone to zero. The platform calls this the zero-caffeine-headroom state.

When no caffeinated candidate can satisfy both ceilings, the platform does not quietly recommend a weaker blend and hope for the best. It recommends a full 100% decaffeinated brew — and, just as importantly, it explains why. The user is told plainly that their current caffeine level, or the nearness of their bedtime, leaves no room for more caffeine right now, and that a decaf brew lets them enjoy the cup without putting the later hours of the day — or the night ahead — at risk.

This is the Advisory Philosophy at its most honest. Even here the platform does not lock the user out; the recommendation is still a starting point, and the user remains free to adjust. But the zero-caffeine-headroom state is the clearest expression of what Caffevolve™ is for — not to push caffeine, and not to forbid it, but to tell the user the truth about where they stand and let them decide with their eyes open.



BN-Decaf — The platform's response to a zero-headroom BrewNow! request. Current residual 90 mg, bedtime ceiling 50 mg, projected residual at bedtime ~69 mg. The platform recommends 100% decaf and explains why — the user retains the option to manually adjust.

2.4 Case Studies

The following three scenarios illustrate the Caffevolve™ approach in action — each showing a different aspect of the system working on behalf of a real user in a real moment. Wayne is six cups in and headed in the wrong direction — the platform keeps him sharp without cutting him off. Claudia has a bedtime ceiling that won't move and an afternoon slump that won't wait — the platform finds her the most it can honestly offer. Diane has a board presentation at nine and a bedtime that's closer than she'd like — the platform finds the strongest brew that protects both.

Wayne on Deadline

A heavy caffeine day collides with an overstimulation ceiling. The platform intervenes before the next cup makes things worse.

It's the back half of a long day and the deadline has gone from "this week" to "tonight." Wayne is six cups in — the last three poured barely an hour apart as the pressure climbed — his knee is bouncing under the desk, and he's just read the same paragraph three times without keeping any of it. He's wide awake, wired tight, and getting almost nothing done.

Here's the part nobody warns you about. The closer the deadline got, the faster Wayne reached for the coffee — more cups, shorter gaps, more fuel in the tank. But somewhere around cup four the coffee quietly stopped helping and started working against him. He isn't tired. He's overcaffeinated — which feels almost exactly like tired from the inside, restless and urgent and unable to settle, except tired you can fix with a cup, and this you only make worse with one.

Because Wayne has a sweet spot, same as everyone does: enough caffeine to be sharp, not so much that he's rattled. For him it's a band that tops out around 300 mg working at once — under it he's locked in and fast; over it he's exactly where he is now, all engine and no traction. Six cups — the last three crammed into three hours as the crunch hit — sail him clean past it to around 375 mg. And the cruel trick is that the overshoot never feels like "too much." It feels like not enough — so he pours another, and digs the hole deeper.

Caffevolve™ knows the difference he can't feel. As his caffeine climbs toward that line, it doesn't cut him off — it recomposes the cup. The morning cups stay full-strength. But as the tempo picks up and the cups come faster, each one comes back blended a little lighter — enough to hold him at the top of his game and not a milligram past. Six cups, same crunch-driven pace, but the sixth lands him at around 285 mg — sharp, planted in the zone, instead of vibrating past it.

The deadline doesn't move. But Wayne meets it from inside his sweet spot — clear, quick, steady — instead of vibrating through the last stretch getting half as much done. For once the coffee does the exact thing he always asked of it: it keeps him at his best, right when his best is due.

Wayne's ceiling ~300 mg of active caffeine at once. Under it, he's sharp. Over it, wired.

Without Caffevolve™ Six full-strength cups — the last three every hour — push his active caffeine to **~375 mg**. Past his ceiling. Jittery, scattered, less productive.

With Caffevolve™ Same six cups, same tempo, blended — active caffeine at **~285 mg**. Under his ceiling. Sharp, steady, productive.

Claudia at Three O'Clock

A sensitive user, a late-afternoon slump, and a bedtime ceiling that won't budge. The platform finds the most it can offer.

Most afternoons, around three, Claudia hits a wall. The screen goes a little blurry, the good ideas stop arriving, and there's a pot of coffee down the hall that would fix all of it in about ninety seconds. She doesn't get up. She already knows how this goes: coffee this late stays with her long past bedtime, and she's lain awake on the wrong side of that decision too many times to chance it. So she stays at her desk and grinds through on nothing — a little slower, a little less productive than she'd like to be — and she's been quietly making that same trade every afternoon for years.

It isn't that Claudia stopped loving coffee. It's that somewhere along the line coffee became something she's forced to budget instead of simply enjoying. Her body sets the limit — much past 50 mg still lingering at bedtime and sleep just won't come — and full-strength coffee spends that budget fast. Two cups before mid-morning and she's already near the edge. So the afternoon cup, the after-dinner cup, the second cup with a friend who's only just sat down — those are the ones she gives away, one polite "no thanks" at a time, until her whole coffee life has shrunk to two functional mugs before the day's even started.

Caffevolve™ doesn't touch her limit — her body still owns that 50 mg line. What it changes is the cup — and the confidence to recompose it as the day goes on, with timely guidance at every brew. Caffeine stops being all-or-nothing and turns into something the machine tunes to the hour she's drinking it. Her morning is strong, the way she likes it. And at three o'clock she gets up and pours herself a proper coffee — the warmth, the ritual, the lift to carry her through the afternoon — built so that by bedtime there's nothing left to cost her the night.

Same body. Same line she's never allowed to cross. But for the first time in years, three o'clock isn't a wall Claudia white-knuckles her way over. It's just coffee.

Claudia's ceiling ~50 mg of residual caffeine at bedtime. Under it, she sleeps. Over it, she doesn't.

Without Caffevolve™ Four full-strength cups leave **~130 mg** at bedtime — more than double her ceiling. To stay under it, she'd have to cut to two cups before mid-morning.

With Caffevolve™ Same four cups, blended — residual caffeine at bedtime **~49 mg**. Under her ceiling. All four cups, including one at 6 PM.

All values reflect residual caffeine at bedtime (10:30 PM) — each cup decayed from brew time (6-hour half-life).

Diane the Night Before

A board presentation tomorrow. A strong cup today. The dual-ceiling architecture doing exactly what it was designed to do.

Diane has a board presentation at nine tomorrow morning and a slide deck that isn't where it needs to be. It's two o'clock, the afternoon is slipping away, and she heads to the Caffevolve™ device the way most people do at two o'clock on a bad afternoon — looking for something to close the gap between where she is and where she needs to be.

The caffeine recommendation comes back lighter than she was planning on. Not decaf, not even close, but not the full-strength cup she had in mind either. She looks at it for a moment. Then she glances at the numbers the screen is already showing her. The projected bedtime residual tells the story plainly — a full-strength cup right now would still be working through her system well past midnight, and midnight is when she needs to be asleep, not running through tomorrow's talking points in the dark. Her Wake Hours ceiling has room. The Bedtime ceiling doesn't. The platform found the strongest cup that clears both, and it's showing her exactly why.

She smiles. Confirms the brew. The afternoon opens back up, the slides come together, and by the time she shuts the laptop the deck is where it needs to be. She sleeps well. The presentation goes well.

The platform didn't give her less. It gave her the most it could — while keeping tomorrow morning intact.

Two o'clock. Presentation at nine.

Wake Hours ceiling — room to spare. Bedtime ceiling — almost full.

The platform found the strongest cup that clears both.

Adjust Caffeine Blend was right there if she'd wanted it. The stronger cup was always her call. She didn't need it.

The Numerical Rationale Principle

Specific published figures cited in this document (e.g., caffeine safety thresholds, half-life ranges, extraction efficiencies) are reference points used to establish the scientific rationale for the platform's methodology — they are not themselves the invention. The protectable core is the process by which the system derives, personalizes, and continuously refines these values from individual user behavior, independent of which literature-sourced constants happen to seed it. Specific citations are maintained separately and should be verified independently before use in patent claim language or regulatory submissions.

3. The Adaptive Feedback System 

The **Caffeine Behavior Instrument** is utilized to set the **Caffeine Comfort Zone** before the first brew. Every answer the user gives during onboarding — how many cups before they feel sharp, how many before they feel wired, the latest they can drink without losing sleep — goes into deriving three personal boundaries. Those boundaries are provisional. They are designed to be close enough to guide the first brew safely, and conservative enough to do no harm while the platform learns. What refines them is the Adaptive Feedback System.

The **Adaptive Feedback System (AFS)** is the mechanism by which the platform learns from the user's actual experience rather than from what they predicted about themselves at setup. Where the Caffeine Behavior Instrument asks what they expect to feel, the AFS asks what they did feel — after a brew that pushed a boundary, after a night that followed a late cup, after a morning when the user reports they didn't sleep well. Each reported outcome is a signal. Each signal either confirms a boundary or prompts a recalibration.

3.1 The Prompt Cadence

The AFS uses a hybrid prompt cadence — not a single trigger mechanism, but three overlapping layers that together ensure every user receives meaningful check-ins, regardless of how closely their brew behavior challenges the Caffeine Comfort Zone boundaries.

The first layer is boundary-triggered: a question fires whenever a brew lands near one of the user's Caffeine Comfort Zone boundaries, at the moment when the response is most informative. A brew that grazes the Wake Hours Ceiling is precisely the brew that reveals whether that ceiling is set correctly. A brew timed so that the projected bedtime caffeine residual level approaches the **Bedtime Ceiling** is the brew that tells the platform most about whether that ceiling needs to move.

The second layer is brew-count-triggered: a proactive floor cadence fires on a rolling basis for each dimension, independent of whether any boundary has been approached. This matters because a pure boundary-triggered system is reactive by design — it only fires when something is near a limit. A well-calibrated user who consistently accepts recommendations and never pushes a ceiling might rarely see a boundary trigger at all. The brew-count cadence closes that gap, ensuring the AFS stays active and continues gathering real-world signal even when the platform appears to be calibrated effectively.

The brew-count cadence has particular importance in the early weeks of use, when the Caffeine Behavior Instrument's provisional boundaries are least refined and the **Caffeine Decay Rate** is still being established. CDR refinement is the AFS's highest early priority — and the brew-count layer ensures that refinement-relevant questions reach the user on a reliable cadence, not only when a boundary happens to be challenged.

The Three-Layer Prompt Cadence

The AFS uses three overlapping trigger mechanisms to ensure that check-ins reach every user — not just those who push boundaries.

Layer 1 — Boundary-Triggered
A question fires whenever a brew lands near one of the user's Caffeine Comfort Zone boundaries. The user is at the exact moment when their response is most informative — the brew that grazed the **Wake Hours Ceiling**, the timing that challenged the Bedtime Ceiling.

> *"Did you feel focused and sharp after that last cup — or a little wired?"*
>
> *"How did you sleep last night?"*

Layer 2 — Brew-Count
A proactive floor cadence fires on a rolling brew-count basis for each dimension, independent of boundary proximity. This ensures that well-calibrated users — those who consistently accept recommendations and rarely graze a ceiling — still receive regular check-ins. The brew-count cadence is also the primary mechanism for CDR refinement in early use, when boundaries are new and the platform needs real-world confirmation regardless of whether any ceiling has been challenged.

Layer 3 — FR-Meta (60-Day)
Every 60 days, a calendar-triggered meta-prompt asks whether the check-in frequency feels right: "Too often," "Just right," or "Not sure." FR-Meta is not tied to any brew event or boundary — it is the long-cycle check that surfaces engagement fatigue and catches slow drift that shorter cadences may miss.

Time-of-Day Anchoring

Both boundary-triggered and brew-count prompts are governed by a second constraint beyond their trigger condition: they only fire at the time of day when the user can answer the question accurately and with minimal friction. Asking a user about the previous night's sleep quality at 3 PM requires them to reconstruct a memory. Asking at the first brew of the morning requires nothing — the answer is already top of mind.

Each of the four feedback dimensions has a designated time-of-day window within which its prompt is eligible to fire. A trigger condition — whether boundary-triggered or brew-count — is only acted upon if the current time falls within the appropriate window for that dimension. Outside that window, the trigger is deferred to the next eligible moment.

The Four Feedback Dimensions

Each dimension captures a distinct aspect of the caffeine experience. Questions are short, one-tap, and timed to the moment in the day when the user's answer is most accurate and most readily available.

D1 — Sleep

"How did you sleep last night?"

→ Refines the Bedtime Ceiling. Posed at the first brew of the morning, when sleep quality is immediately top of mind.

D2 — Focus

"Did this brew get you going?"

→ Refines the **Peak Zone Floor**. Posed mid-morning, after the user has had time to feel the first brew's effect.

D3 — Jitteriness

"Feeling wired or jittery?"

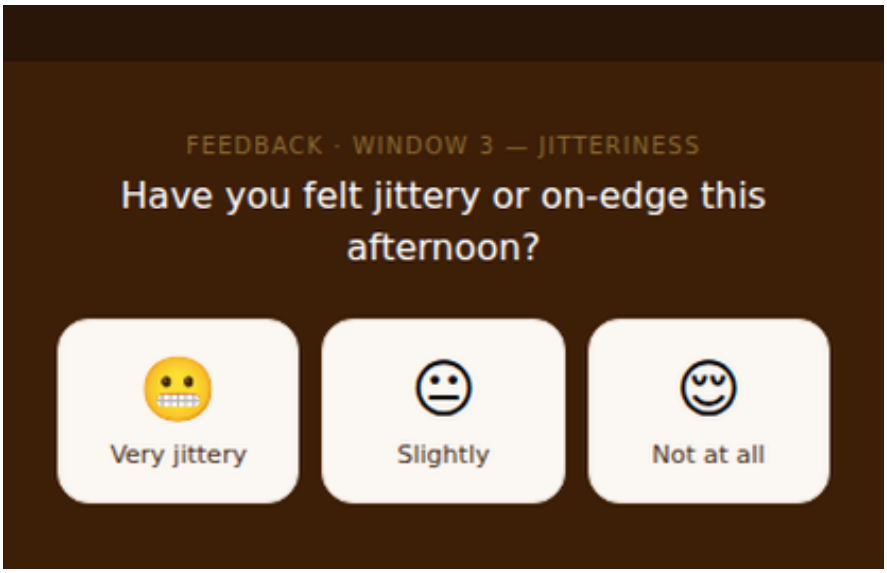
→ Refines the Wake Hours Ceiling. Posed mid to late afternoon, when overstimulation effects are most likely to have surfaced.

D4 — Bedtime

"Did caffeine affect how easily you fell asleep?"

→ Refines the Bedtime Ceiling directionally. Posed at the evening brew or last brew of the day, when the user is approaching the window the question is about.

D1 and D4 both inform the Bedtime Ceiling — D1 from aggregate sleep quality, D4 from sleep onset directly. Together they triangulate the boundary from two angles.



AFS — Boundary-Triggered Feedback Prompt (D3 — Jitteriness)

3.2 What the Responses Do

Each feedback response is not a direct instruction to move a boundary. It is a signal — one data point that the platform weighs alongside all prior signals for that dimension. The AFS applies an exponential moving average (EMA) methodology: recent responses carry more weight than older ones, so the platform adapts to changes in the user's habits, tolerance, or life circumstances without overreacting to a single outlier response.

In plain terms: the EMA works like a rolling memory that forgets gradually. A response from last week counts more than a response from three months ago, and a cluster of consistent responses in the same direction moves a boundary faster than a single isolated report.

The EMA weighting is also sensitive to boundary proximity: a response collected after a brew that closely grazed a ceiling carries more signal value than a response collected after a brew well within range. Responses gathered near boundaries are therefore weighted more heavily than those gathered far from them, concentrating the refinement signal where it is most diagnostic.

The boundary movement itself is incremental. The AFS moves boundaries in small steps, verifies the effect at the next boundary-grazing brew, and adjusts again if needed. This stepwise convergence avoids the jarring experience of a dramatically different recommendation from one day to the next while still getting the user to a well-calibrated Caffeine Comfort Zone within a few weeks of normal use.

3.3 CDR Refinement

Among the four feedback dimensions, Caffeine Decay Rate refinement is treated as the AFS's highest early priority. CDR refinement begins during Phase 1 Orientation — before any formal prompting starts — as the system passively collects brew timing and residual data from the shared daily log to begin converging the user's decay rate toward their actual biology. The CDR underlies every time-based projection the platform makes — residual level, caffeine headroom, projected bedtime residual — so an inaccurate CDR compounds across every recommendation. The Caffeine Comfort Zone boundaries can be wrong in one direction and the platform still produces useful guidance; a miscalibrated CDR systematically biases every output. The AFS therefore prioritizes decay-relevant signals in the early weeks of use, gathering D1 (Sleep) and D4 (Bedtime) responses on an accelerated cadence so that CDR convergence happens before boundary refinement fully engages.

3.4 Boundary Update Rules

Two rules govern every boundary update. First, a minimum response threshold: no boundary moves on fewer than three consistent signals in the same direction. A single response, however clear, is treated as a data point rather than a mandate — the platform waits for a pattern before acting. Second, a maximum step size: no single update moves a boundary by more than a defined increment, regardless of how strong the signal cluster is. This prevents a dramatic shift from one anomalous session and ensures the Caffeine Comfort Zone converges steadily

rather than lurching. Together, these constraints mean the platform adapts to the user's actual biology without exposing them to abrupt changes in their daily caffeine experience.

The AFS treats reported outcomes — not observed outcomes — as its input. What the user reports experiencing is the signal, not what the platform's model predicts they should have experienced. This distinction matters: if the model and the reported experience diverge, the model is questioned, not the user.

AFS enrollment, phase status, and dimension response history are stored at the profile level — not the device level. A household with an Advancara device and five profiles has five completely independent AFS states. One user's opt-out has no effect on another's calibration, and one user's well-refined boundaries have no bearing on the starting point of a new profile added to the same device months later.

The boundary update rules govern how the platform refines the Caffeine Comfort Zone on its own. They do not constrain the user. The Wake Hours Ceiling and the Bedtime Ceiling can be manually adjusted by the user at any time, independent of where the AFS has set them. The platform's refinements inform the user's judgment — they never override it.

The Three-Phase Enrollment Arc

The AFS enrolls the user gradually — first observing, then validating and offering, then calibrating — so that participation is earned, not assumed.

Phase 1 — Orientation (approx. Week 1)
The system observes baseline brew behavior silently. No prompts are shown and no user involvement is required. The user is not notified that Orientation is underway. During this period, the system passively collects from the shared daily log: brew count per wake cycle, serving sizes, brew timing patterns, first and last brew times, and any boundary proximity events that occurred naturally during normal use. This data establishes the pre-prompt behavioral baseline — the picture of this user's real brew behavior before any AFS signal has been collected — against which early feedback signals are calibrated. The Caffeine Decay Rate refinement process also begins during Orientation: every brew contributes timing and residual data that the AFS uses to begin converging the CDR toward the user's actual biology, even before formal prompting starts.

Phase 2 — Introduction (approx. Start of Week 2)
At the first brew of Week 2, the AFS presents the user with a structured Introduction sequence. This is not simply an enrollment offer — it is a full validation moment comprising four steps presented in sequence:

Step 1 — Observation Summary (P2-OBS). The platform presents the behavioral baseline it derived from Week 1 silent observation: the user's typical brew count, serving patterns, timing habits, and whether any Caffeine Comfort Zone boundary proximity events occurred. The user sees, for the first time, what the platform has been learning about them.

Step 2 — CBI Validation Questions (P2-VAL). The platform presents a short set of targeted questions mapped directly back to the original Caffeine Behavior Instrument inputs, asking whether the user's reported onboarding thresholds matched their actual lived experience during the observation week. These include at minimum:

- *"Over the past week, did the number of servings it took before you felt alert and focused match what you reported when you set up your profile?"*
- *"Did your overstimulation threshold — the point where you felt wired or jittery — feel consistent with what you told us?"*
- *"Was your reported caffeine cutoff time accurate, or did you find yourself brewing later without it affecting your sleep?"*

The answers to these questions are used immediately to flag any CBI-derived boundary that the user's lived experience suggests may be miscalibrated — marking those boundaries as higher-priority refinement targets for Phase 3.

Step 3 — Boundary Review and Adjustment (P2-BDY). The platform presents the user's current Caffeine Comfort Zone parameters — Peak Zone Floor, Wake Hours Ceiling, Bedtime Ceiling, and Caffeine Decay Rate — and invites the user to review them and, if anything feels off, to adjust the two ceilings before AFS prompting begins. This is the user's opportunity to correct a ceiling that their first week of experience has revealed as inaccurate. Adjustments made here are applied immediately. Phase 3 Calibration then begins from this user-validated baseline rather than the raw CBI provisional prior.

Step 4 — Enrollment Offer (P2-ENR). With the observation summary reviewed, validation questions answered, and boundaries confirmed or corrected, the platform presents the AFS enrollment offer: a brief explanation of what check-in prompts involve, and two options — "Count me in" or "Not right now." The user opts in with complete visibility into what the platform has observed and with boundaries they have confirmed or corrected. If the user declines, the three-stage opt-out arc begins.

Phase 3 — Calibration (ongoing)
Once enrolled, the user receives boundary-triggered and brew-count-triggered prompts for each feedback dimension according to the cadences described above. Refinement proceeds from the user-validated baseline established in Phase 2 — not from the raw CBI provisional prior. Questions fire only when a brew lands near a relevant Caffeine Comfort Zone boundary, or on the brew-count proactive floor cadence. As responses accumulate, the provisional boundaries give way to boundaries shaped by how this specific person actually experiences caffeine.

The Opt-Out Arc

AFS participation is always voluntary. The three-stage opt-out arc is designed to distinguish temporary disengagement from genuine objection. The first opt-out pauses prompts and presents a re-enrollment offer — the user may simply have found the timing inconvenient. The second opt-out pauses prompts for a longer window. The third opt-out is permanent: prompts are disabled for the profile and do not resume. The device operates normally under the stored Caffeine Comfort Zone boundaries regardless of AFS enrollment status. The platform does not require ongoing feedback to function — it benefits from it.

Opt-out is reversible at the first and second stages — the user can re-enroll at any time, and all accumulated signal history is preserved when they do. No prior data is discarded on opt-out; only the prompting is paused. A user who opts out after six weeks of calibration and re-enrolls a month later returns to their accumulated history intact, with refinement resuming from where it left off rather than starting over.

> ***The Science Advises. The User Decides.***
>
> Where reported outcomes diverge significantly from what the platform's model predicts — a gap the system recognizes as a 50% variance trigger — the AFS does not quietly recalibrate. It surfaces the discrepancy as an informed-choice advisory. The user is told, plainly, that the platform's working model and their reported experience don't match, and is invited to reflect. This is the AFS expression of the platform's core Advisory Philosophy: the science advises. The user decides.
>
> *The platform does not override the user's lived experience with a model. It flags the discrepancy honestly and leaves the next step to the user.*

When Answers Need More Context

Some feedback responses touch more than one dimension. A user who reports poor sleep and high jitteriness on the same day is providing correlated data that touches both the Bedtime Ceiling and the Wake Hours Ceiling. The AFS is designed to recognize these co-occurrences and to weight them accordingly rather than routing each answer in isolation to a single dimension. In cases where a response is ambiguous — for example, poor sleep that could be explained by caffeine or by any number of other factors — the AFS assigns lower confidence to the update and requires more confirming responses before moving the boundary.

When the Caffeine Behavior Instrument's initial answers were themselves contradictory or at the edge of what the derivation can resolve — for example, a user who reports needing very few cups to feel alert but also very few cups before feeling wired — the AFS enters a higher-sensitivity early-calibration mode. Boundaries that were set with lower confidence at onboarding are marked as higher-priority refinement targets, and the AFS prioritizes questions in those dimensions first.

3.5 The Role of Scientific Research

The Caffevolve™ platform is grounded in peer-reviewed pharmacokinetic research on caffeine metabolism and its effects on sleep architecture. That science informs the decay model, the bedtime ceiling values, and the Reality Range variance band. But in the AFS — as in the rest of the platform — science is consultative, not constraining. The research tells the platform what to expect. The user's reported experience tells it what is actually happening for this specific person.

When a user's reported outcomes diverge from what the pharmacokinetic model would predict — when a 50% variance from model expectations is triggered — the AFS surfaces this as an informed-choice advisory rather than silently overriding either the model or the user. The user is given the information they need to make a considered decision: the platform's model, what their reported experience suggests, and the option to explore whether a different boundary setting would serve them better. This is not a fault condition. It is the system working as

designed — honest about uncertainty, respectful of individual variation, and always deferring the final call to the person holding the mug.

> ***FR-Meta: The 60-Day Check-In***
>
> Every 60 days, FR-Meta — the third layer of the AFS prompt cadence — presents a single meta-prompt asking whether the check-in frequency feels right. Three options: “Too often,” “Just right,” “Not sure.” The purpose is simple: surface engagement fatigue before it leads to opt-out. FR-Meta fires on a calendar cadence — not tied to any brew event or boundary trigger.

4. The User Profile 

A significant amount of information about the user is needed for Platform Intelligence to build a caffeine recommendation each time a brew is initiated. The responses the user provides to the **Caffeine Behavior Instrument (CBI)** are a large contributor to this information base. If the user was required to input all of this information every time they wished to brew a cup, the process would likely be quite time consuming and tedious. Given that virtually all of this information is relatively stable — and not changing often — it can be trusted for relatively long periods of time, and recalled with a single tap each time the user commences a brew process. This is a much more positive user experience and eliminates the possibility of inaccurate information being entered. This is the purpose of the User Profile.

When a user decides to initiate a brew, they simply select their Profile, enter an expected bedtime for that day, and a caffeine recommendation can be generated. It requires about five minutes to initially set up the User Profile, but once completed, requires only about five seconds of the user's time to get a brew started. Efficient and convenient.

Included in the **Resident User Profile** are the parameters of the user's personal **Caffeine Comfort Zone** — their **Peak Zone Floor**, **Wake Hours Ceiling**, and **Bedtime Ceiling** — along with their personal **Caffeine Decay Rate**. In addition, a preferred grind setting and list of preferred additives are saved.

The Resident User Profile also supports two optional personalization layers that tailor the device to the user's routine without reconfiguring every brew — **Sub-Profiles** and a **BrewNow!** Profile — described in 4.2, below.

4.1 Resident and Guest Profiles

There are two types of Profile recognized by the Caffevolve™ Platform: a Resident User Profile and a **Guest Profile**. The preceding description is that of a Resident User Profile — a resident of the household or office, or a frequent user of the device, who utilizes the Caffevolve™ brewing device on a regular basis. Given that the number of user profiles that can be created is limited, it is prudent to restrict these profiles to frequent users of the brewing device. The number of Resident User Profiles supported scales by tier: Originara supports 2, Advancara 5, and Culminara Office 20.

Because these slots are limited, the device administrator controls who receives one. A new Resident User Profile is started from **Administrator Functions** — the single, Admin PIN–gated entry on the device home screen — where the administrator enters only the basics: the user's first name, initials, and profile color. That reserves the slot. Everything personal is then completed by the user, on their own time and without the administrator PIN: the Caffeine Behavior Instrument, additive preferences, grind setting, and any optional Sub-Profiles or BrewNow! Profile. The administrator cannot answer the CBI on the user's behalf, and is never asked to.

Until the user finishes that setup, the profile is *unfinished*. It appears on the profile carousel marked *Ready for you*, and the profile-select screen greets the user with a welcome card ("Your profile is set up and waiting") offering *Finish my profile* or *Not now*. Meanwhile the resident brewing paths — Home, Away (CaffevolveGo), and Plan my Day — remain locked, because a recommendation built on an uncalibrated profile would be misleading. In the meantime the user is free to brew through any guest channel. The user may pause setup and resume it later, but cannot discard the reserved slot; only the administrator can remove a Resident User Profile.

A Guest Profile is automatically established when a guest completes the Quick Profile Path — name, initials, a self-selected caffeine sensitivity level, and a declared bedtime. The Guest simply selects a caffeine sensitivity level based on their own gut feeling rather than having it determined through their responses to the Caffeine Behavior Instrument (CBI); the Guest Profile is established with known average parameters to establish a personal Caffeine Comfort Zone, while the Resident User Profile benefits from the personalized parameters set by completing the CBI process. At the end of the Quick Profile Path, the guest is asked whether they are likely to brew again: if so, the profile is saved as a **Returning Guest Profile**, loading and saving settings from the guest's last completed brew (size, strength, additives) for reuse via the Returning Guest Path; if not, the profile is used for that single brew only and then discarded.

Guest Sensitivity Presets — Guest profiles only

Selected on the Quick Profile Path. Not applied to Resident Users.

SENSITIVITY	WAKE HOURS CEILING	BEDTIME CEILING	PEAK ZONE FLOOR	CAFFEINE DECAY RATE
Higher	200 mg	30 mg	40 mg	7.0 hr
Moderate	300 mg	50 mg	40 mg	6.0 hr
Lower	400 mg	70 mg	60 mg	5.0 hr

These fixed presets are a deliberate conservatism shortcut for guests, who do not complete the Caffeine Behavior Instrument. Higher sensitivity pairs the lowest ceilings with the slowest assumed clearance (7.0 hr) — both the more protective setting — so the two move in the same direction. Resident Users receive no tier: their boundaries are computed individually by the CBI from reported behavior.

4.2 Sub-Profiles & the BrewNow! Profile

Beyond the base Resident User Profile, two optional layers let a user tailor the device to their daily routine without reconfiguring every brew: **Sub-Profiles** and the **BrewNow!** Profile. Both are conveniences layered on top of the base profile; neither changes how the caffeine blend is calculated, which is always computed fresh at brew time.

A **Sub-Profile** is a time-windowed, day-patterned variant of the profile — a saved brew recipe that activates on its own during the part of the day it covers. Each Sub-Profile captures a name, a time window, a day pattern, a serving

size, a serving count, a brew strength, and additives. It does not store a caffeine blend: the blend is always recomputed at brew time from the user's Caffeine Comfort Zone and residual level, so a Sub-Profile shapes the cup around the recommendation rather than overriding it. A user can build and save up to 3 individual Sub-Profiles on Originara and Culminara Office, or up to 6 on Advancara — routines tied to a particular day and time window, such as a weekday-morning boost or a weekend wind-down blend.

Sub-Profiles are built to disappear into the user's routine. Once set, an active Sub-Profile applies silently at brew time — no selection required — so the user simply brews and the right serving, strength, and additives are already in place. The platform imposes no lifestyle template: each Sub-Profile is assigned to whatever combination of days the user chooses, and the same day can carry more than one Sub-Profile as long as their time windows don't overlap. Setup is conversational — a name, a time window, the days, the strength, and the additives — and a Weekly Picture shows the full week of coverage before anything is saved, with any uncovered hours falling back to the standard brew flow. Sub-Profiles are entirely optional; a user can skip them at setup and add them later.

The **BrewNow!** Profile is the express-brew counterpart: a single saved set of defaults — serving size, serving count, brew strength, and additives — applied whenever the user taps BrewNow!. Its purpose is a one-tap brew, so it is deliberately independent of the time of day. When a brew is initiated while a Sub-Profile window is open, the platform applies that Sub-Profile's serving size, serving count, brew strength, and additives on the QuickBrew path. The express BrewNow! path is the one exception: it always uses the BrewNow! Profile's own defaults, even inside an open Sub-Profile window. As everywhere else in the platform, every one of these values remains adjustable at brew time.

Each of these three additive stores — the profile's My Additive List, each Sub-Profile, and the BrewNow! Profile — records additives by reference to what is loaded in the device. Because a device's loaded additives can be changed at any time (see Section 6, *When a Loaded Additive Changes*), a saved selection can come to name an additive the device no longer carries. When that happens the platform flags the affected profile and surfaces a one-time review notice before that user's next brew, so the saved additives are reconciled against the current device loadout rather than silently swapped. The caffeine recommendation is never affected — additives and the blend ratio are independent.

4.3 Profile Security

A user's profile is not a set of cosmetic preferences — it holds the calibrated boundaries the platform reasons from: the user's Caffeine Comfort Zone, which is itself the quantified expression of that user's sensitivity, together with their personal Caffeine Decay Rate. The integrity of that profile matters, because acting under it, or altering it, changes the very basis on which the platform advises. On a personal device this is a non-issue: the device belongs to one person, and there is nothing to guard against. Profile security becomes necessary the moment a device is shared.

The Culminara Office device is exactly that — a single device used by many people in a workplace — and its handling of profiles reflects it. A profile intended for use at work carries a user-chosen four-digit PIN, established

when the profile is set up; on the profile-confirmation screen the PIN is shown masked and is entered and confirmed there. Selecting a profile is the mandatory first action at the device: nothing can be brewed until one is chosen. The PIN is therefore required at the moment of selection. That single authentication establishes the active user: once a profile has been selected with its PIN, brewing as that profile and reviewing or changing its settings follow without further prompts. Because a profile's sub-profiles are variants of the same person's settings, that one authentication carries down to them as well — the gate admits the user to their profile and to every sub-profile beneath it. Authenticating the selection thereby protects everything the selection unlocks — brewing as that person, and access to their profile and its boundaries — with one gate at the point of entry. Within a profile, one field is the exception: revealing or changing the PIN itself requires re-entering the current PIN, so that an already-open profile never exposes the credential that guards it.

This profile PIN is deliberately separate from the device's administrator PIN, which governs the machine's own configuration — its network, beans, canisters, and hardware settings — and its roster of users: adding and removing Resident User Profiles. All of these tasks are reached through a single entry, **Administrator Functions**, with one administrator PIN entry per visit. The two PINs protect different things and never overlap: the administrator PIN secures the device and decides who holds a profile on it; the profile PIN secures the person.

The home device, being personal rather than shared, carries no such gate. Profile security is a property of the shared setting: the platform requires it where a device serves many people, and leaves the personal device unencumbered.

Guest profiles likewise carry no PIN. A guest declares only conservative, self-selected values — a sensitivity level and tonight's bedtime — and runs on known average parameters rather than the CBI-derived calibration a profile PIN exists to protect. A one-time guest's profile is discarded when the session ends, and a Returning Guest Profile retains nothing more sensitive than the convenience of its last brew. With no personalized reasoning basis at stake, the gate that guards a Resident User's profile would add friction without protecting anything, and is deliberately absent.

5. Plan my Day 

Every other path in the platform resolves a single cup in the moment. **Plan my Day** is the platform's whole-day layer: the user lays out an entire day's drinks in advance and is able to adjust the caffeine of each one, until the whole day sits inside their **Caffeine Comfort Zone** before the first cup is brewed. It draws the platform's foundations — the user's Caffeine Comfort Zone and **Caffeine Decay Rate**, and the device's ability to blend caffeinated and decaffeinated coffee to order — into a single instrument aimed at the day ahead. It is a planning tool, not a brew path: it commits nothing to the device and brews nothing itself. As everywhere else, the platform advises; the user decides.

Plan my Day works in two parts. The *Planner* (5.1–5.3) is where the day is laid out and shaped in advance. **Plan Check** (5.4–5.5) carries that plan through the real day: each time a planned drink comes due, it re-solves the rest of the day against what the user has actually had, so the plan stays true as the day drifts.

5.1 Two boundaries, one projection

A single **Brew Session** answers a local question — the strongest cup that is safe right now. Plan my Day answers the whole-day version: what the entire day can look like if every drink is strong yet safe. A day of caffeine is a budget spent against two boundaries at once — the *Wake Hours Peak*, checked against the ***Wake Hours Ceiling***, and the *bedtime residual*, checked against the ***Bedtime Ceiling***. The Planner projects them by summing every planned dose, each decayed by the user's personal Caffeine Decay Rate (a typical profile clears at roughly a six-hour half-life), together with whatever remains in the system from the night before, across the full waking day, and reading both values off the resulting curve. Every change the user makes re-projects immediately. Because caffeine clears slowly, the day resolves into a clean budget problem: spend earlier, taper and blend down toward evening — so a late-afternoon or after-dinner cup becomes a deliberate choice, shown inside both ceilings before it is brewed. The bedtime the user declares before the Planner opens becomes that plan's planned bedtime, the time against which the Bedtime Ceiling is checked.

5.2 Shaping the day

The Planner is a what-if simulator, and this is where the user takes control of the whole day at once. Every planned drink is adjustable — its time, its place, what it is, and how much of it there is. What can be tuned follows the device: a coffee made on a Caffevolve device, at home or at an equipped office, is tuned by serving size *and* caffeine blend (a dial from full strength to decaffeinated); a drink from a place without a device — a café, a shop, a workplace counter — is shaped by size, or shot count for espresso. The user's profile already records which workplaces are equipped, so the Planner offers the right controls without asking. Each adjustment re-reads the peak and bedtime figures live.

Up to two drinks can be *protected* — held exactly as planned while the rest of the day is shaped around them; none is protected by default. When the plan lands over either ceiling, *Rebalance for me* adjusts every unprotected drink to bring the day back under both. Drinks from places without a device size down first, the latest first. Device cups then share the remaining room in a gentle taper: earlier cups, which have longer to clear, stay stronger, and no cup is cut by more than fifteen points beyond the cup before it, so the day steps down toward evening rather than leaving the afternoon at little or no caffeine. Rebalance applies its result directly, and each changed drink carries a back arrow that returns it to its earlier setting; the user can keep adjusting by hand from there. A user who wants most of the day's caffeine in the morning protects the morning cups. When even every unprotected drink at its minimum cannot clear the day, the Planner says so, and names what would have to give: unprotecting a drink, moving it earlier, or taking it decaf.

5.3 Routines and one-off days

A shaped day can be saved as a Routine and set to repeat on any chosen days. On a day a Routine covers, it becomes that day's active plan automatically, projected fresh against the current clock and the user's real state before any brewing begins. On days when a Routine doesn't apply, the Planner lets the user build today's plan from any starting point — a blank day, a saved Routine, or a recent day — and commit it with "This is my plan for today," taking precedence for that day only before the Routine resumes. Whether built for an empty day or opened partway through one, the projection always starts from the user's real state: opened mid-day, it folds the drinks already had into a locked baseline drawn from the device log and any logged drinks, along with what remains from the night before. A plan built for tomorrow starts from tonight's projected bedtime residual. Changes made during the day through Plan Check apply to today's plan only; a saved Routine changes only when the user saves it again from the Planner.

5.4 Plan Check: the plan at the point of brewing

Once a day's plan is active, **Plan Check** brings it forward on its own. Each time the user goes to make a drink — on the Make a Drink screen of a Caffevolve device at home or at an equipped office, or in the **CaffevolveGo** app when away — the platform looks for the planned drink due within about ninety minutes and re-solves the rest of the day against what the user has actually had. Drinks already had are frozen as actuals; every planned drink still to come is recomputed; both ceilings are checked. The notice shows the planned drink and where the day now lands, in one of three states:

- **On plan.** The notice confirms the planned drink and the bedtime figure, and the user brews as usual.
- **Off plan, fixable.** An unplanned drink or a change earlier in the day has pushed the day over a ceiling. *Rebalance* adjusts the remaining unprotected drinks by the same rule the Planner uses (5.2).
- **Off plan, beyond Rebalance.** The notice says the day cannot be brought back under, and what would have to give.

Once the current drink has been adjusted, *Brew this cup* takes the user straight into the normal **BrewNow!** path with the adjusted settings, under all the usual brew rules; *Confirm & Brew* opens the full picture first, where anything can still be edited. *View plan* expands the notice into the rest of the day, each drink editable for time, size and blend, with the figures re-solving on every change. Nothing is brewed without the user's tap, and the notice appears only for a **Resident User Profile**. Away from the device the check comes earlier: a drink weighed in the Offsite Advisory System that would push the planned day over a ceiling is flagged before the user commits to it, and a drink logged in External Caffeine Entry is checked against the plan as it is logged, so the next planned drink opens with the day already re-solved.

5.5 When the real day differs from the plan

A plan is a forecast; the real day drifts. Plan Check reconciles the two with a small set of rules, so the projection always reflects what was actually had:

- **A drink within ninety minutes of a planned drink is that drink**, before or after its planned time. Where two windows overlap, it is the nearest one not yet had; on a tie, the earlier. The drink takes the planned drink's place at its actual time and settings.
- **Outside every window, the platform asks** "Does this replace a planned drink, or is it an extra?", offering the nearest drinks not yet had. This applies equally to a drink logged in CaffevolveGo, which can stand in for a drink planned at the device.
- **A planned drink whose window passes with nothing brewed or logged** keeps counting until the user says otherwise — the safe assumption, since leaving out a drink that was had would understate the day. At the next interaction the platform asks what happened: *Had it somewhere else* or *Skipped it*. When more than one planned drink in a row passes this way, the earlier ones are removed automatically and only the most recent is asked about.
- **A different bedtime declared mid-day** re-solves the whole day against it at once.

The one gap no rule can close is caffeine had away and never logged; the platform can only count what it is told.

(The screen-by-screen walkthrough — building a plan and Diane's worked example — is in Section 8, The User Experience.)

6. The Device 

Section 6 covers the Caffevolve™ device hardware — the components that physically carry out the brew, turning Platform Intelligence's blend recommendation into the cup. All of it is made possible by a single foundational architectural decision: blend-then-grind. Before any hardware component is described, that decision deserves its own introduction — because everything the device does flows from it. The Hardware subsection then covers the components that execute the blend-then-grind architecture: the dual-hopper system, gravimetric dispensing, blending chamber, grind system, water reservoir, carafe, and firmware. The touchscreen interface through which the consumer interacts with the device — the profile-driven brew path a registered user follows from the Brew Gateway to a completed cup, and the guest path for users without a stored profile — is covered separately in Section 8, The User Experience.

The Blend-Then-Grind Architecture

The foundational Caffevolve™ invention is blend-then-grind: caffeinated and decaffeinated beans are blended in the system-computed ratio before they reach the grinder, not after. This single architectural decision is what makes precise caffeine management from a single brewing device possible at all.

Pre-grinding blending is the conventional alternative — mixing pre-ground caffeinated and decaffeinated coffees in some proportion. It is also an unreliable one. Caffeine distribution in a pre-ground blend depends on how uniformly the discrete caffeinated and decaffeinated particles mix before and during brewing, a variable that cannot be controlled with precision. Blend-then-grind eliminates that uncertainty at the source.

Why Blend-Then-Grind Changes Everything

The blend-then-grind method establishes the caffeinated-to-decaffeinated ratio at the whole-bean level, before the grinder sees the beans. Every particle of ground coffee that results contains both bean types in the target proportion — the blend is homogeneous at the particle level. This matters because caffeine extraction is a particle-level event. If the blend is established after grinding — by mixing pre-ground caffeinated and decaffeinated coffees — the result is a physical mixture of discrete fully-caffeinated and fully-decaffeinated particles. Caffeine distribution then depends on how well those particles mix before and during brewing, which is inherently variable and unpredictable. Blend-then-grind eliminates that variable entirely. Because every gram of ground coffee carries the same caffeine load, the Blend Ratio Formula's output is reliably accurate: the caffeine yield is consistent regardless of particle settling, channeling, or brewing variation. The math and the cup agree.

This is the foundational Caffevolve™ invention — and the reason precise caffeine management from a single device is possible at all.

6.1 The Paths to a Perfect Cup

Whichever path a cup takes, the intelligence is in the preparation — and the preparation is the same. The platform computes a caffeine blend recommendation for the cup. The dual hoppers then dispense caffeinated and decaffeinated beans gravimetrically in the exact ratio it calls for. The blending chamber combines them into a single homogeneous whole-bean blend, and the burr grinder grinds that blend to the chosen coarseness. From that point the platform offers two paths to a finished cup: the device can brew it — the *Brew for Me* path — or stop at the grind and hand off the grounds to finish by hand — *Fresh Grounds Only*.

Brew for Me

On the brew path the device finishes the cup itself: the ground blend passes into the brew chamber, where the device extracts it, dispenses any chosen additives into the brew, and delivers the drink — a single cup or a full carafe — carrying the exact caffeine dose the recommendation called for.

The brew path offers two options. **QuickBrew** walks the brew-time selections — size, servings, strength, and additives — pre-populated from an active sub-profile when one is open and editable before the cup is made; it is the path for a cup that differs from the everyday. **BrewNow!** is the express, one-tap option that brews straight from the profile's saved BrewNow! Profile, for the daily ritual that doesn't change often.

Fresh Grounds Only

Fresh Grounds Only is the path for coffee the device does not brew. It runs the identical preparation — recommendation, dispensing, blending, grinding — then stops one step short of extraction and delivers the fresh

grounds in the basket for the user to finish by hand, as a French Press, Pour-Over, or Cold Brew. Additives, dispensed into the brew, aren't part of a Fresh Grounds order.

Nothing intelligent is given up. The grounds carry the same precise, computed dose a brewed cup would; the grind is matched automatically to the method (coarse for French Press, medium for Pour-Over, extra-coarse for Cold Brew); and the result is projected against the user's ceilings and logged to the shared daily log exactly as a device brew is — so the caffeine it carries folds into the running residual model and shapes every recommendation that follows. Fresh Grounds Only exists because not every coffee ritual is a machine brew. It lets the manual methods a coffee drinker already loves inherit the same caffeine intelligence as everything else the platform makes.

6.2 The Hardware

Brew Components

The Dual-Hopper System

The device stores caffeinated and decaffeinated beans in two separate reservoirs: Hopper A, which holds the caffeinated beans, and Hopper B, which holds the decaffeinated beans. The dual-hopper architecture enables real-time blend ratio adjustment without any manual bean mixing — the system independently controls how much it draws from each hopper on every brew.

During device setup, the user declares the Bean Family Classification (Arabica or Robusta) and Roast Level Classification of the beans loaded in each hopper. Bean Family Classification determines the bean family caffeine multiplier applied to all subsequent caffeine calculations: Arabica is the baseline; Robusta carries approximately 1.8 times the caffeine content. Roast Level Classification is advisory for pairing consistency and has no impact on caffeine calculations. These declarations are stored as the Hopper Content Configuration and require Admin PIN access to modify after initial setup.

Twin Bean Certified Pairings

A Twin Bean pairing is a caffeinated and decaffeinated bean matched and certified together through the Caffevolve™ Twin Bean Program — a planned Caffevolve™ initiative developed in partnership with premium coffee roasters. Rather than treating the two beans as independent, the Program selects and characterizes them as a set: chosen wherever possible from a shared origin — the same growing region, ideally the same farm — so the caffeinated and decaffeinated components begin from a common flavor foundation, then each roasted to its own calibrated profile to account for the different way the two behave under heat. The result is a pair engineered to taste as though it belongs together at every blend ratio, so that a fully caffeinated morning cup and a decaf-leaning evening cup read as two expressions of the same coffee rather than two different ones.

Certification also captures the pairing's measured characteristics as a set: bean family, roast level, and caffeine density for both the caffeinated and decaffeinated component. Those values are what the blend calculation consumes, and a certified pairing supplies them directly rather than relying on the user's declaration at hopper

load. A Twin Bean Certified pairing is therefore the configuration in which cup consistency across the blend range is fully optimized, and in which hopper setup requires no manual entry at all.

Gravimetric Dispensing

The device dispenses beans by weight, not by time or volume. A precision scale measures the mass dispensed from each hopper in real time, stopping the auger the moment the target weight is reached. This matters because bean density varies — by roast level, by fill level, by ambient humidity — and a time-based or volume-based dispenser would introduce error that compounds across every brew.

Gravimetric dispensing is the mechanical link between the Blend Ratio Formula and the physical cup. The formula computes exact gram weights for the caffeinated and decaffeinated fractions; gravimetric dispensing delivers them. Without it, the mathematical precision of the blend calculation would not survive contact with the real world.

The Blending Chamber and Agitator Arm

After the hoppers dispense their respective bean loads, the beans drop into the blending chamber — the internal cavity where the blend ratio is physically realized before grinding. Passive mixing alone is insufficient: bean density and size differences between caffeinated and decaffeinated varieties mean gravity would produce an uneven distribution. The blending chamber addresses this with an active mechanical solution.

A stepper motor drives a reciprocating agitator arm through a controlled 180° arc, completing approximately five to six sweep cycles over two to three seconds. The result is a uniformly intermixed bean load in which caffeinated and decaffeinated beans are distributed evenly throughout the chamber before the combined mass drops into the grinder. The agitator arm design is standardized across all product tiers.

The Grind System

Ground particle size governs two distinct extraction measures at once: the solids yield that determines flavour (the SCA 20–22% band) and the caffeine extraction efficiency that the Blend Ratio Formula uses as a direct input — the proportion of the caffeine contained in the dosed beans that reaches the cup (0.6277 at the medium-grind reference). The two move together with grind but are denominated differently and are not interchangeable. Because the formula depends on caffeine extraction efficiency specifically, grind consistency is not just a quality concern; it is a calculation-accuracy concern. The device controls grind through a calibrated burr mechanism whose settings map to known values for both measures.

Grind setting is stored at the profile level and applied automatically at brew time. The user selects their preferred grind during profile setup; the device references the appropriate caffeine extraction efficiency at every subsequent brew without requiring the user to manage the relationship between grind and caffeine output.

The grind setting alone determines caffeine extraction efficiency; the blend ratio does not affect it. Because blend-then-grind establishes the caffeinated-to-decaffeinated ratio at the whole-bean level, both bean types enter the

burrs together and fracture identically under the same grind setting. The resulting grounds are homogeneous at the particle level, so the particle-size distribution — and with it the extraction efficiency the Blend Ratio Formula assumes — is the same at any blend ratio. A single stored grind setting holds across the full range of Caffeine Blend Recommendations the day may call for, with no per-brew adjustment required.

This same grind-and-dispense pipeline can also stop one step short of a brew. In the platform's Fresh Grounds Only mode (Section 8), the device runs the full calculation, blends, and grinds exactly as it would for a brew, then delivers the fresh grounds in the basket instead of extracting them — so a French Press, Pour-Over, or Cold Brew made by hand still carries a precise, computed caffeine dose.

The Water Reservoir

Every brew begins with water, and how much water the device holds — and how fresh it is — is a deliberate part of the design rather than an afterthought. On each consumer tier the reservoir is sized to hold an estimated full day of brewing on a single fill, and it is removable: the user carries it to the sink to refill and returns it to the device. Two goals shape it at once. It must hold enough for the day's estimated brew load, and it must stay light enough to carry comfortably when full.

Sizing the Reservoir

Reservoir capacity is built up from a single unit: the 12 oz serving. Producing a 12 oz cup draws more than 12 oz from the reservoir, because the spent coffee bed retains water it never releases and a small amount is lost to evaporation over the brew. Accounting for both, the device draws approximately 14 oz of water for every 12 oz served. Capacity from there is simply the number of servings a tier is estimated to deliver in a day, multiplied by that per-serving draw.

Reservoir Capacity by Tier

- **Originara** — 56 oz (approximately 4 servings), roughly 4.3 lb when full
- **Advancara** — 84 oz (approximately 6 servings), roughly 6.0 lb when full
- **Culminara Commercial** — 90 oz batch capacity, sized for continuous throughput rather than a single daily fill

Filled weight is treated as a design constraint, not just an outcome. Originara's full reservoir carries at roughly 4.3 lb and Advancara's at roughly 6.0 lb — both kept within comfortable one-handed carry — and each is supported by a positioned handle and a wide fill opening so the tank is easy to grip and quick to fill.

Culminara Commercial is built for cafés, offices, and hospitality venues. Its reservoir is a throughput batch replenished continuously across a service day rather than filled once each morning, so the daily-fill sizing logic of the consumer tiers does not apply to it, and constant turnover keeps its water fresh as a matter of course.

Why Fresh Water

Water that sits for days in a reservoir goes stale. It picks up off-flavors from the air and from the tank itself, and standing water is simply not the starting point a good cup deserves. The remedy is a reservoir designed to be refilled at the start of each day, and it also reminds the user when water has been standing too long. Fresh in, fresh out.

Two Reservoir Reminders

The reservoir carries two independent reminders. They answer different questions and behave differently — one gates a brew, the other only advises.

Water Sufficiency — a hard gate. Before every brew, the device compares the water in the reservoir against what the requested brew requires, reading level from the reservoir's water-level sensor. If the reservoir holds enough, the brew proceeds. If it does not, the device does not begin — it alerts the user that water must be added first. This is the platform's one hard water gate: a brew that cannot be completed with the water on hand does not start, so the user is never left with a half-filled carafe.

Water Age — a soft advisory. Independently, the device tracks how long the current water has been standing. The moment a reservoir is refilled, an age clock starts; if that water is still in the tank 36 hours later, the device surfaces a freshness reminder — a suggestion to empty and refill, never a block. The user can always brew with the water they have. The platform advises; the user decides.

The two reminders do not interact, and that independence is deliberate. Water level never suppresses the age reminder. A reservoir can be completely full and still be flagged as old — which is exactly the case that matters most. A user who has been away for several days returns to a full tank of water that has been standing the entire time; because only a refill resets the age clock, not the water level, that full, stale tank is correctly flagged for a refresh. A full reservoir is not evidence of fresh water. Only a recent refill is.

The First-Brew Freshness Reminder

At the first brew of the user's day — whenever that falls, since the platform is schedule-agnostic and a user's day may begin at any hour — the device offers a brief reminder that fresh water makes a better cup. When that first brew also lands past the 36-hour mark, the two messages combine into a single, stronger prompt rather than two separate nudges.

What Counts as a Refill

An age-clock reset requires a genuine refresh, not a splash. Topping a small amount of water onto a tank of older water does not restart the clock — if it did, a few ounces added to two-day-old water would misrepresent the whole reservoir as fresh. Using the level sensor already present, the clock resets only when a fill brings the reservoir up to near-full. A true refill resets the clock; a partial top-up does not.

Filling to Any Level

The reservoir is sized for a full day, but the user is never required to fill it to the top. Any fill level below capacity brews normally, because the pre-brew sufficiency check confirms adequate water before every brew — a user who prefers a lighter reservoir can fill partway and rely on the device to catch any shortfall before a brew begins. Graduated capacity markings on the tank let the user fill to a known serving level on purpose.

The Carafe

The carafe is the final stage of the brew path and the vessel in which in-stream additives — dispensed after extraction — complete their dissolution. It is where the day's coffee is held, and its design reflects two priorities: keeping that coffee hot without degrading it, and making the additive mixing that happens inside it easy to keep clean.

An Insulated Carafe on Every Tier

Every Caffevolve™ carafe is vacuum-insulated — double-walled, with no glass carafe and no warming plate on any tier. The reason is quality. A glass carafe holds temperature by sitting on a heated plate, which slowly scorches the coffee and flattens its flavor the longer it sits. Vacuum insulation holds temperature passively instead, keeping the coffee hot for hours without a burner and without cooking it. This also draws no standby power, and it reinforces what the platform is built around: a cup brewed earlier is still a good cup later, so an afternoon pot is still worth pouring in the evening.

Carafe capacity is matched to each tier and enforced by the device at brew time, which prevents a serving-count selection that would exceed the installed carafe:

> **Carafe Capacity by Tier**
>
> - **Originara** — approximately 40 oz insulated
> - **Advancara** — approximately 60 oz insulated
> - **Culminara Commercial** — 90 oz insulated server

The Removable Baffle System

As brewed coffee flows into the carafe carrying its dispensed additive, the carafe's internal baffle geometry sets the coffee spinning — a turbulent, helical path that mixes the additive through the full volume and drives its dissolution. Every tier carries this passive baffle; it is the mechanism that turns a dose dropped into the stream into an evenly integrated cup.

What makes the Caffevolve™ baffle different is that it lifts out. A baffle fixed to the carafe wall traps exactly what this device dispenses — coffee oils, and the proteins, collagen, and oil-based wellness additives that dry into films a brush can't reach in a fixed corner. So the baffle is built as an independent part: a central column with helical vanes

that seats in the carafe, structurally separate from the carafe wall, and lifts out as a single piece for cleaning. The vane edges stand off from the wall rather than meeting it, so nothing is trapped between the two — lift the column out and the carafe interior is a smooth, open vessel that wipes clean in one pass, while the column itself rinses clear under the tap.

The column is designed to be effortless in daily use. It locks into a socket in the carafe floor with a quarter-turn — twist to seat it before brewing, twist to release it for cleaning — so the incoming brew stream can't unseat it. It carries a finger loop at the top for lifting. And because the helical column is symmetric about its axis, there is no wrong way to put it back: dropped in at any rotation, the mixing geometry is the same. Coffee pours around the seated column to the spout, so the user removes it only to clean, never to serve.

Flex-Vane Vanes for the Insulated Neck

An insulated carafe has a narrower neck than a glass one — the double wall requires it — and a mixing column wide enough to work well in the body would be too wide to pass back out through that neck if its vanes were rigid. The Caffevolve™ column solves this with flexible vanes on a rigid central spine. As the column is drawn up through the neck, the neck presses the vanes inward and they bow toward the spine, narrowing enough to clear the opening; once past the neck, they spring back to full span. Reinsertion reverses it. The neck does the work — there is no lever, collar, or fold step for the user to operate, and because the vanes flex rather than pivot, the column has no hinges or joints of its own to trap residue. One baffle design serves every tier this way.

Culminara Commercial: Ultrasonic Assist

Culminara Commercial adds an active 40 kHz ultrasonic pulse to the same removable baffle geometry. The higher additive volumes and continuous throughput of a commercial setting benefit from the extra dissolution energy; the ultrasonic system lives in the carafe base, beneath the seated column, and works alongside the passive baffle rather than replacing it.

Care

The carafe and its baffle column are kept clean with a warm-water rinse after daily use, with a periodic warm-water-and-vinegar soak — column expanded and in place — to clear accumulated oils and films. Because the column lifts out and its vanes carry no joints, both the carafe interior and the column present only smooth, open surfaces to that routine.

Firmware

The device's onboard firmware is where Platform Intelligence runs on the machine itself. As Platform Intelligence running on the device, the firmware performs the full eight-step dual-ceiling calculation at every brew event — retrieving the current residual caffeine level, applying the Blend Ratio Formula, and translating the result into precise dispense instructions for the gravimetric system. It then drives the machinery that carries the result out: the

brew-cycle sequence of hopper dispensing, blending chamber agitation, grinding, brewing, and synchronized additive dispensing — and it maintains the running record that the companion app reads and writes against.

Firmware updates are delivered over-the-air and are managed at the device level. The calculation engine, the additive library, and the platform's pharmacokinetic constants are all firmware-resident, ensuring that every device in the field operates from a consistent and current version of the platform logic.

Every calculation the platform makes is only as good as the machine that executes it. The blend recommendation produced by the calculation engine specifies a precise caffeinated-to-decaffeinated ratio — and the device's job is to deliver that ratio in the cup, brew after brew, with the accuracy the math demands.

6.3 The Additive System

The hopper system manages caffeine. The additive system manages everything else the user chooses to put in their cup. Together they make the Caffevolve™ device more than a precision caffeine brewer — they make it a configurable daily supplement and flavor platform, all operating from a single brew cycle.

Canisters and Slot Codes

Where the device uses hoppers for beans, it uses canisters for additives. The two terms are not interchangeable: hoppers are the large dual bean reservoirs; canisters are the smaller refillable containers that hold powdered additives. Every device accommodates two canister families, each with its own dedicated slot code prefix: Flavor canisters occupy F-coded slots, and Wellness canisters occupy W-coded slots.

The number of canister slots scales with device tier, while the two-family Flavor / Wellness organization is retained across every tier.

> **Canister Configuration by Tier**
>
> - **Originara** — 3 canisters (2 Flavor: F-1, F-2; 1 Wellness: W-1)
> - **Advancara** — 6 canisters (3 Flavor: F-1–F-3; 3 Wellness: W-1–W-3)
> - **Culminara Office** — 12 canisters (6 Flavor: F-1–F-6; 6 Wellness: W-1–W-6)
> - **Culminara Café** — a greater number of canisters for broader self-serve service; count being finalized

Each canister carries a permanent alphanumeric slot code imprinted at manufacture — for example, F-1 for the first Flavor slot or W-2 for the second Wellness slot. The slot code is the canister's identity within the device and is used to register, track, and display its contents. Slot codes do not change when a canister is refilled or re-registered; the code identifies the physical slot, not the ingredient currently loaded in it.

Canister family segregation is enforced through three reinforcing layers. First, physical keying: F-family and W-family canister slots are physically shaped to accept only their corresponding canister family — a W canister

cannot be inserted into an F slot by mechanical design alone, requiring no sensor or firmware intervention. Second, color coding: F-family canisters and W-family canisters are manufactured in distinct colors, providing immediate visual differentiation before insertion is attempted. Third, alphanumeric slot codes: each canister bears a permanent human-readable slot code comprising a category prefix (F for Flavor, W for Wellness) and a sequential slot number, imprinted at manufacture, providing precise identification during registration and on the device's canister shelf display. Together these three layers ensure that cross-contamination between Flavor and Wellness additive families is prevented at the physical design level, independently of any software or firmware control.

The Two-Auger Dispensing System

The device uses a two-auger dispensing system: one auger dedicated to the Flavor canister family, one to the Wellness canister family. The separation is intentional — keeping Flavor and Wellness on independent auger pathways prevents cross-contamination between additive categories, regardless of which specific canisters are installed or how frequently they are swapped.

Within each family, a microcontroller-driven canister selection mechanism routes the appropriate canister's output to the shared auger pathway. Selection is entirely automatic — the user chooses their additive on the touchscreen, and the device handles the mechanical routing without any manual valve operation.

Dispensing is synchronized with the active brew flow. After extraction, as hot brewed coffee moves from the brew chamber toward the carafe, the auger meters a calibrated dose from the selected canister and delivers it to a dispensing port positioned in the coffee flow path. The hot, actively moving coffee entrains the powder and carries it into the carafe, where the baffled carafe geometry promotes turbulent mixing and dissolution. Dispensing into an active flow — rather than into a finished, static beverage — maximizes both dissolution efficiency and ingredient integration.

The Additive Library

The device does not permit unrestricted additive loading. Every additive that can be registered to a canister must appear in the Additive Library — a curated, Caffevolve™-validated on-device database of approved Flavor and Wellness ingredients. Library entries have been tested for compatibility with the dispensing system across four physical characteristics: particle size, density, flow rate, and moisture behavior. An ingredient that cannot be reliably metered, dispensed, and dissolved under in-stream conditions is not listed.

The Additive Library is filtered by canister family at the point of registration: when a user registers an F-slot canister, they see only Flavor entries; when registering a W-slot canister, only Wellness entries appear. The library is a Caffevolve™-managed asset, updatable by Caffevolve™ or its licensees as new validated ingredients are approved. Custom or user-added additives outside the library are not permitted.

Canister Registration

Registration is triggered automatically on canister insertion. Each canister slot incorporates a slot-level insertion sensor that detects when a canister has been successfully seated. Because the physical keying ensures that only a correct-family canister can be seated in a given slot, successful insertion confirms correct-family placement without any electronic verification. The device identifies which slot was triggered by the sensor and immediately presents a category-filtered registration prompt on the touchscreen — showing only additive library entries in the family corresponding to that slot. The user selects the specific ingredient loaded from the filtered library view. Once confirmed, the registration is stored and displayed on the canister's digital label.

The slot code is a human-readable label, not a machine-readable identifier. The device does not read or scan the slot code — it detects insertion via the slot sensor and relies on the physical keying to confirm family correctness. The slot code's role is to identify the canister to the user and to the device's display system, not to communicate with the device electronically.

Registration does not expire on its own. If the canister is refilled with the same ingredient, no re-registration is needed — only a refill confirmation to reset the fill-level counter. If the ingredient changes, the user initiates a new registration, which replaces the prior record for that slot.

Compound Categories and the Dissolution Hold Timer

Different additive compounds dissolve at different rates in hot coffee. A flavoring powder engineered for rapid dissolution in a hot beverage application behaves very differently from a protein supplement or an oil-based extract under the same conditions. If the device dispensed every additive the same way and moved on immediately, undissolved powder would reach the carafe — affecting both the beverage and the user's experience of the additive's intended effect.

The platform addresses this through the additive compound category system. Each Additive Library entry is assigned to one of six compound categories based on its chemical composition and physical dissolution characteristics. Each category carries a scientifically derived minimum hold time — the duration required for that compound type to fully dissolve in hot coffee under active flow conditions. When an additive is dispensed, the Dissolution Hold Timer reads the compound category of the selected ingredient and applies the appropriate hold duration automatically. The user never manages hold times directly.

The Six Additive Compound Categories

Each entry in the Additive Library is assigned to one of six compound categories. The category determines the dissolution hold time applied automatically after dispensing.

Proteins & Peptides — Longest hold time. Dense molecular structures require extended exposure to hot liquid for full dissolution.

Fats & Oils — Extended hold time. Emulsification into hot coffee is time-dependent; turbulence from baffled carafe assists.

Plant Extracts — Moderate hold time. Dissolution varies by extract density and particle fineness.

Vitamins & Minerals — Moderate hold time. Water-soluble vitamins dissolve readily; mineral compounds may require slightly longer.

Amino Acids — Short hold time. Highly water-soluble; dissolve rapidly in hot liquid under active flow.

Flavor Additives — Shortest hold time. Fine-particle powders engineered for rapid dissolution in hot beverage applications.

Hold times are stored as properties of each Additive Library entry and applied automatically — the user selects the additive, the system determines the appropriate hold duration.

Fill-Level Monitoring

The device tracks the remaining content of each registered canister through a user-declared initialization and firmware dispense-event decrement system. Each additive canister incorporates six molded fill lines on the canister body, labeled 1 through 6 from bottom to top, providing a user-readable reference scale. At registration or refill confirmation, the user selects the fill line nearest to the powder level in the canister — declaring the starting fill level for that slot. No weighing mechanism is required.

From the declared starting level, the device tracks remaining content by decrementing the fill-level counter by the known dose weight of the registered ingredient on each dispense event. The three-state indicator — Green, Amber, Red — is computed independently from this running estimate as a percentage of full canister capacity: Green above 25%, Amber between 10% and 25%, Red below 10%. The fill-level scale and the indicator thresholds are independent — a user who declares level 3 of 6 starts in Green and crosses into Amber and Red at calculated thresholds, as does a user who declares level 6 of 6, just later.

The fill-level counter resets upon re-registration following a refill of the same ingredient, or initializes at the user-declared starting level when a new ingredient is registered. This ties fill-level tracking directly to the registration events established by the slot-triggered registration system.

Fill status is surfaced at two points: on the canister's digital label during normal operation, and as a pre-brew refill alert on the brew configuration screen when an Amber or Red additive is selected for a brew. The alert appears before the brew is confirmed, giving the user the opportunity to refill before the session begins.

Canister Fill-Level Indicators

Green — >25% estimated remaining **Amber** — 10–25% estimated remaining **Red** — <10% estimated remaining

Note: these thresholds apply to the firmware's running estimate based on user-declared starting level and cumulative dispense-event decrement — not a direct weight measurement.

Filling Funnels and Refill

Canisters are refillable, and refilling is designed to be clean and unambiguous. Two dedicated filling funnels are included with every device — one for the Flavor family, one for the Wellness family — each color-matched to its canister family. Both are stored onboard the device in a dedicated compartment adjacent to the canister shelf, so the correct funnel is always at hand at the point of use.

Two Filling Funnels

- **Flavor funnel** — warm tone, color-matched to Flavor canisters; seats into the open sliding top of a Flavor canister.
- **Wellness funnel** — cool tone, color-matched to Wellness canisters; seats into the open sliding top of a Wellness canister.
- **Onboard storage** — both funnels are held in a dedicated compartment adjacent to the canister shelf.

Each canister carries a sliding top that seals its contents during transport and slides open to expose the fill opening. Canisters are removable and are filled on the counter rather than in place: the user pulls the canister from its slot with the sliding top closed, retrieves the matching funnel, sets the canister on a flat surface, slides the top open, and seats the funnel — which seats positively into the open top — before pouring. The color-matched funnel and the family-specific keying together keep Flavor and Wellness additives from crossing during a refill.

When the canister is returned to its slot, the slot's insertion sensor detects the reinsertion and the device prompts the user to confirm the existing contents or register a change — the same registration flow triggered on first insertion.

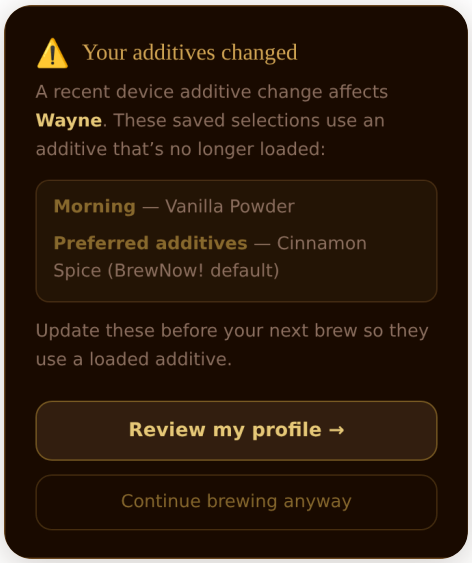
Additives and the Sub-Profile System

Additive selections are a dimension of personalization entirely separate from caffeine management. They are stored at the sub-profile level — each sub-profile holds its own additive canister selections and dose levels, independent of the blend ratio, which is always computed fresh at brew time from the caffeine calculation. A user can configure one sub-profile for a morning wellness routine with a specific Wellness additive and a separate sub-profile for an afternoon brew with a different Flavor additive, without either configuration affecting the other or the underlying caffeine recommendation.

When a sub-profile window is active at brew time, its additive selections become the default for that brew on the QuickBrew path. The one exception is **BrewNow!** — the express path always uses the BrewNow! Profile's own additives, even inside an open sub-profile window. Additive selections can always be adjusted at brew time regardless of sub-profile defaults. The additive system is designed to be present when the user wants it and invisible when they don't.

When a Loaded Additive Changes

The additive(s) a user profile relies on are chosen from what is loaded into the device. This selection of available additives can be changed at any time by an administrator, through Administrator Functions. If changes are made to what is currently loaded, an existing sub-profile may be calling for an additive the device no longer carries. Because user additive preferences live in three places — the My Additive List, one or more **Sub-Profiles**, and the BrewNow! Profile — a single additive change at the device level can reach into a number of routines the user set up, resulting in an unexpected additive in the cup when any of them is executed.



Additive-Change Review Notice

The platform reconciles this automatically. The moment an additive is added or changed in the device, it checks the My Additive List, every stored Sub-Profile, and the BrewNow! Profile against the new additive inventory and notes which of them will be impacted. Then, when an affected profile is next used to brew, the user is informed of the additive change(s) made at the device level and allowed to select an alternative(s) before it ever impacts their drink.

The change that is made is then stored as a new component of that My Additive List, Sub-Profile, or BrewNow! Profile moving forward.

7. CaffevolveGo 

CaffevolveGo is the platform's companion app — the component that extends **Platform Intelligence**'s reach beyond the device, into wherever the day actually takes the user. Every cup consumed where there is no Caffevolve device is a cup it would otherwise never know about; CaffevolveGo closes that gap, through two components: **External Caffeine Entry (ECE)** and the **Offsite Advisory System (OAS)**.

Both are reached through a single entry point: opening CaffevolveGo presents ECE and OAS together — record a drink already consumed, or weigh what to consume next.

7.1 External Caffeine Entry (ECE)

ECE lets the user quickly log every external drink consumed, keeping the platform's model of their caffeine intake accurate and up to the moment. Every external caffeine source is logged and fed back into the **Caffeine Comfort Zone** model, keeping the platform's intelligence complete across the full day. This ensures that when the user looks to brew or purchase another cup later in the day, the caffeine recommendation is always based on a complete picture of all they have consumed since waking. The device manages the cup at home; CaffevolveGo, through ECE, manages it everywhere else — closing the 24-hour window by capturing everything Platform Intelligence needs to know about the user's day, wherever it happens.

7.2 The Offsite Advisory System (OAS)

OAS is the second component of CaffevolveGo, and the one that puts the user's current caffeine standing directly in their hand when they are away from home — displayed in real time, so every decision away from home is as informed as every decision at it. Rather than simply logging what has already been consumed, OAS presents actual drink options against the user's current standing, turning the record of what's already been consumed into guidance for what to consume next.

OAS organizes these options by venue, since what is available — and how much control the user has over it — changes with the setting. At an equipped office this does not apply: the device brews to the user's profile and logs automatically, and OAS steps aside entirely. Where the workplace has no Caffevolve device of its own, though, it is the venue users encounter most consistently, and OAS treats it as its primary setting — the *Worksite*. The drinks there tend to be a shared office pot, single-serve pods, tea, and packaged or canned drinks, and the strength of a communal brew is fixed rather than chosen. OAS accounts for this by making serving volume the lever at the workplace — a full, two-thirds, or one-third pour — and by letting a user save the specific drinks they return to as personal workplace options, so that on each visit those recurring choices are presented and measured against their current standing.

Away from the workplace, OAS draws on its own database of venue-specific beverages for cafés and convenience stores — presenting the estimated caffeine content of all likely available drinks at that venue alongside the user's real-time caffeine and headroom status. The screen-level behavior of all three venues is detailed in Section 8.

Both components draw on the same Platform Intelligence that drives the brewing device. Whatever the user chooses is sent immediately back to it, logged and folded into that day's caffeine calculations, so the next recommendation at home remains accurate.

8. The User Experience 

The Caffevolve™ device serves two types of users: **Resident Users**, with stored profiles and personalised blend recommendations, and **Guest Users**, who brew through a quick-profile or express path without a permanent record. This section walks through every screen in sequence — device setup, profile creation, brew paths, Guest paths, and External Consumption.

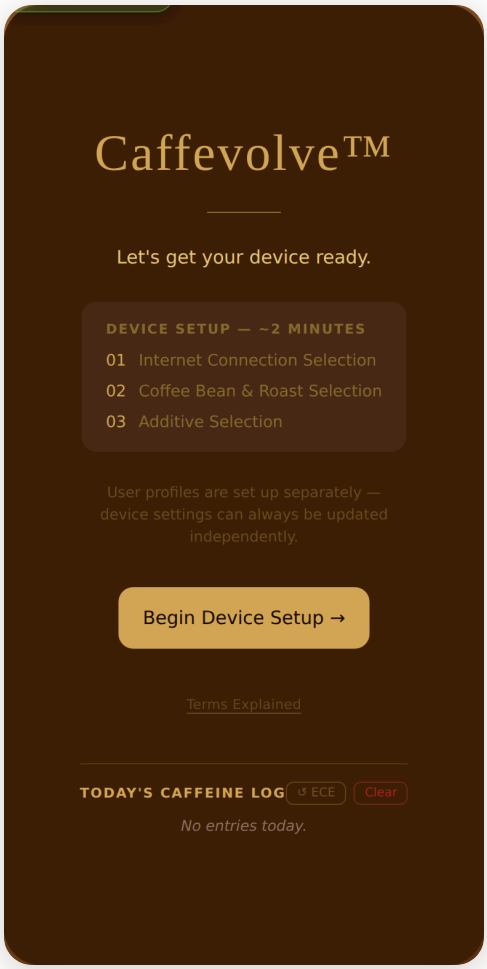
The device offers two ways to make coffee, and a way to plan ahead. From the Home menu, **Brew for Me** has the device brew the cup or carafe itself (the **BrewNow!** and QuickBrew paths), while **Fresh Grounds Only** has the device grind a computed blend into the basket for a French Press, Pour-Over, or Cold Brew they finish by hand. Separately, **Plan my Day** lets a user lay out a full day of drinks in advance to see where the caffeine level lands, and adjust the plan to keep themselves within their **Caffeine Comfort Zone**. All three are reached from the profile screen (PS1a) and are covered in this section.

Anatomy of a Brew Session

A **Brew Session** is the complete sequence from device wake to a finished cup, and it comprises two phases. Phase 1 of the Brew Session — Pre-Brew Configuration — begins at the Brew Gateway, the device's brew entry point, and ends at hold-to-brew confirmation: the user selects a profile, declares tonight's bedtime, makes the brew-time selections, and reviews the caffeine recommendation, with the option to adjust additives or caffeine level. Every residual calculation, dual-ceiling evaluation, and blend-ratio computation happens in this phase; no physical machine operation has yet begun. Phase 2 of the Brew Session — the Active Brew Cycle — is initiated by hold-to-brew confirmation and carries out every physical operation in sequence: gravimetric bean dispensing, blending, grinding, brewing, and synchronized additive dispensing. No profile changes or recalculations occur once the Active Brew Cycle has begun.

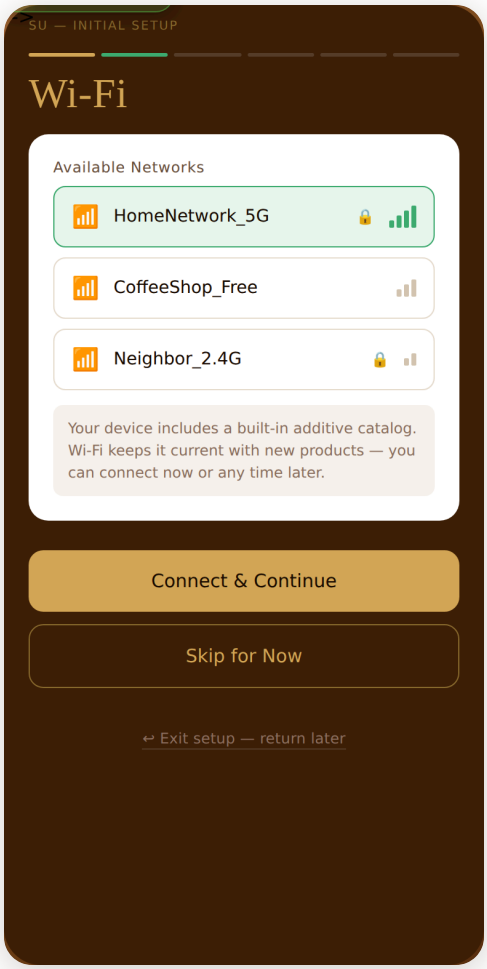
These Brew Session phases are distinct from the **AFS** enrollment phases of Section 3 — Orientation, Introduction, and Calibration — which describe the feedback system's rollout across weeks of use, not the two stages of a single brew.

8.1 Set-up



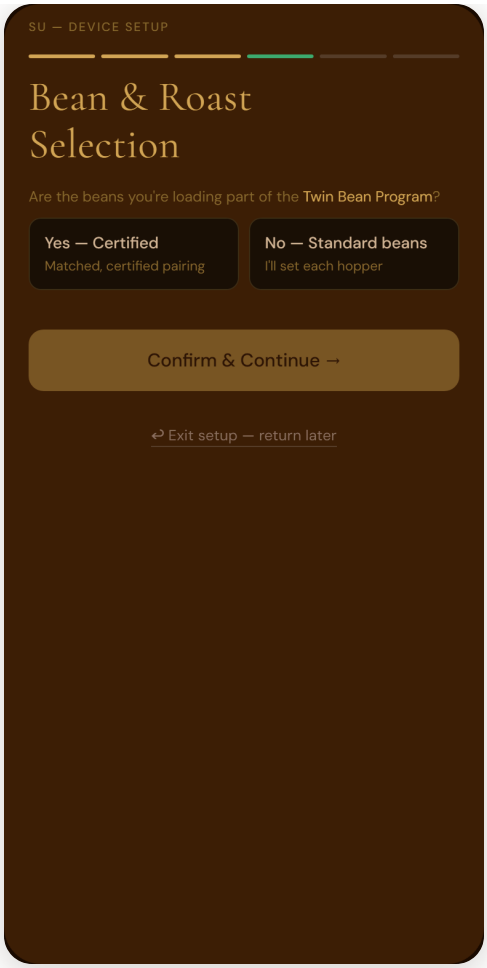
SU0 — Device Welcome

SU0 — Device Welcome. Lists the three setup steps: *01 Internet Connection Selection, 02 Coffee Bean & Roast Selection, 03 Additive Selection.* User profiles are set up separately — the first during initial setup, any additional profiles later by the administrator. Tap *Begin Device Setup* to start.

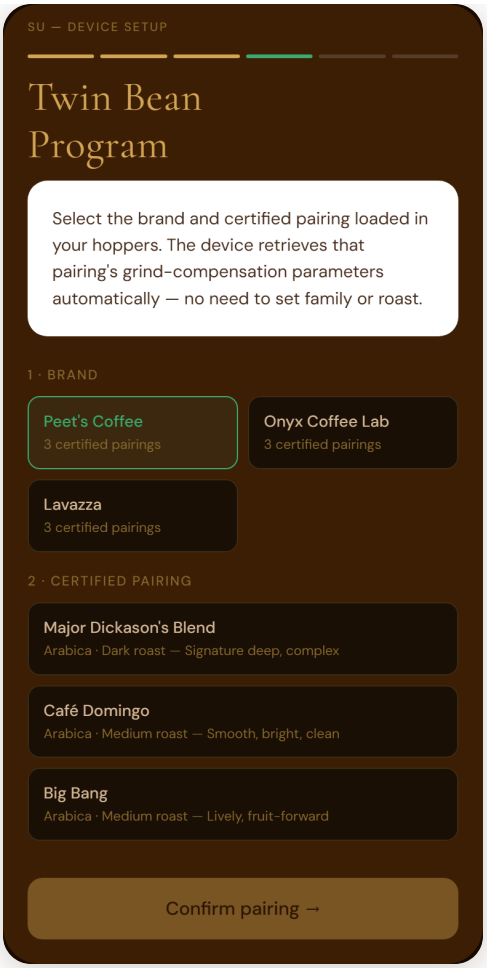


SU2 — Internet Connection Selection

SU2 — Internet Connection Selection. Available Wi-Fi networks are listed. The device ships with a built-in additive catalog — Wi-Fi keeps it current with new products and seasonal additions. The user selects a network, or taps *Skip for Now* to connect later.



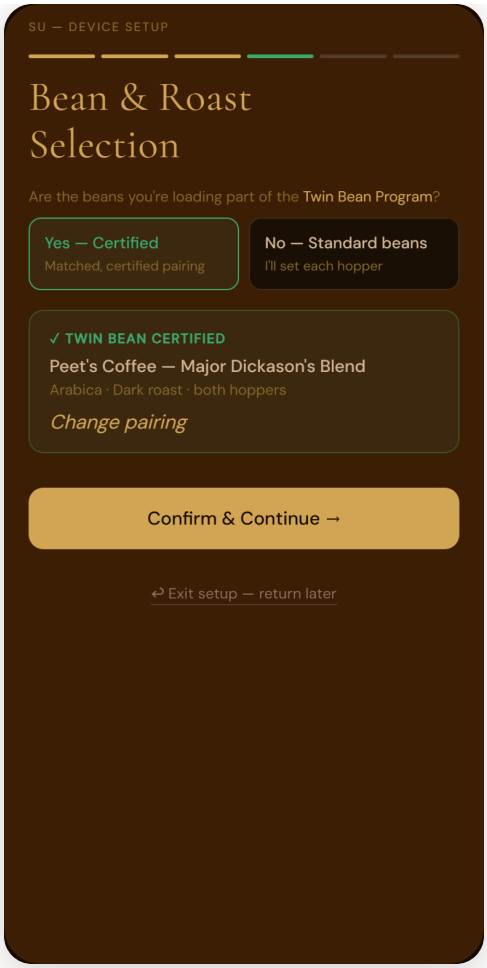
SU-BEAN — Bean & Roast Selection



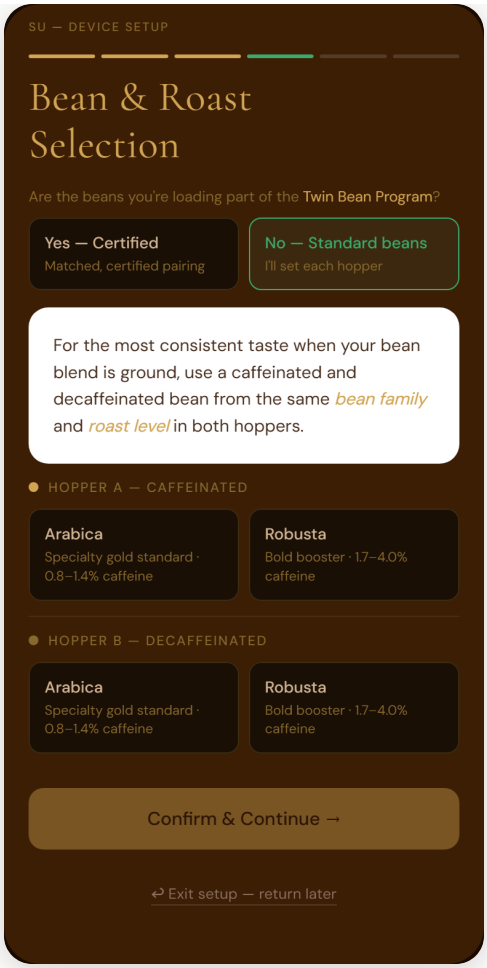
SU-BEAN-TBP — Twin Bean Program

SU-BEAN — Bean & Roast Selection. Setup step 02. The device first asks whether the beans being loaded are part of the *Twin Bean Program* — Caffevolve's certified-pairing program, in which a partner roaster supplies a matched caffeinated and decaffeinated pair. The answer routes the user down one of two paths: *Yes — Certified* or *No — Standard beans*.

SU-BEAN-TBP — Twin Bean Program. The certified path. The user selects the partner brand and the specific certified pairing loaded in the hoppers. The device retrieves that pairing's bean family, roast level, and caffeine density automatically from the certification record — so no hopper values need to be entered by hand. Shown: Peet's Coffee, with three certified pairings available. *Shown for illustrative purposes only: the Twin Bean Program is a planned program.*



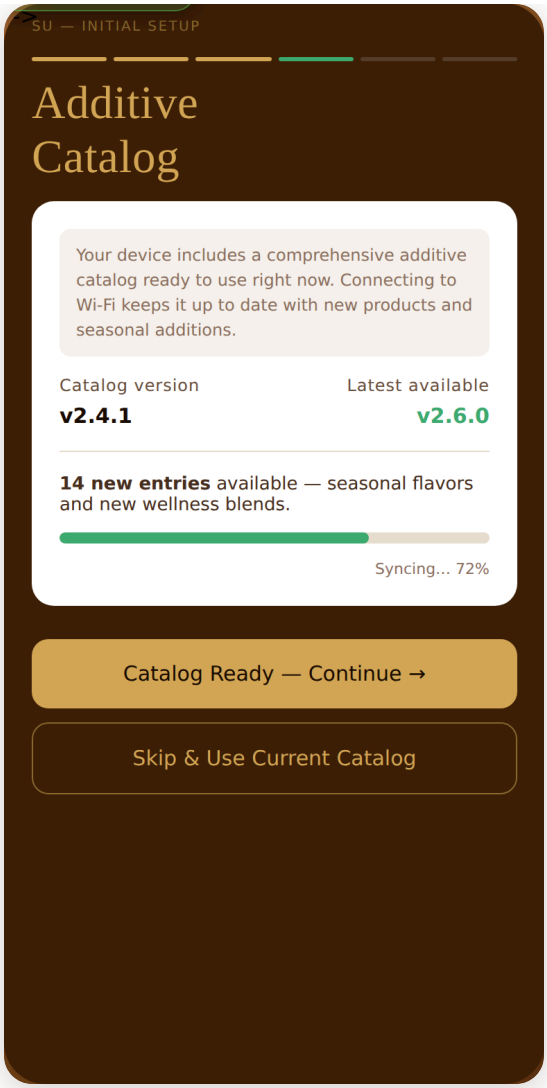
SU-BEAN — Twin Bean Certified



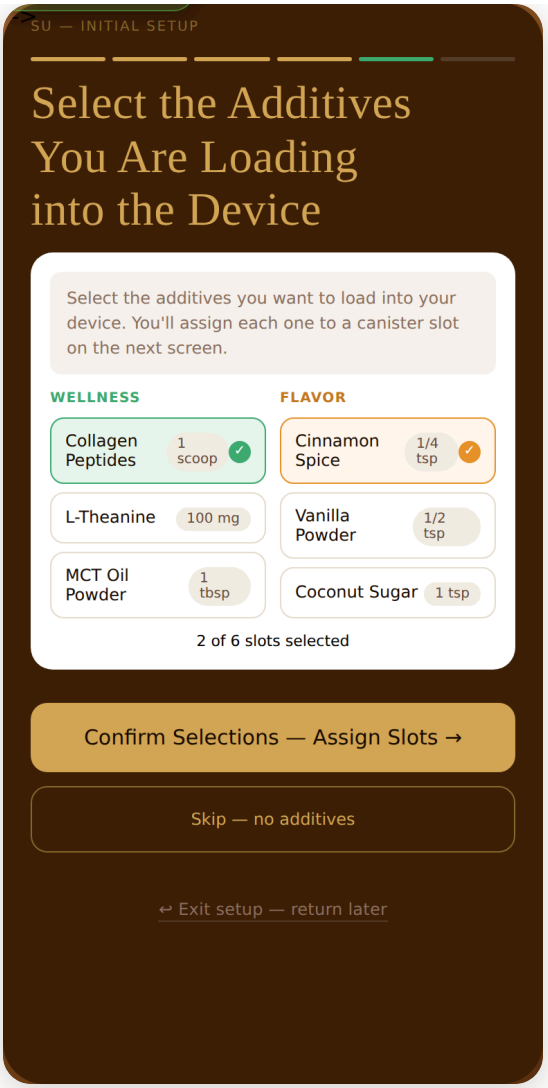
SU-BEAN — Standard Bean Path

SU-BEAN — Twin Bean Certified. Returning from the pairing picker, the selection is confirmed inline: *Peet's Coffee — Major Dickason's Blend, Arabica · Dark roast · both hoppers.* A *Change pairing* link reopens the picker, and *Confirm & Continue* is now enabled. *Shown for illustrative purposes only: the Twin Bean Program is a planned program.*

SU-BEAN — Standard Bean Path. Choosing *No — Standard beans* reveals manual hopper configuration. Guidance recommends a caffeinated and decaffeinated bean from the same *bean family* and *roast level* in both hoppers for the most consistent grind. The user configures *Hopper A — Caffeinated* and *Hopper B — Decaffeinated* independently, selecting Arabica or Robusta and then a roast level for each.



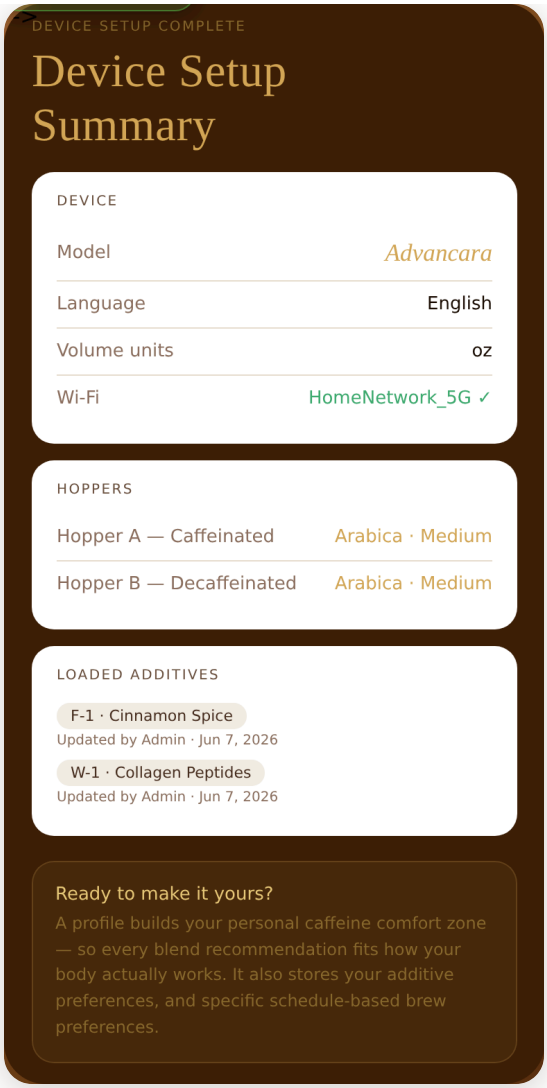
SU5 — Additive Catalog



SU4a — Additive Loading

SU5 — Additive Catalog. Once Wi-Fi is connected the device syncs its catalog. Shown: current version v2.4.1, latest v2.6.0, with 14 new seasonal flavors and wellness blends syncing at 72%. Continue once ready, or skip and use the current catalog.

SU4a — Select the Additives You Are Loading into the Device. The user selects additives across Wellness and Flavor categories, each showing its default dose. Selections are confirmed and assigned to canister slots on the next screen. Two of six slots selected here.

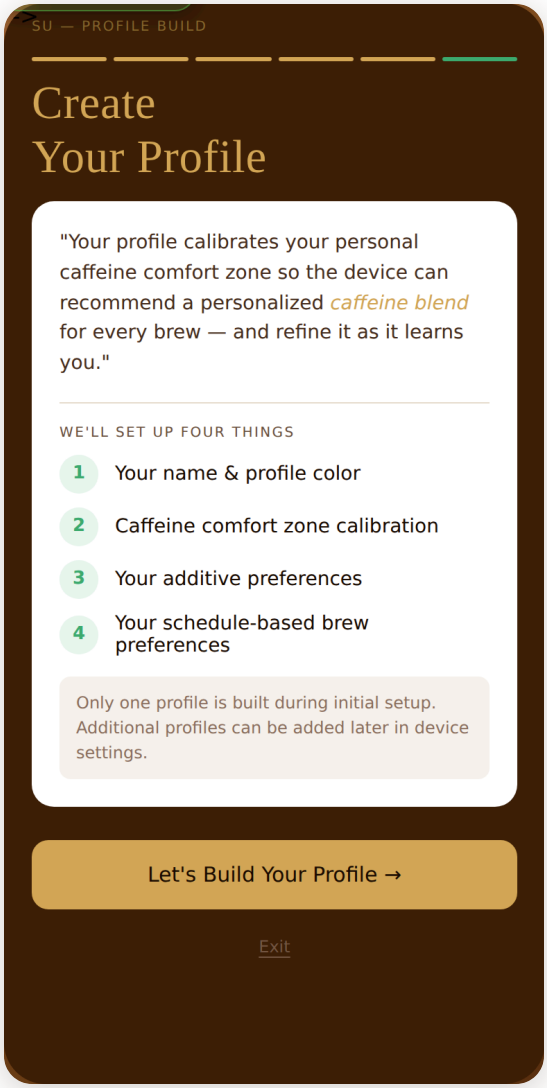


SU5a — Device Setup Summary

SU5a — Device Setup Summary. Full summary of completed device configuration: Device (model Advancara, English, oz, Wi-Fi connected), Hoppers (Hopper A — Caffeinated: Arabica · Medium; Hopper B — Decaffeinated: Arabica · Medium), and Loaded Additives (F-1 · Cinnamon Spice; W-1 · Collagen Peptides). A prompt invites the user to create a Resident User profile.

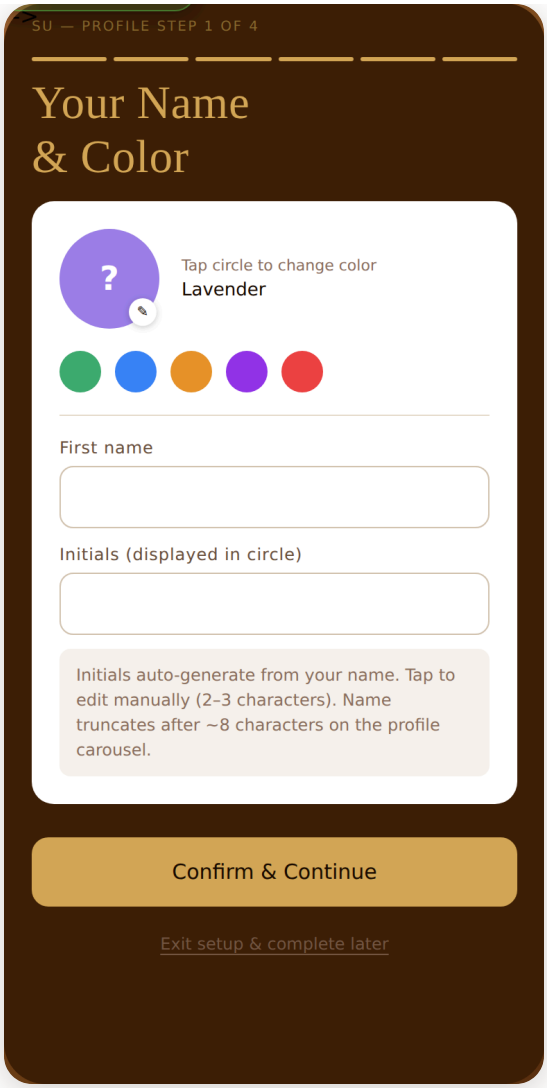
Resident User — User Profile Set-up

A Resident User is a regular user of the device with a stored profile holding their calibrated Caffeine Comfort Zone, brew preferences, additive selections, and grind setting.



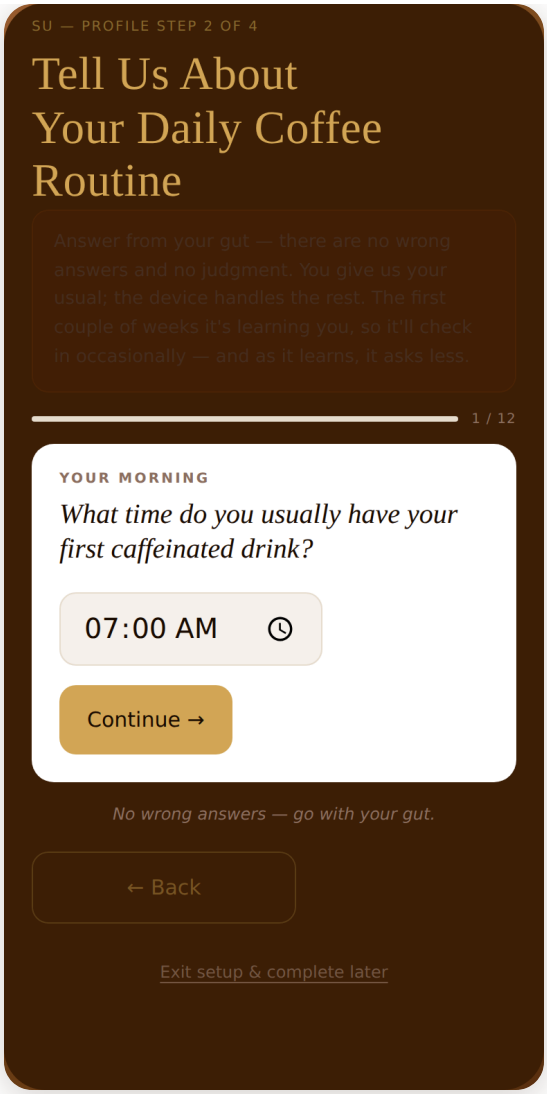
SU6 — Create Your Profile

SU6 — Create Your Profile. Introduces four setup steps: *1 Your name & profile color, 2 Caffeine Comfort Zone calibration, 3 Your additive preferences, 4 Your schedule-based brew preferences.* Only one profile is built during initial setup; additional profiles are started later by the administrator from Administrator Functions, described at the end of this section.



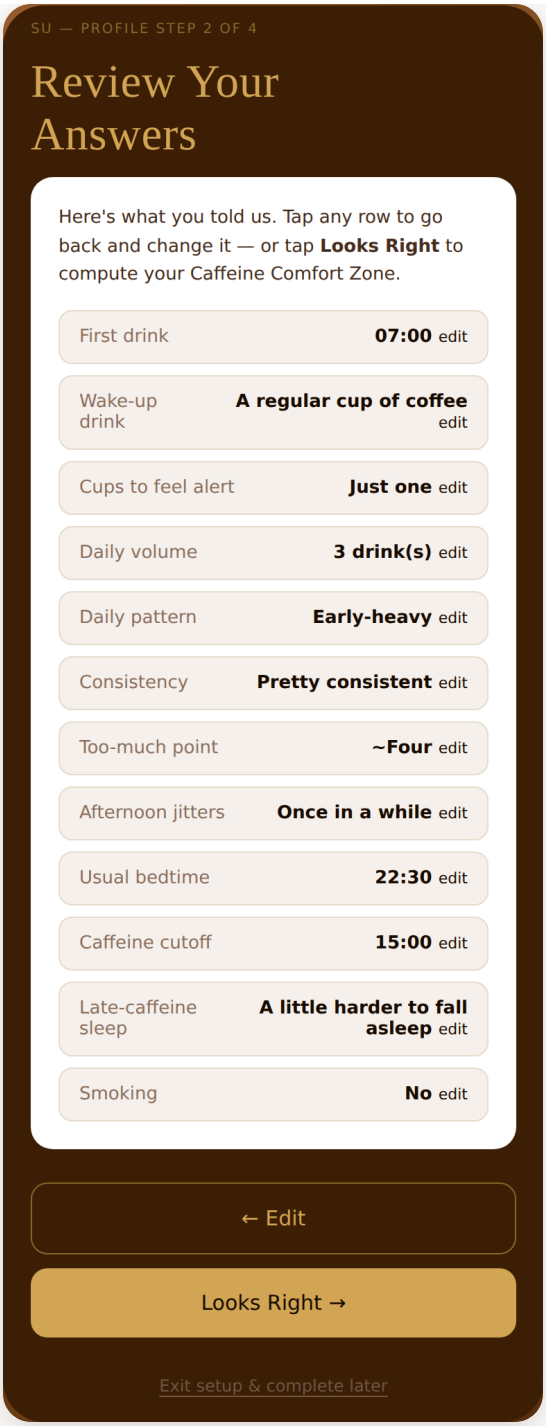
SU7 — Your Name & Color

SU7 — Your Name & Color. Profile Step 1 of 4. The user enters their first name and initials (auto-generated from the name, editable to 2–3 characters) and chooses a profile color. The initials circle previews the chosen color. Name truncates at approximately 8 characters on the profile carousel.



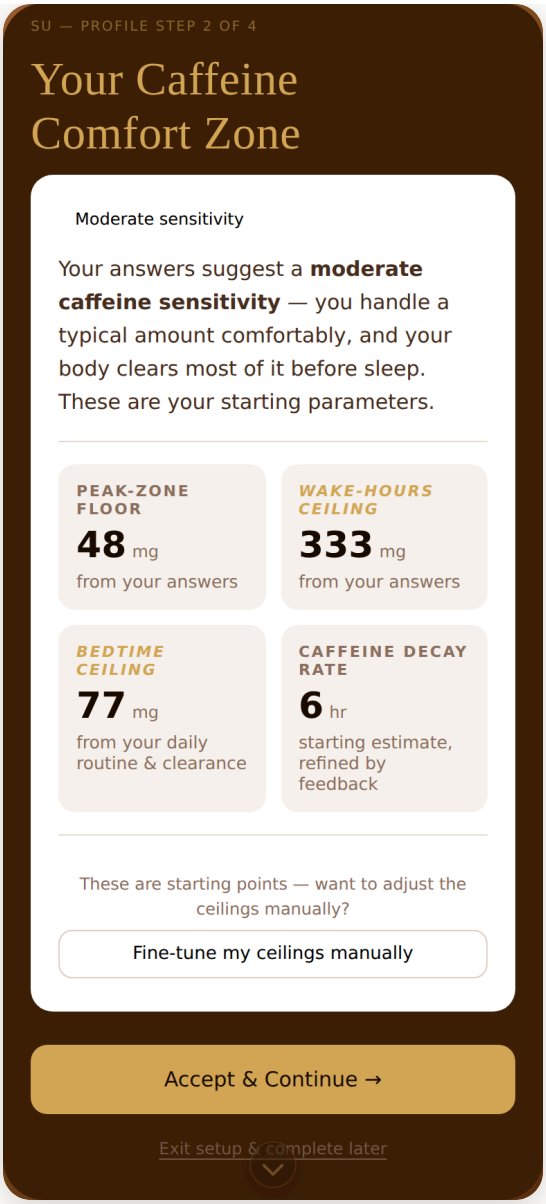
SU8 — Tell Us About Your Daily Coffee Routine

SU8 — Tell Us About Your Daily Coffee Routine. Profile Step 2 of 4. A twelve-question sensitivity calibration instrument (thirteen when the espresso-shots step applies), presented one question at a time to establish initial sensitivity parameters for the user. Over the next few weeks the device presents outcome-related questions (FR System) to the user to refine its understanding of the user's sensitivity and adjusts the sensitivity parameters if needed.



SU8-Confirm — Review Your Answers

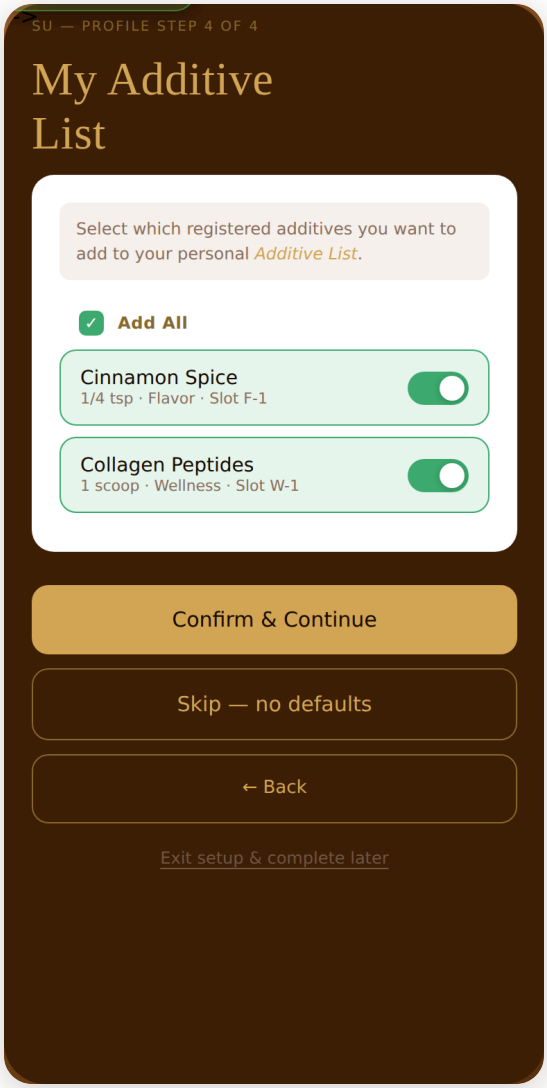
SU8-Confirm — Review Your Answers. All twelve instrument answers displayed together (thirteen when the espresso-shots step applies): First-drink time, First drink, Espresso shots (when applicable), Cups to feel it, Daily volume, When most caffeine falls in the day, Routine consistency, Too-much point, Afternoon jitters, Usual bedtime, Caffeine cutoff, Late-caffeine sleep effect, and Smoking status. Tap any row to edit, or tap *Looks Right* to compute the Caffeine Comfort Zone.



SU9 — Your Caffeine Comfort Zone

SU9 — Your Caffeine Comfort Zone. Four computed parameters: *Peak Zone Floor 48 mg* (from answers), *Wake Hours Ceiling 333 mg* (from answers), *Bedtime Ceiling 77 mg* (CBI-derived — the reconstructed typical-day load projected forward to bedtime at the user's Decay Rate), *Caffeine Decay Rate 6 hr* (a research-anchored prior — a 6 h baseline adjusted for smoking status and afternoon overstimulation, refined by feedback). Accept and continue, or fine-tune ceilings manually.

Right to compute the Caffeine Comfort Zone.



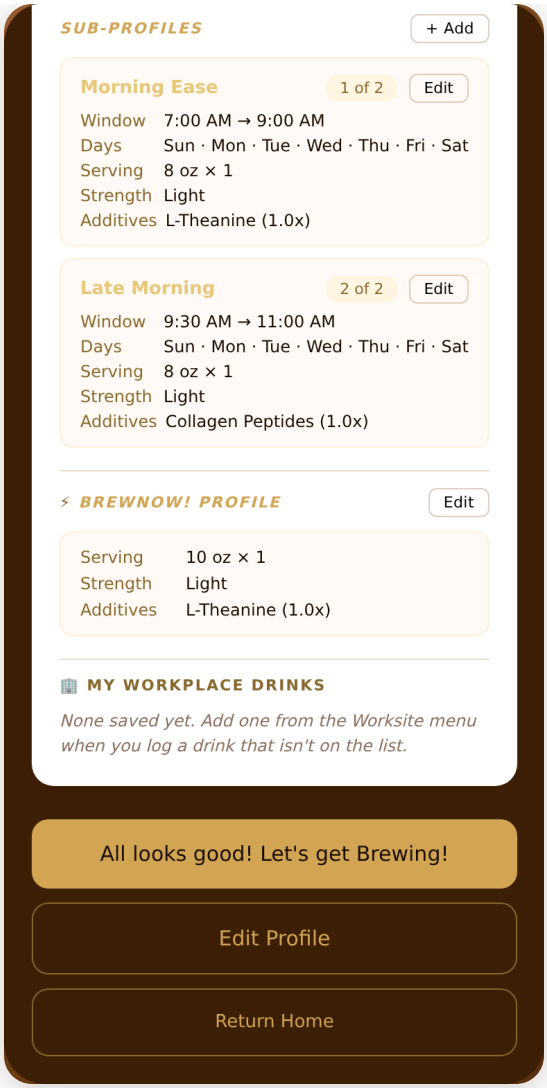
SU12 — My Additive List



SU-GRIND — Grind Setting

SU12 — My Additive List. Profile Step 3 of 4. The user selects which device-registered additives to include in their personal Additive List. Shown: Cinnamon Spice and Collagen Peptides have been selected. Only these two additive choices will appear in the brew-time carousel for the user.

SU-GRIND — Grind Setting. Six settings from Coarse to Extra Fine, each described by taste impact: *1 Coarse — light, bright, clean; 2 Med-Coarse — smooth and mellow; 3 Medium — balanced and approachable; 4 Med-Fine — fuller body, richer flavor; 5 Fine — intense and concentrated; 6 Extra Fine — maximum intensity, dense, syrupy.*



SU13 — Profile Summary

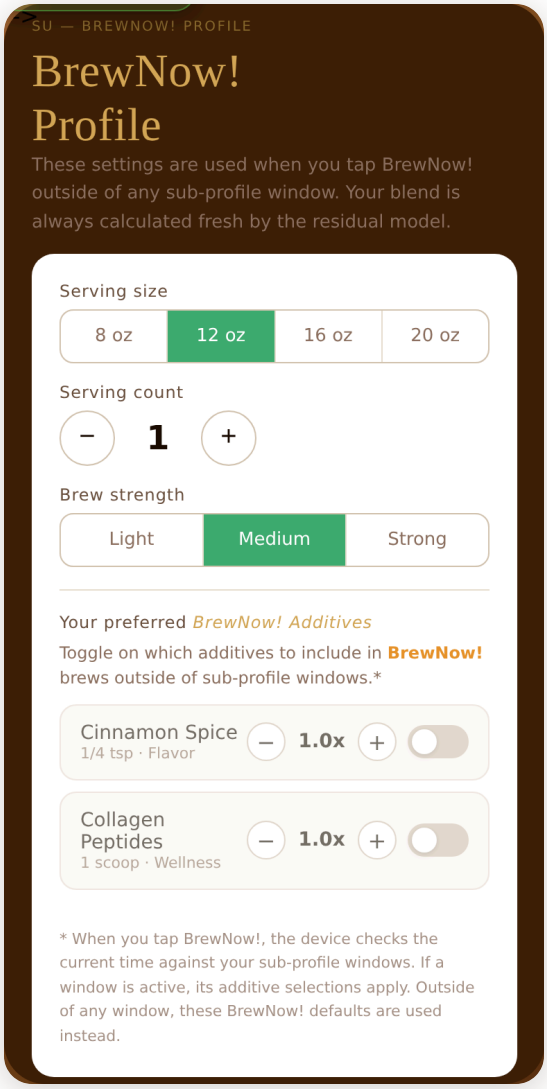
SU13 — Profile Summary. Setup complete. Example displays: Calibrated Caffeine Comfort Zone (a moderate-sensitivity profile, shown for illustration — WHC 333 mg, Bedtime Ceiling 77 mg, Peak Zone Floor 48 mg, Caffeine Decay Rate 6 hr, all Calibrated), Grind Setting 3 — Medium, Preferred additives (Cinnamon Spice, Collagen Peptides). Sub-profile status and BrewNow! Profile status (not set up). This is covered in the following section. When the Profile Summary is opened later to review an established profile — by selecting the profile and choosing *Review my Profile* — it additionally presents a **Today's Consumption** summary and a **My Workplace Drinks** section, both described below. Tap *All looks good! Let's get Brewing!*



SU13 — Today's Consumption

Today's Consumption. Reached by selecting a profile and choosing *Review my Profile*, the Profile Summary presents a live, per-profile account of the day: each caffeine event logged since midnight — device brews, off-site drinks logged through **CaffevolveGo**, and any residual carried forward from the prior day — each shown with its time, type, and caffeine amount, together with a running total and the residual caffeine level active now. The summary reflects only what the user has logged; it does not certify that the day is complete. It is therefore framed as a confirmation — an *anything missing?* prompt with a one-tap path to log an omission — so the user can attest that the day is fully accounted for before a late-day decision to brew or purchase.

Optional Profile Additions — BrewNow! Profile Set-up

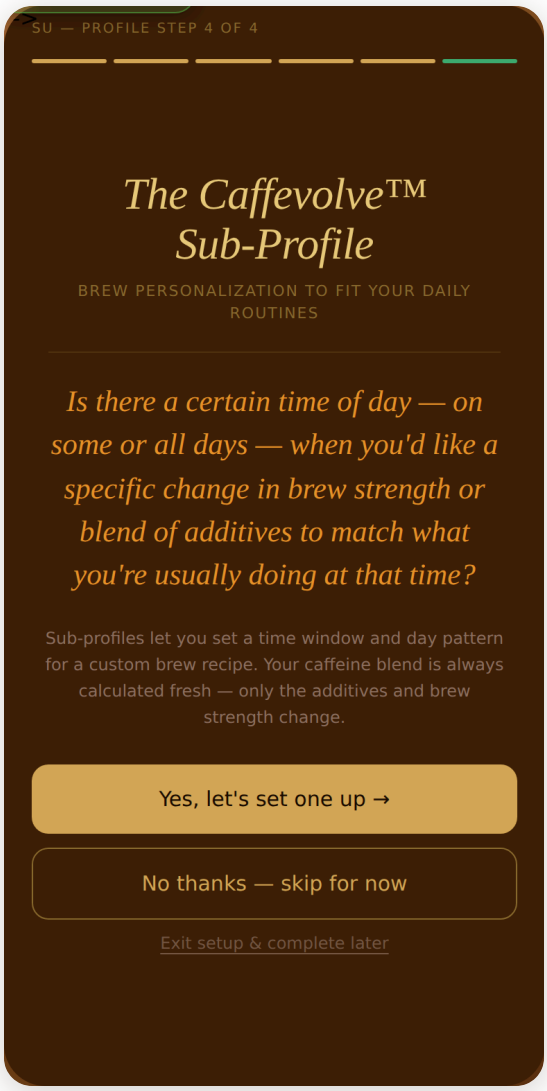


SU12c — BrewNow! Profile

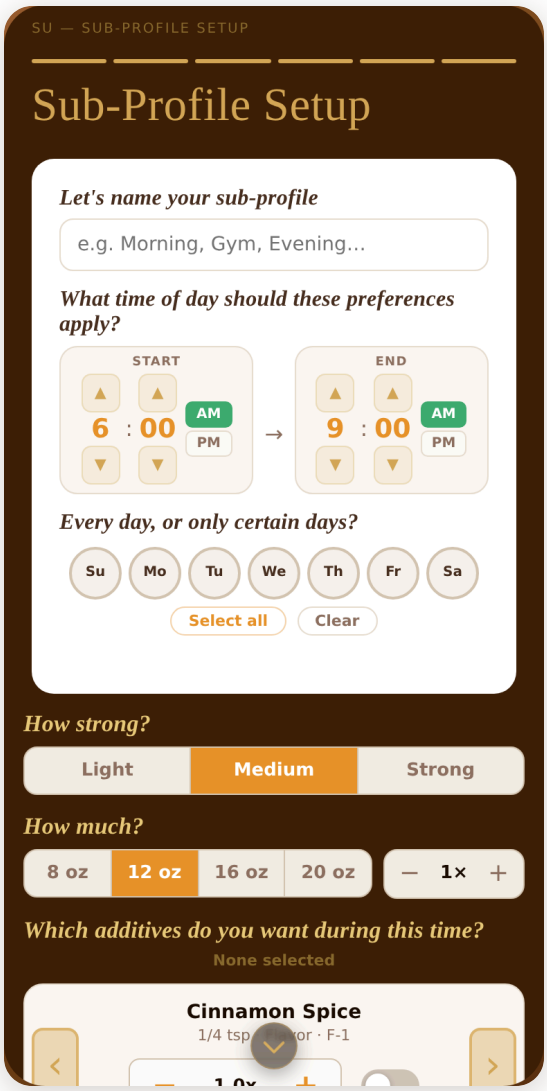
SU12c BrewNow! Profile. Configures express brew defaults applied when the user chooses the BrewNow! path. Profile includes; serving size (12 oz selected), serving count, brew strength, and additives to add to the brew automatically — Cinnamon Spice and Collagen Peptides were chosen in the example shown. The recommended caffeine blend is never a user profile element and is always calculated at the time of the brew.

Optional Profile Additions — Sub-Profile Set-up

Sub-profiles define a user-chosen time window and day pattern for a custom brew recipe. The caffeine blend is always calculated fresh — only the serving size, serving count, additives, and brew strength change. Up to 3 sub-profiles per profile on Originara and Culminara Office, and up to 6 on Advancara.



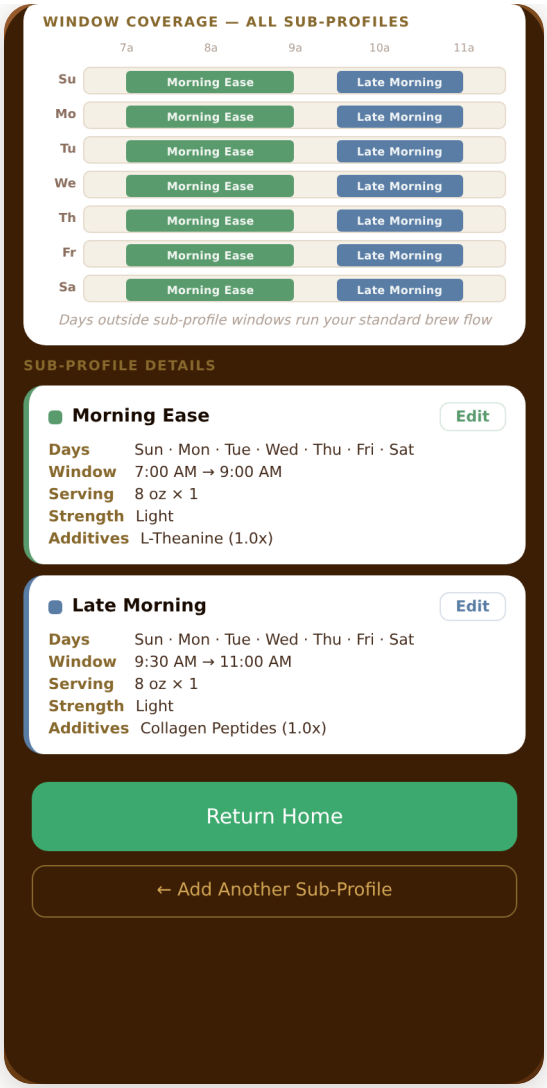
SU12a — The Caffevolve™ Sub-Profile



SU12b — Sub-Profile Setup

SU12a — The Caffevolve™ Sub-Profile. Introduces the sub-profile concept with a question: *Is there a certain time of day — on some or all days — when you'd like a specific change in brew strength or blend of additives to match what you're usually doing at that time?* The user can proceed to set one up now, or skip and come back to it later.

SU12b — Sub-Profile Setup. Captures: a name (e.g. Morning, Gym, Evening), a time window (6:00 AM → 9:00 AM shown), and the day(s) it will apply on. User sets the serving size and count (How much?), the brew strength, and which additives to add to the brew during this window. Tap *See Complete Weekly Picture* to preview complete sub-profile coverage before confirming.

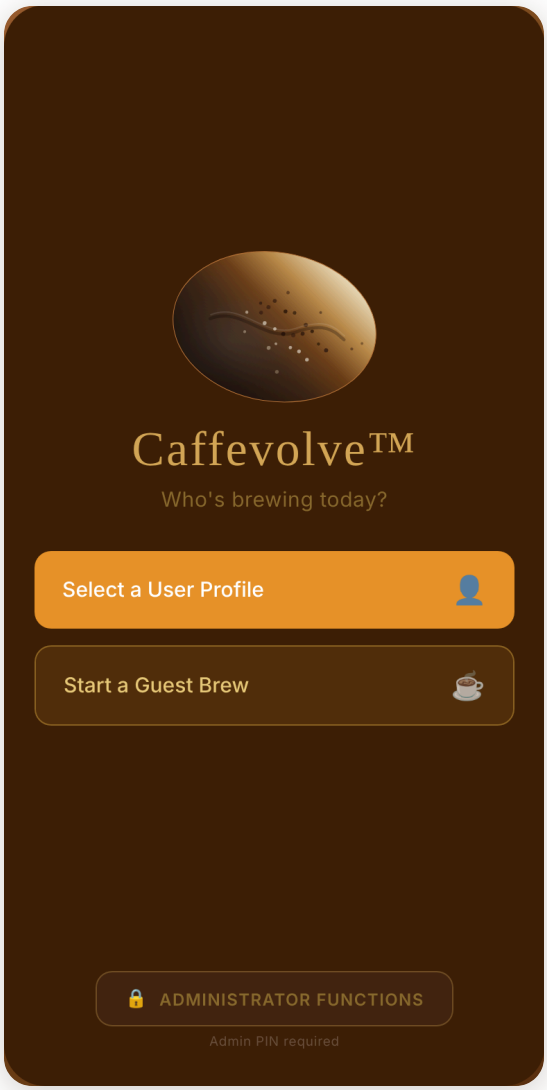


SU12b-CONFIRM — Your Weekly Sub-Profile Picture

SU12b-CONFIRM — Your Weekly Sub-Profile Picture. A calendar-style view showing window coverage across all sub-profiles. Days outside sub-profile windows run the standard brew flow. Confirm with *Confirm All Sub-Profiles* or return to add another.

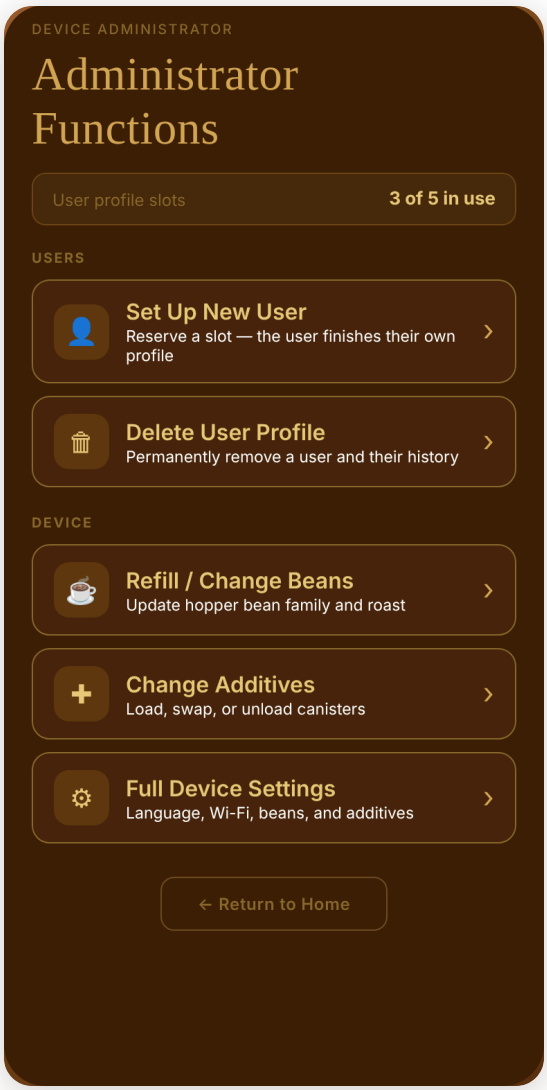
Administrator Functions & Adding a Resident User

Administrative tasks are gathered behind a single button at the bottom of the device home screen, *Administrator Functions*, so the home screen itself offers only the two choices most people need. Tapping it asks for the administrator PIN once; every task on the screen that follows is then available without further prompts.



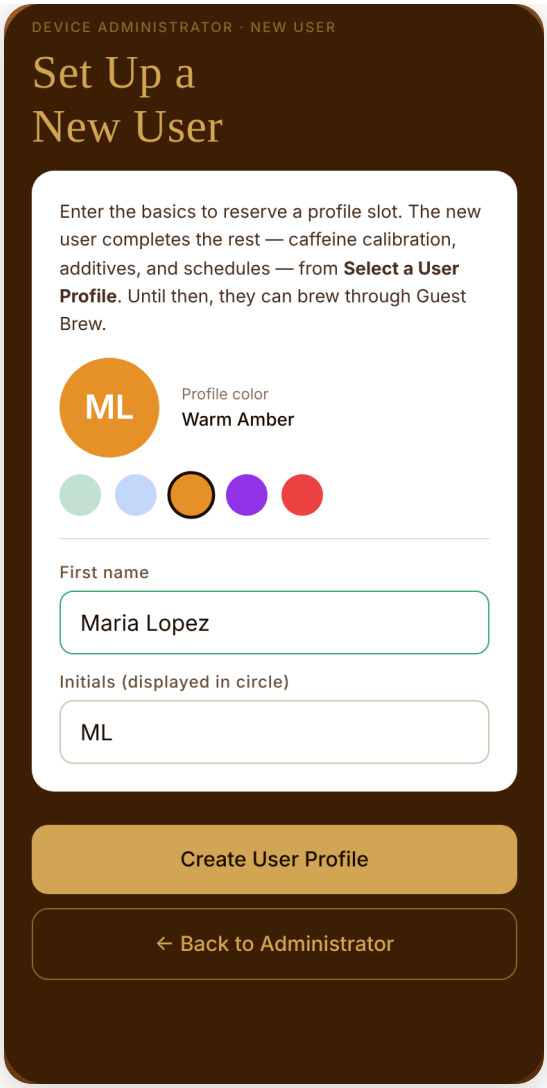
PS0a — Brew Gateway

PS0a — Brew Gateway. The device home screen: *Select a User Profile* and *Start a Guest Brew*. The former scattered administrator links — profile deletion, bean and additive changes, and device settings — are consolidated into the single *Administrator Functions* button at the bottom, marked as requiring the Admin PIN.

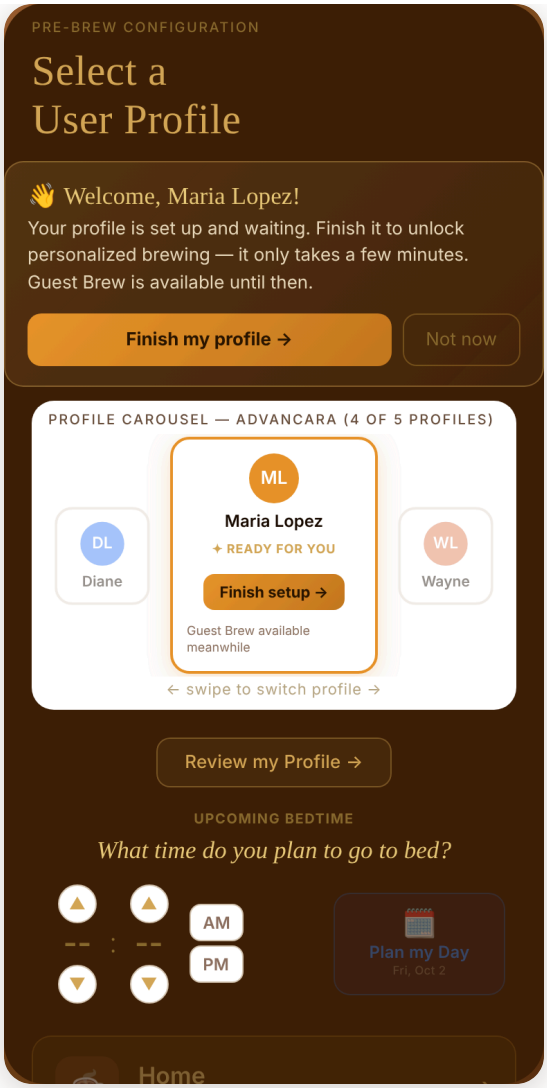


ADMIN — Administrator Functions

ADMIN — Administrator Functions. Shows profile-slot usage (*3 of 5 in use* on Advancara) and groups the administrator's tasks in two sets. *Users:* Set Up New User and Delete User Profile. *Device:* Refill / Change Beans, Change Additives, and Full Device Settings. When every slot is in use, Set Up New User is disabled until a profile is deleted. Completing a bean or additive change, or deleting a profile, returns the administrator to this screen.



ADMIN-NEWUSER — Set Up a New User



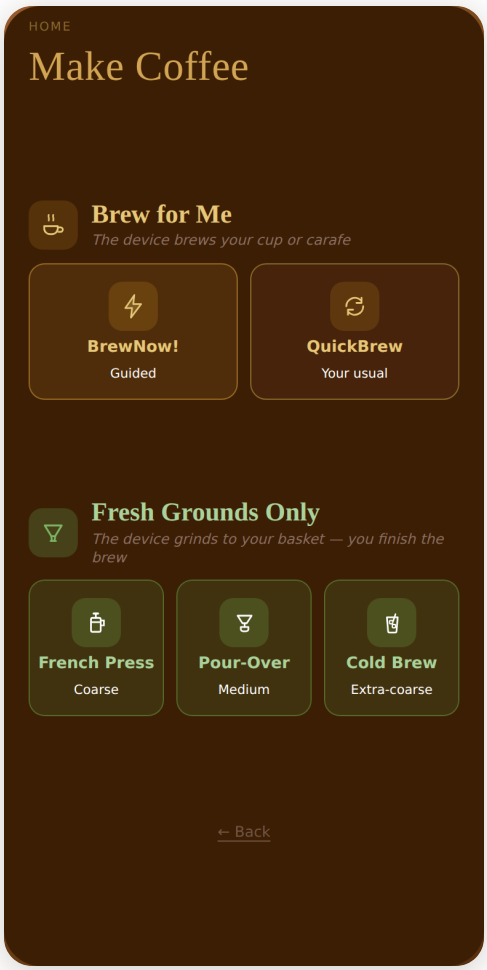
PS1a — Ready for You

ADMIN-NEWUSER — Set Up a New User. The administrator enters the new user's first name and initials (auto-generated from the name, editable) and chooses a profile color; colors already used by other profiles are shown faded and cannot be chosen. *Create User Profile* reserves the slot and returns to Administrator Functions with a confirmation. No calibration happens here — that belongs to the user.

PS1a — Ready for You. The reserved profile appears on the carousel marked *Ready for you*, and a welcome banner greets the new user by name. *Finish setup* opens the profile build at the Caffeine Comfort Zone calibration (SU8), since the name and color are already in place, and continues through the same steps a Resident User follows above. Until it is complete, Home, Away, and Plan my Day are locked; tapping any of them explains that personalized brewing is locked and offers *Finish setup* or *Start a Guest Brew*. Leaving setup partway saves progress — the option to discard is withheld, because the slot belongs to the administrator to release.

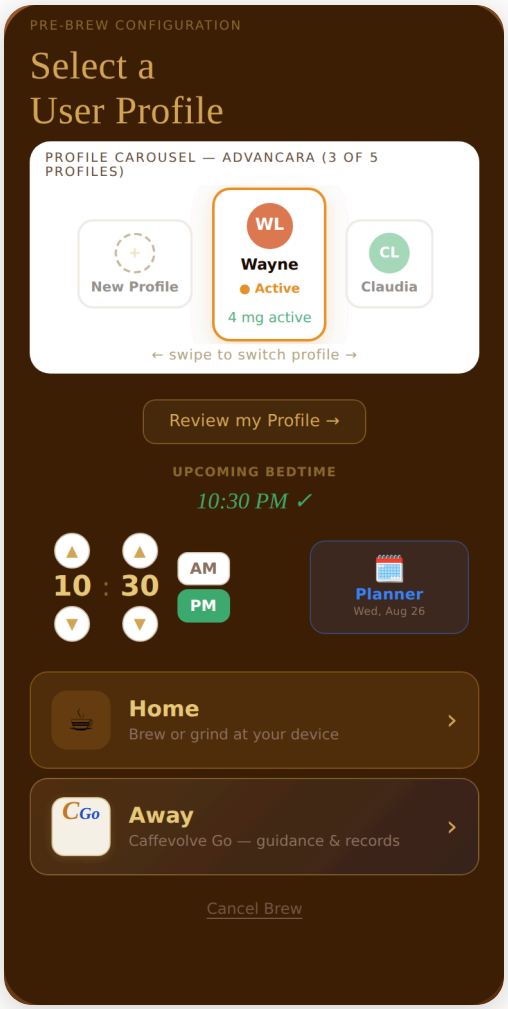
8.2 The Brew Session

Every device brew begins at the *Make Coffee* menu, reached by tapping *Home* on the profile screen. It groups the device's options under two headings: **Brew for Me**, where the device brews the cup or carafe (BrewNow! and QuickBrew, both below), and **Fresh Grounds Only**, where the device grinds a blend into the basket to finish by hand (covered later in this section).



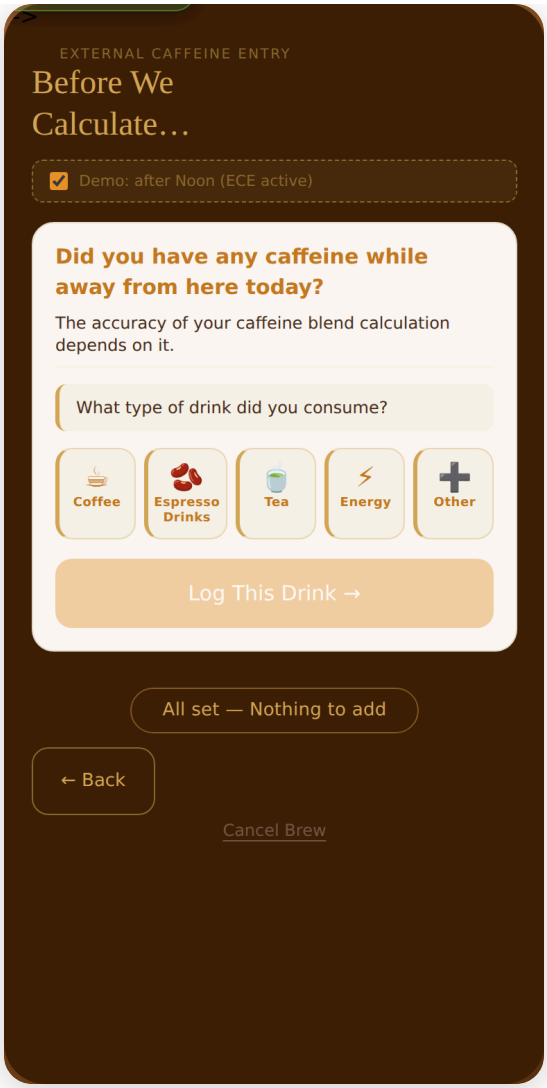
HOME — Make Coffee

HOME — Make Coffee. The device's brew home: under *Brew for Me*, the two brew options — BrewNow! and QuickBrew — alongside *Fresh Grounds Only*, the grounds-to-basket path. What each does is covered in 6.1.

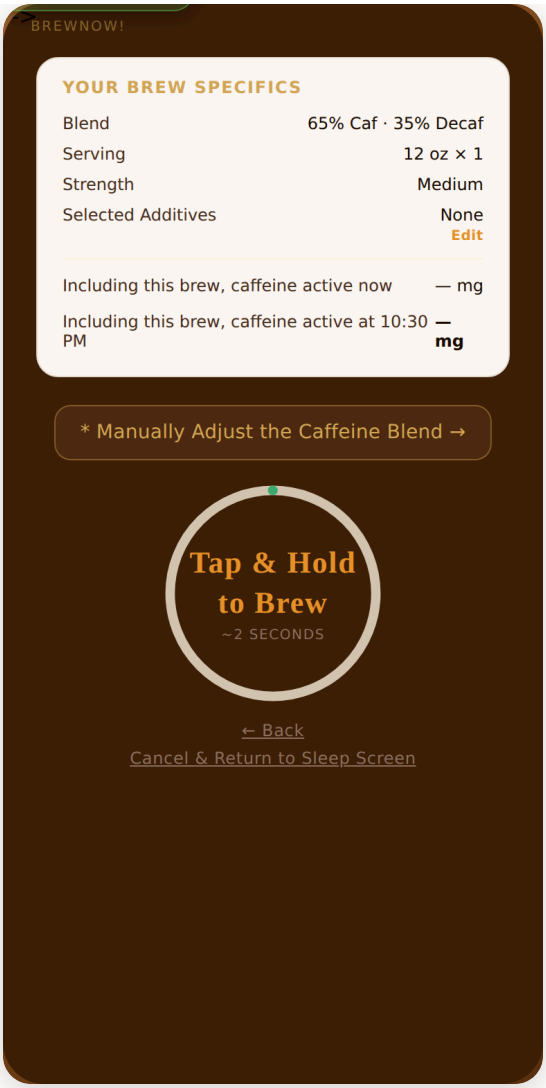


PS1a — Profile Selection

PS1a — Profile Selection. The Resident User selects their profile from the carousel and declares tonight's bedtime. From here the screen branches three ways: *Home*, which opens the *Make Coffee* menu on the device (Brew for Me — BrewNow! or QuickBrew — and Fresh Grounds Only); *Away*, which opens CaffevolveGo for off-device guidance and logging (ECE and OAS); and the *Planner*, which opens Plan my Day. Bedtime gates every branch — Home, Away, and the Planner stay disabled until a bedtime is set, because each depends on the bedtime projection.



ECE — External Caffeine Entry

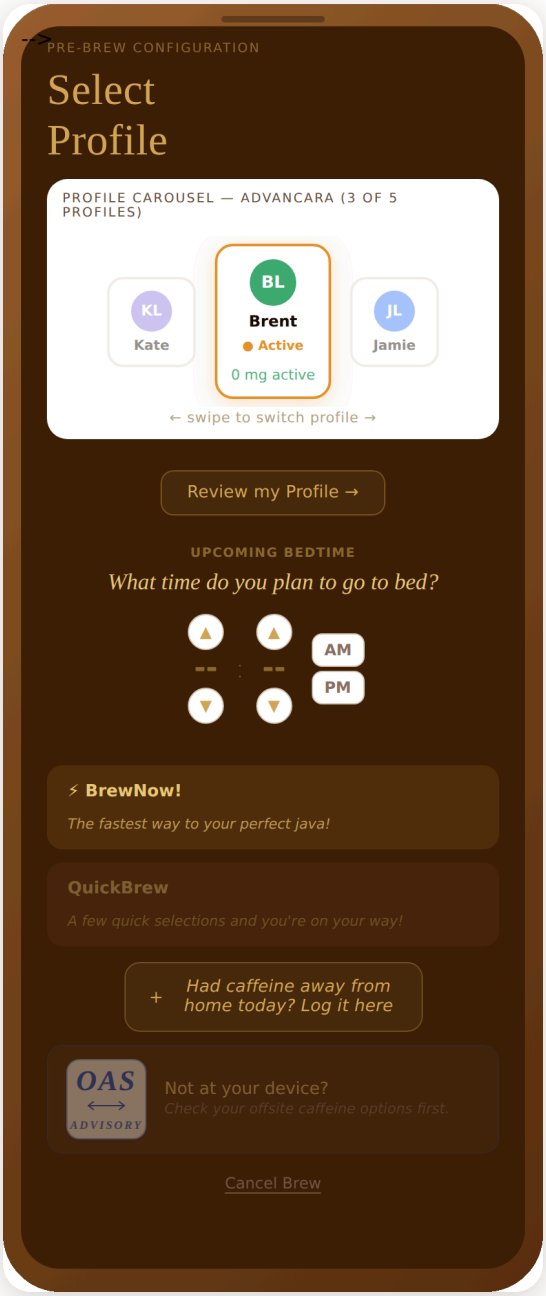


BN-HTB — Hold to Brew

ECE — External Caffeine Entry. A couple of times each day, this ECE screen will surface automatically before a new brew can be initiated to ensure that all external caffeine consumption has been captured up to that point. This inquiry is necessary to ensure the caffeine recommendation for the next cup is based on accurate and up to date consumption information. If this does not apply, the user taps “All set - Nothing to add”. But if the inquiry does apply, the user logs the drink type, amount, and approximate time. The residual model is updated and all subsequent recommendations reflect the logged intake.

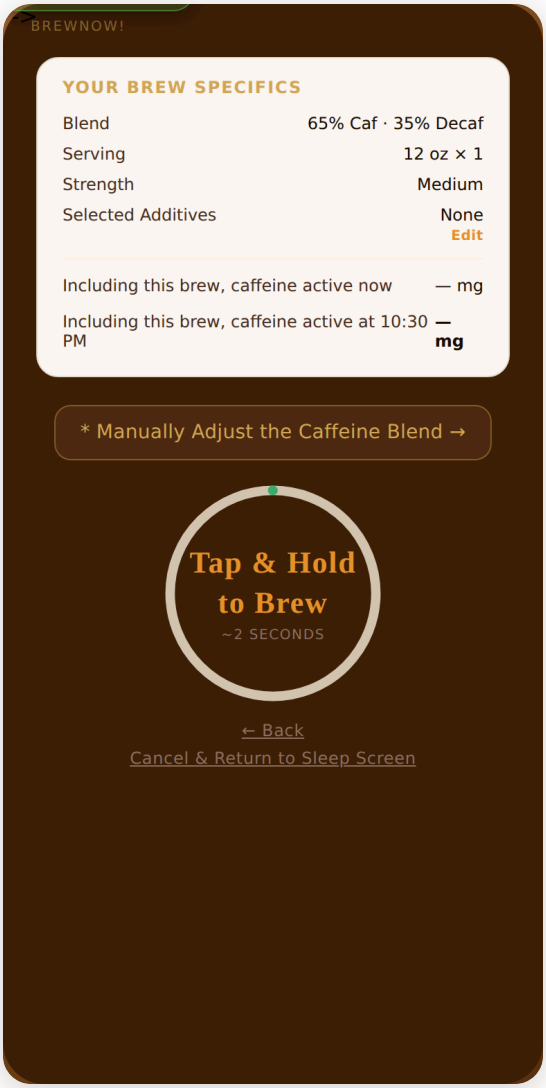
BN-HTB — Hold to Brew. Once the user selects the BrewNow! option, a confirmation screen shows stored express defaults — size, strength, and preferred additives. The Blend recommendation is calculated once this option is chosen. Tap and Hold the brew button to initiate the brew cycle, or tap *Adjust manually* to open the Adjust Caffeine Blend slider.

BrewNow! — With Caffeine Adjustment



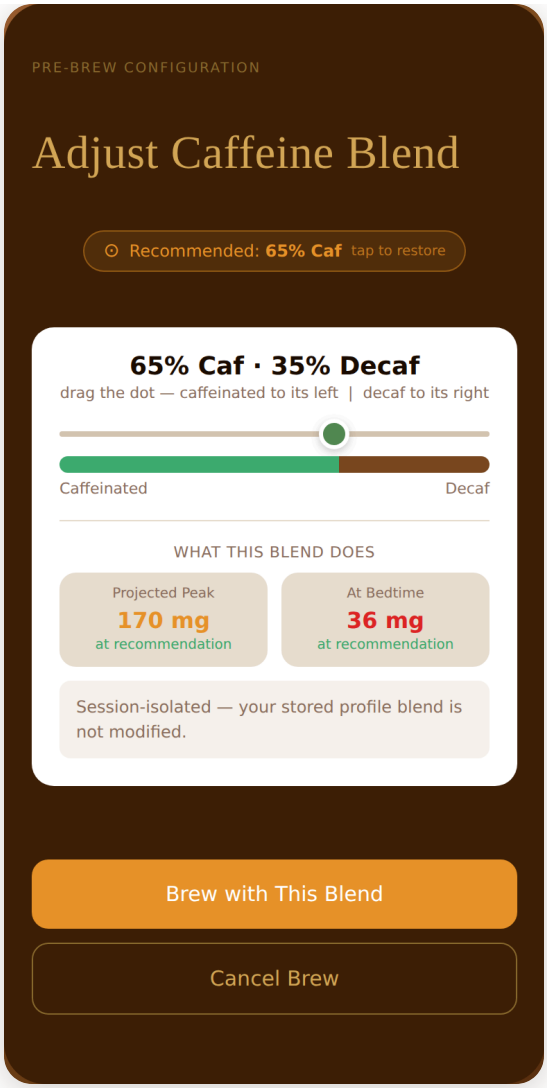
PS1a — Profile Selection

PS1a — Profile Selection. Profile selected, bedtime declared, BrewNow! initiated.

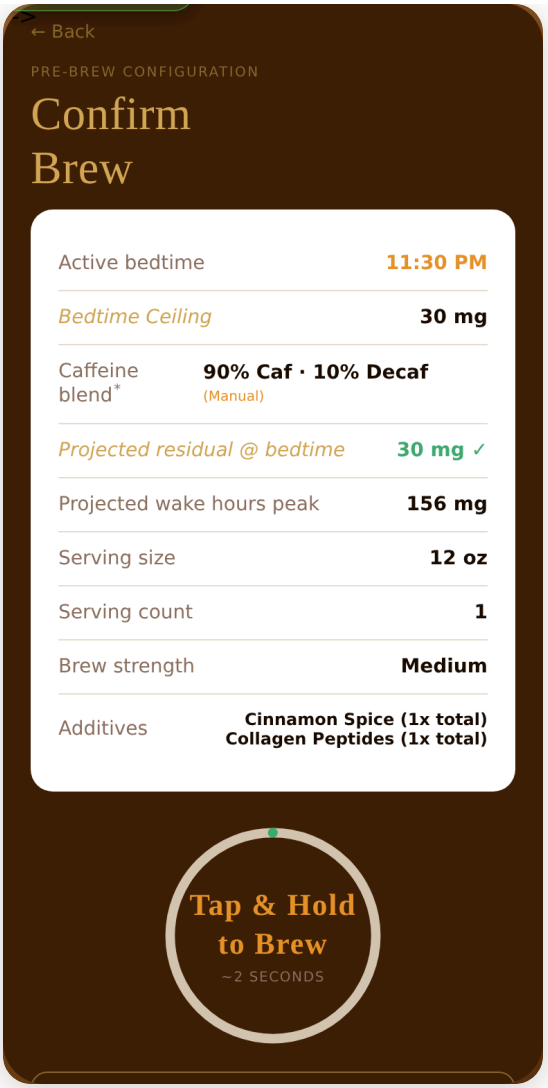


BN-HTB — Hold to Brew

BN-HTB — Hold to Brew. The user taps *Adjust manually* instead of holding to brew, opening the Adjust Caffeine Blend slider.



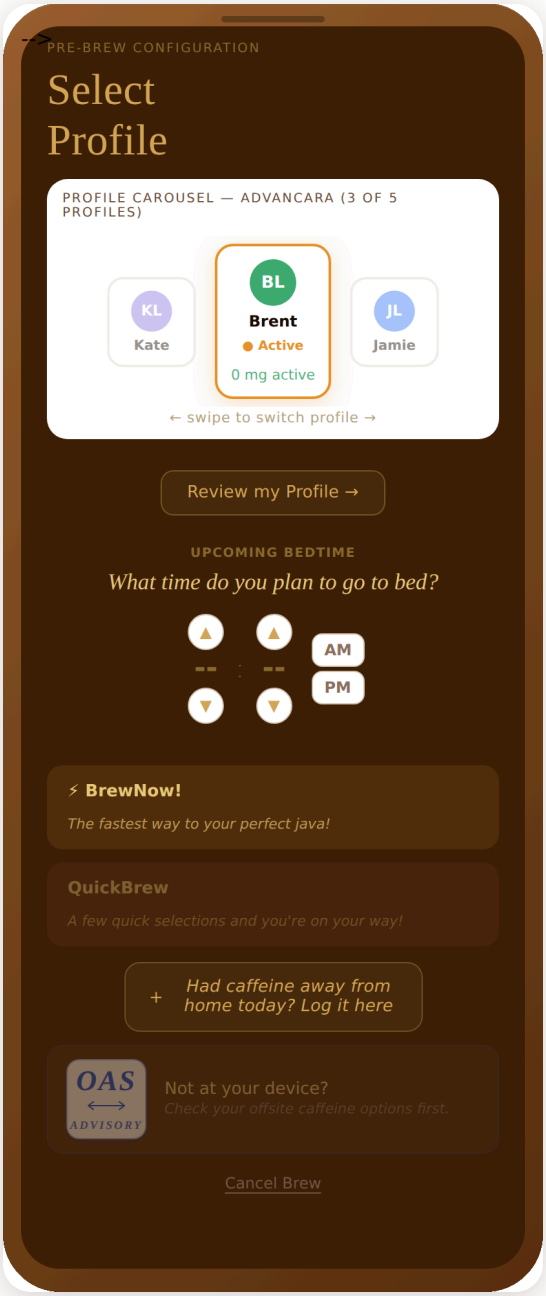
PS4 — Adjust Caffeine Blend



PS6-A — Confirm & Brew

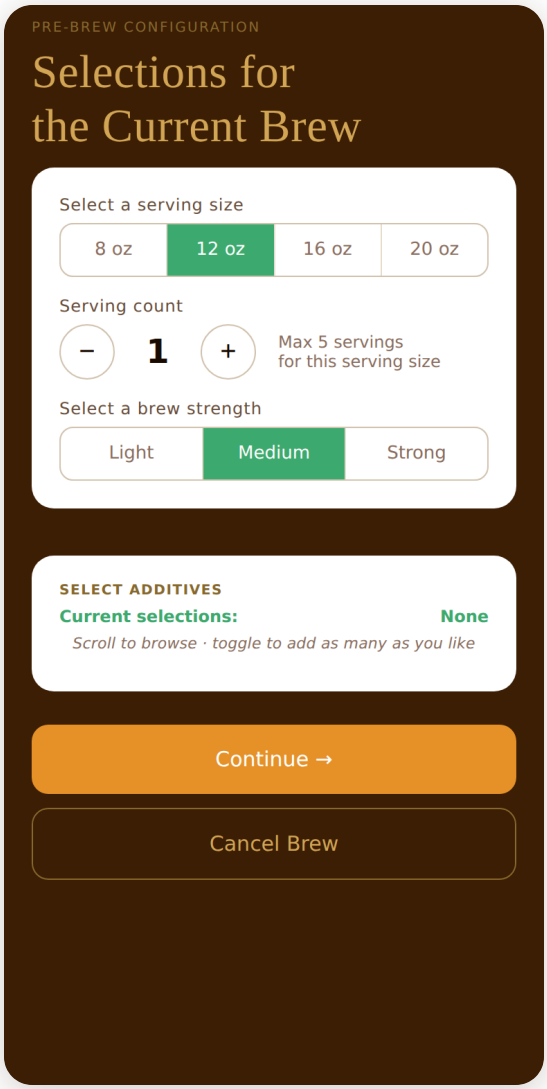
PS4 — Adjust Caffeine Blend. A live slider shifts the caffeine blend percentage. Projected Peak and Projected Bedtime Residual update in real time. Ceiling lines and the Peak Zone band are displayed. Ceiling violations trigger a warning banner.

PS6-A — Confirm & Brew. The adjusted brew confirmation screen. Displays the final blend and all parameters before the hold-to-brew gesture initiates the active brew cycle.



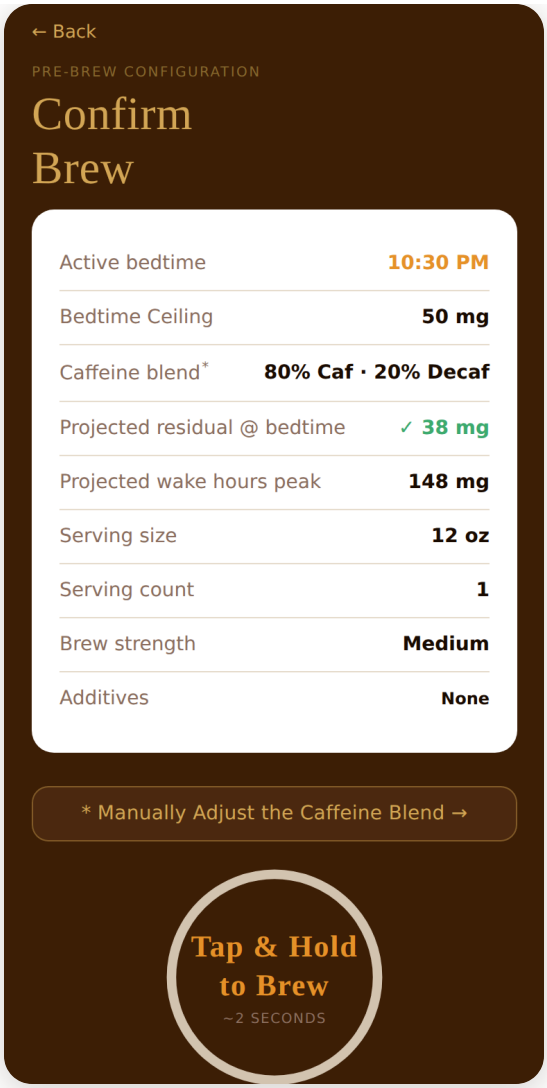
PS1a — Profile Selection

PS1a — Profile Selection. Profile selected, bedtime declared. The user taps QuickBrew to enter the full pre-brew configuration path.

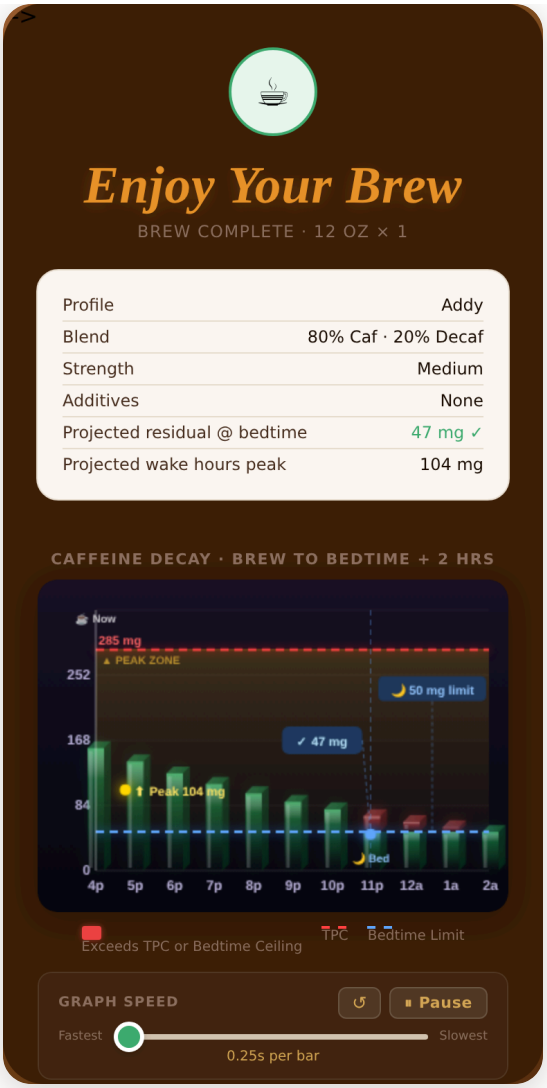


PS2a — Brew-Time Selections

PS2a — Brew-Time Selections. The user configures serving size and number of servings for this brew. If a sub-profile window is active, brew strength and additive selections are pre-populated from that sub-profile but remain editable; otherwise, brew strength and additive selections are made fresh at brew time, with additives selected from the user's Additive List. These parameters are always session-level — never stored to the profile.



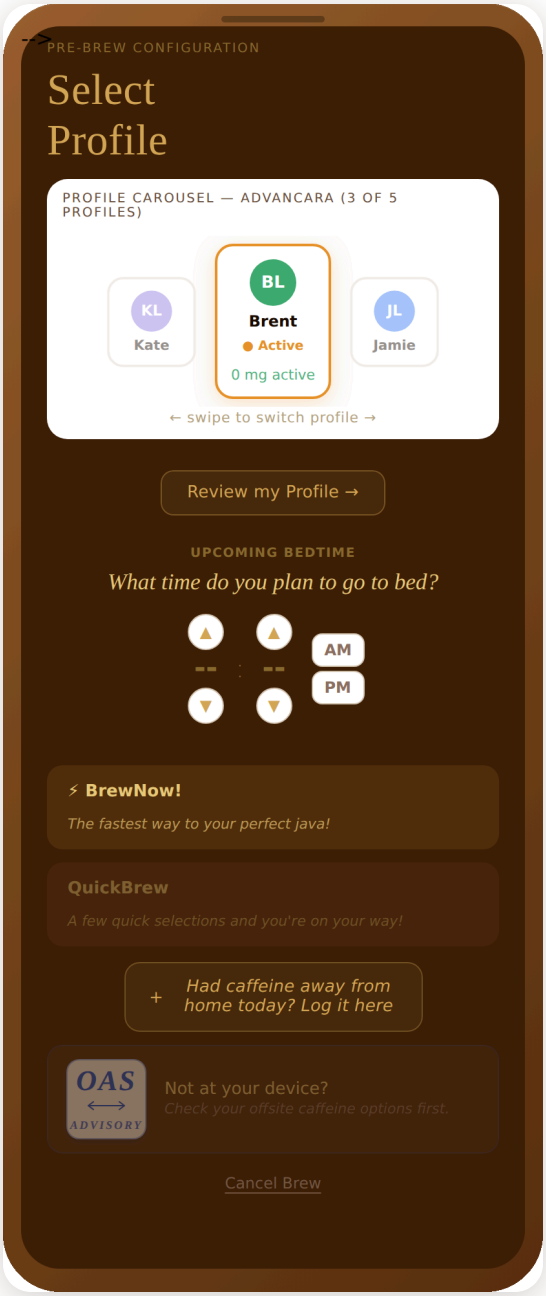
PS6-A — Recommendation & Brew



PS8 — Enjoy Your Brew

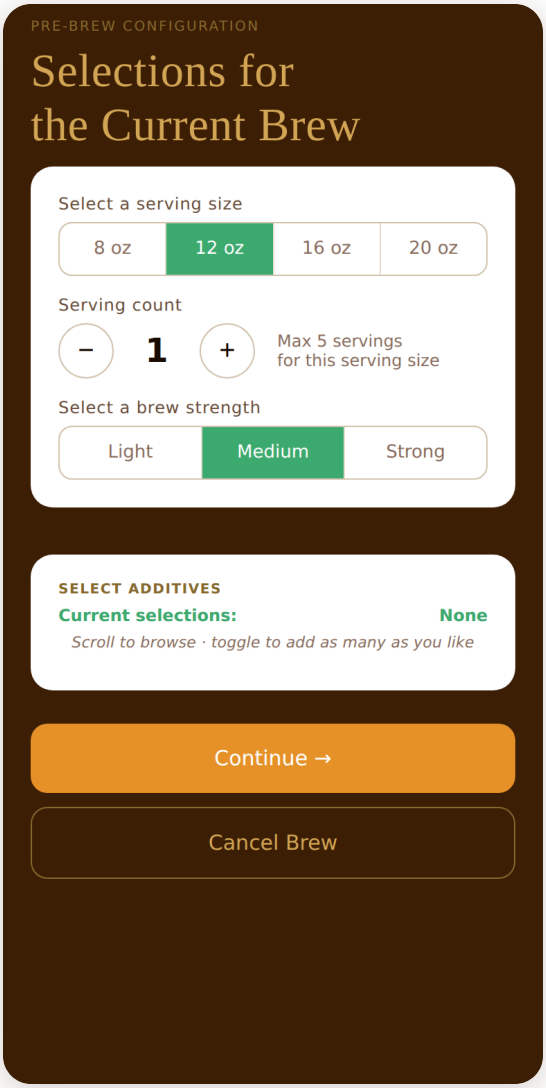
PS6-A — Recommendation & Brew. The caffeine blend percentage is presented alongside the Projected Peak and Projected Bedtime Residual values. The user accepts the recommendation and taps and holds to brew.

PS8 — Enjoy Your Brew. The brew complete screen. A session summary card confirms the profile, blend, strength, additives, projected residual caffeine at bedtime, and projected wake-hours peak for the brew just completed. A Caffeine Decay chart animates the projected caffeine curve from brew time through bedtime plus two hours, with reference lines marking the Wake Hours Ceiling and Bedtime Ceiling. The screen auto-advances to the home screen after a short countdown; the user may also tap *All set — Thank you!* to return immediately.



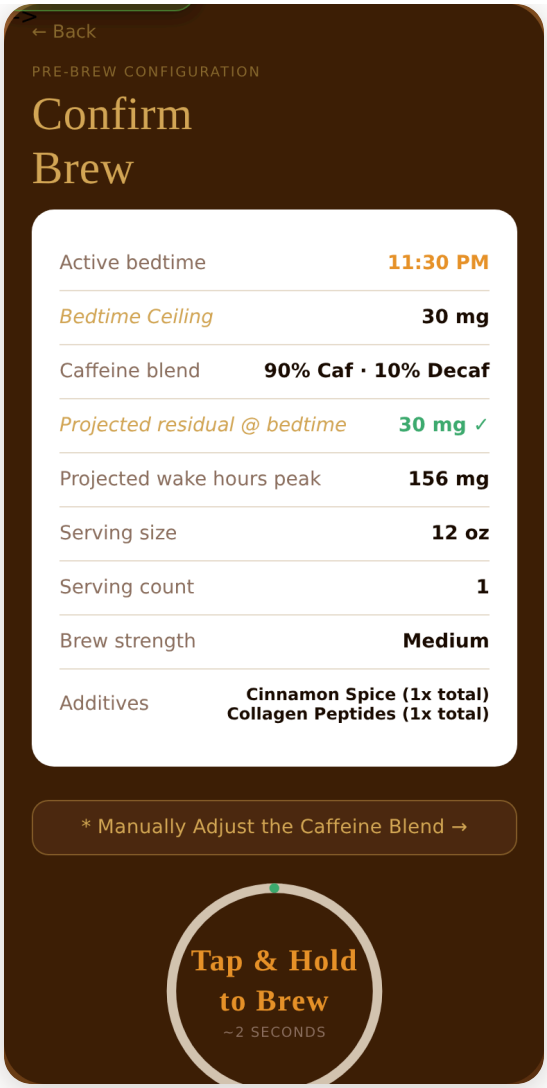
PS1a — Profile Selection

PS1a — Profile Selection. Profile selected, bedtime declared, QuickBrew initiated.

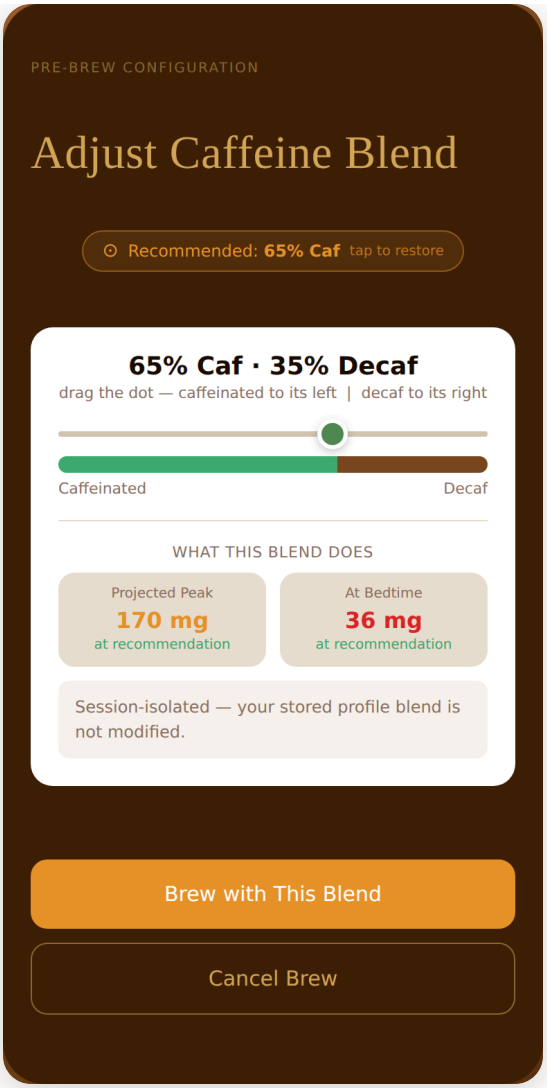


PS2a — Brew-Time Selections

PS2a — Brew-Time Selections. Size, strength, and additive selections for this brew.



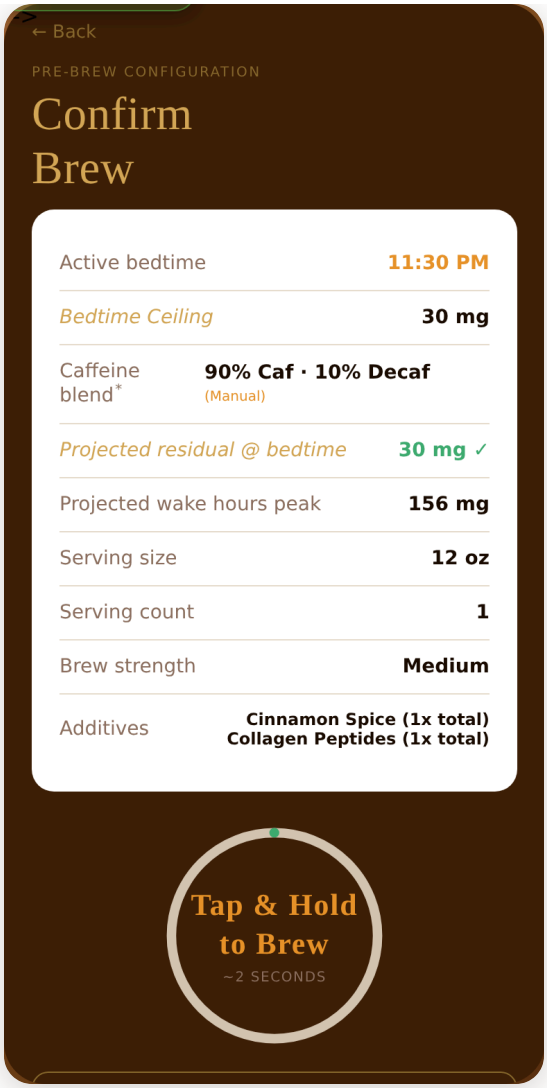
PS6-A — Recommendation



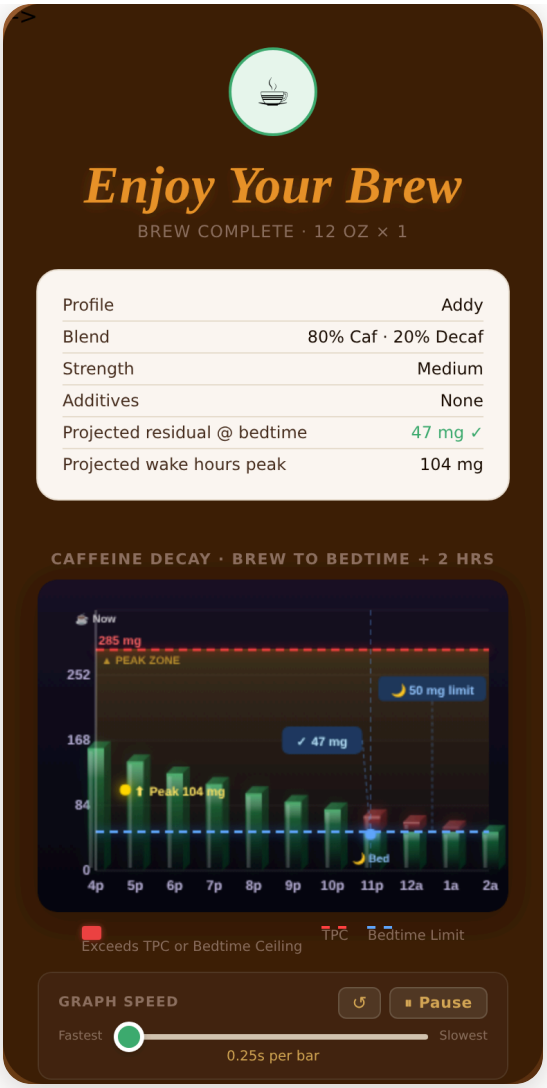
PS4 — Adjust Caffeine Blend

PS6-A — Recommendation. The system blend recommendation is presented. If the user chooses to not accept the recommendation, a tap on the *Adjust manually* rather than tap and hold to brew will allow the adjustment to proceed.

PS4 — Adjust Caffeine Blend. A live slider shifts the caffeine blend percentage. Projected Peak and Projected Bedtime Residual update in real time. Ceiling lines and the Peak Zone band are displayed. Ceiling violations trigger a warning banner.



PS6-A — Confirm & Brew



PS8 — Enjoy Your Brew

PS6-A — Confirm & Brew. Final confirmation with the adjusted blend. Hold to brew initiates the active brew cycle.

PS8 — Enjoy Your Brew. The brew complete screen. A session summary card confirms the profile, blend, strength, additives, projected residual caffeine at bedtime, and projected wake-hours peak for the brew just completed. A Caffeine Decay chart animates the projected caffeine curve from brew time through bedtime plus two hours, with reference lines marking the Wake Hours Ceiling and Bedtime Ceiling. The screen auto-advances to the home screen after a short countdown; the user may also tap *All set — Thank you!* to return immediately.

Fresh Grounds Only

The concept and mechanics of Fresh Grounds Only are covered in 6.1; below is the user-facing flow, across three screens. As with any brew, a proactive External Caffeine Entry check may surface first, so the projection starts from a complete picture.



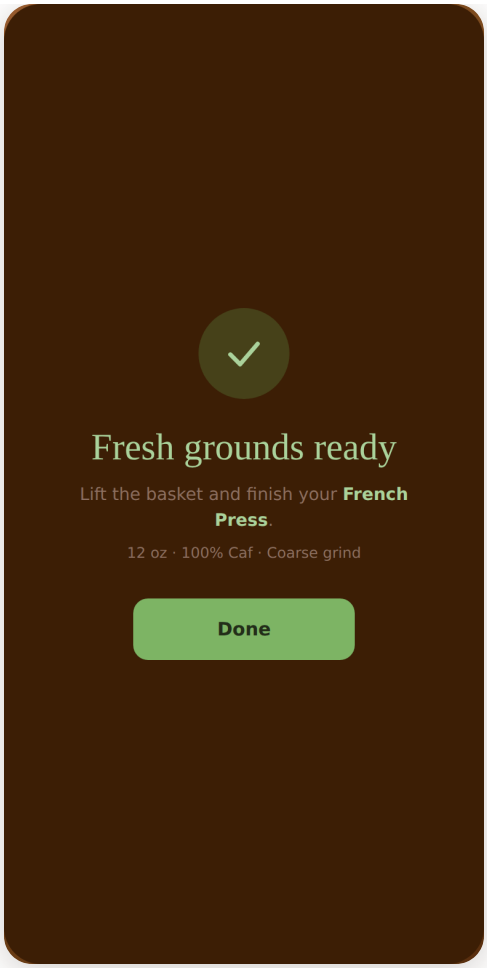
FGO — Fresh Grounds

FGO — Fresh Grounds. The configuration screen. The user picks a brew method — French Press, Pour-Over, or Cold Brew — and the grind follows automatically to match (Coarse, Medium, or Extra-coarse); the grind here is advisory, a taste pairing for the method rather than a caffeine input. There is no brew-strength selector — Medium is taken as the baseline — and the user sets the amount and the caffeine blend on the same slider the brew paths use. The screen shows the caffeine in the grounds and what it will leave at bedtime. A resident's declared bedtime carries in; a returning guest confirms it here.



FGO-CONFIRM — Confirm Grounds

FGO-CONFIRM — Confirm Grounds. The confirmation screen lays out the whole decision before the grind: tonight's bedtime and the Bedtime Ceiling, the method and grind, the amount, the caffeine blend, the caffeine in the grounds, the projected residual at bedtime, and the projected wake-hours peak. When the projected bedtime residual sits above the ceiling, a *Bedtime Ceiling Advisory* appears — the same honest, non-blocking warning the brew paths use: the user can still grind, but is told plainly what it will cost and invited to add more decaf. *Adjust the caffeine blend* reopens the slider; *Tap & Hold to Grind* commits.



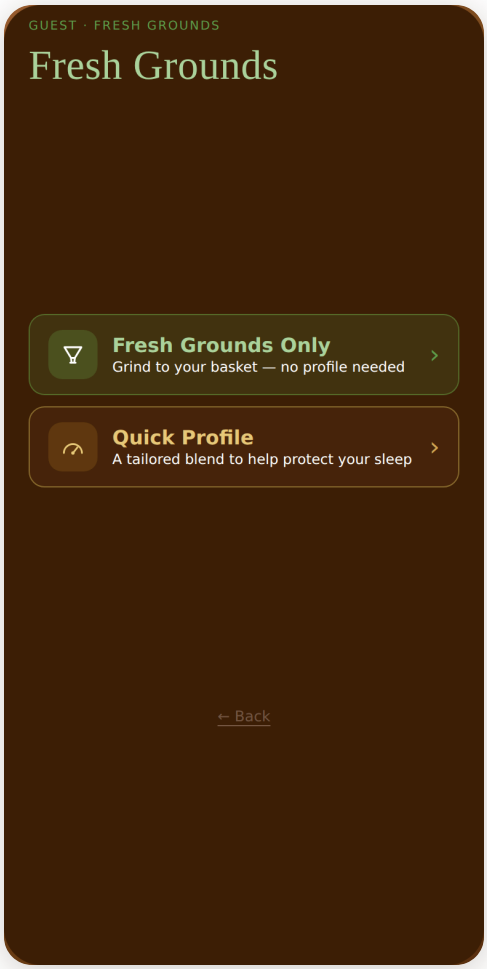
FGO-DONE — Fresh Grounds Ready

FGO-DONE — Fresh Grounds Ready. The completion screen. The grind is done and the grounds are in the basket; the screen confirms the amount, blend, and grind and tells the user to lift the basket and finish their chosen method. The event is logged to the shared daily log exactly as a device brew would be, so the caffeine it carries is folded into the running residual model and every recommendation that follows.

Guest User

A guest brews without a stored **Resident User Profile**. Four paths are available for guests: **Fresh Grounds Only** for grinding a blend to the basket with no profile at all, **BrewNow!** for immediate manual-blend brewing, the **Quick Profile Path** for a personalized caffeine blend recommendation that establishes a **Guest Profile**, and the **Returning Guest Path** for guests whose Guest Profile was saved as a **Returning Guest Profile** within the past 72 hours.

Guest Brew Paths — Fresh Grounds Only

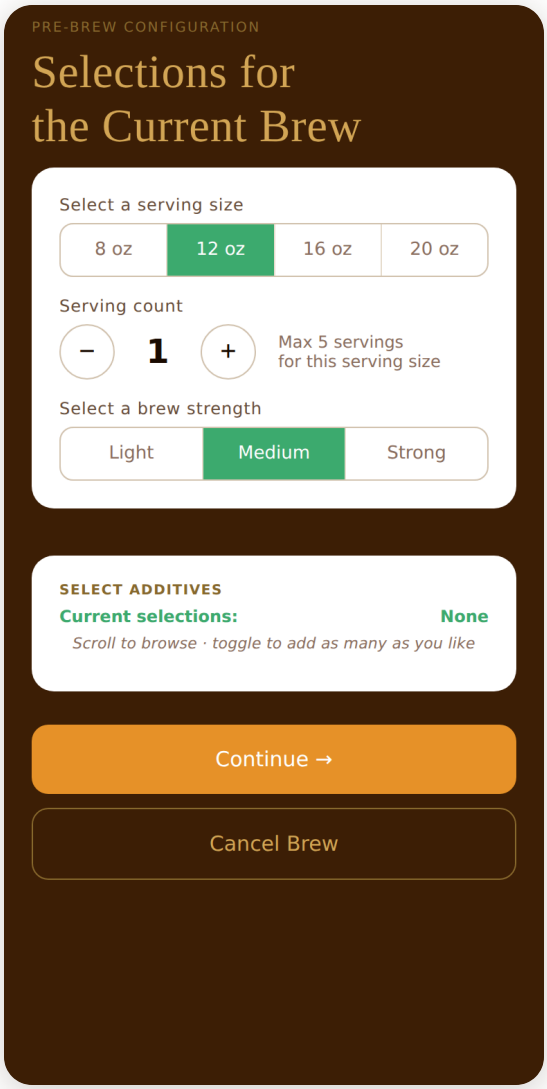


GFGO — Guest Fresh Grounds

GFGO — Guest Fresh Grounds. A guest who chooses Fresh Grounds Only at the entry screen lands here, where two routes are offered: **Fresh Grounds Only** — grind to your basket, no profile needed — or **Quick Profile**, a tailored blend to help protect sleep. Choosing Fresh Grounds Only opens the same grounds-to-basket configuration a resident uses — method, amount, and caffeine blend — still projecting the drink forward to bedtime and surfacing the Bedtime Ceiling advisory. Because a guest has no stored bedtime, the flow confirms tonight's bedtime first, since the projection depends on it.



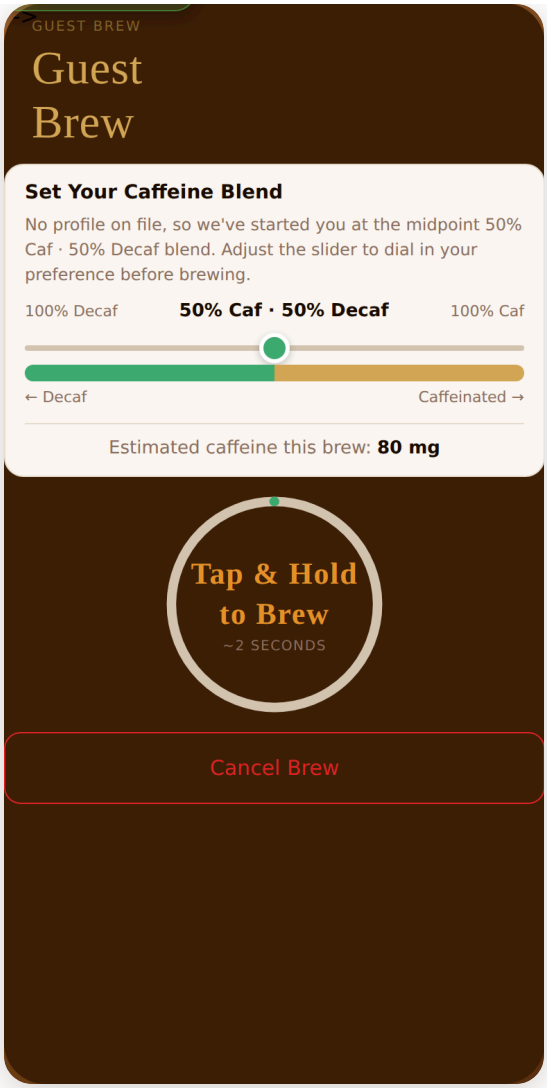
PS1b — Guest Entry



PS2a — Brew-Time Selections

PS1b — Guest Entry. The guest entry point. Returning guests (within 72 hours) appear as cards with a direct ▶ Brew route. New guests tap either *BrewNow!* for the fastest path forward or *Quick Profile* for a limited scope profile with a complete caffeine blend recommendation.

PS2a — Brew-Time Selections. Once the Guest taps *BrewNow!* on the entry screen, they are advanced to select the size, serving count, and strength of the brew. All device-loaded additives are available — no profile-level filtering applies.



PS3a-C2 — Guest Blend

PS3a-C2 — Guest Blend. The guest manually sets their caffeine blend percentage with the slider. No recommendation is generated. Hold to brew confirms the selected blend and initiates the brew.

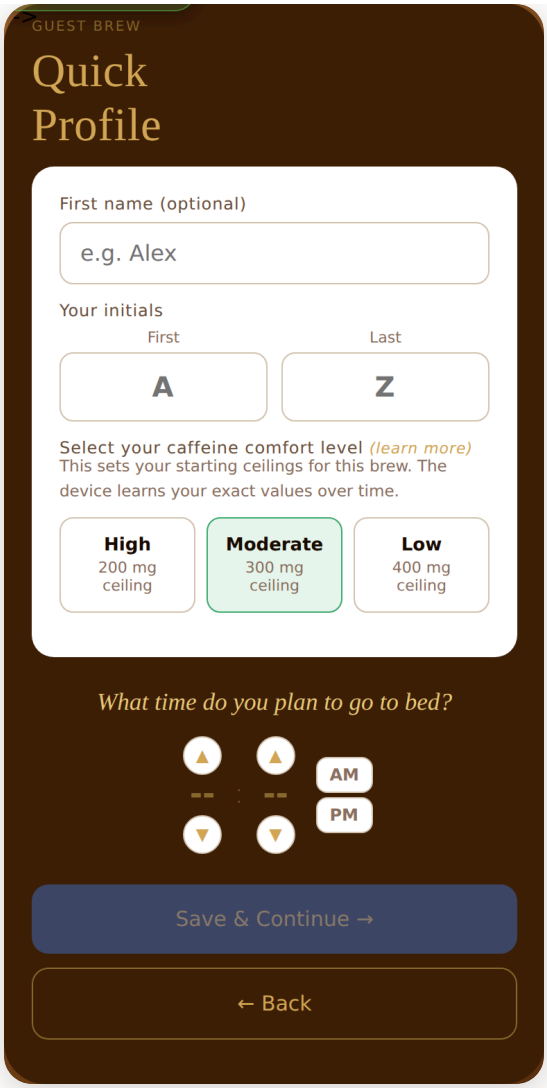
Guest Brew Paths — Quick Profile Path

A new guest can receive a personalized blend recommendation via the Quick Profile Path, which establishes a Guest Profile. At the end of the flow the guest is asked whether they are likely to brew again: if so, the Guest Profile is saved as a Returning Guest Profile for reuse via the Returning Guest Path within the next 72 hours; if not, the Guest Profile is used for this brew only and then discarded.

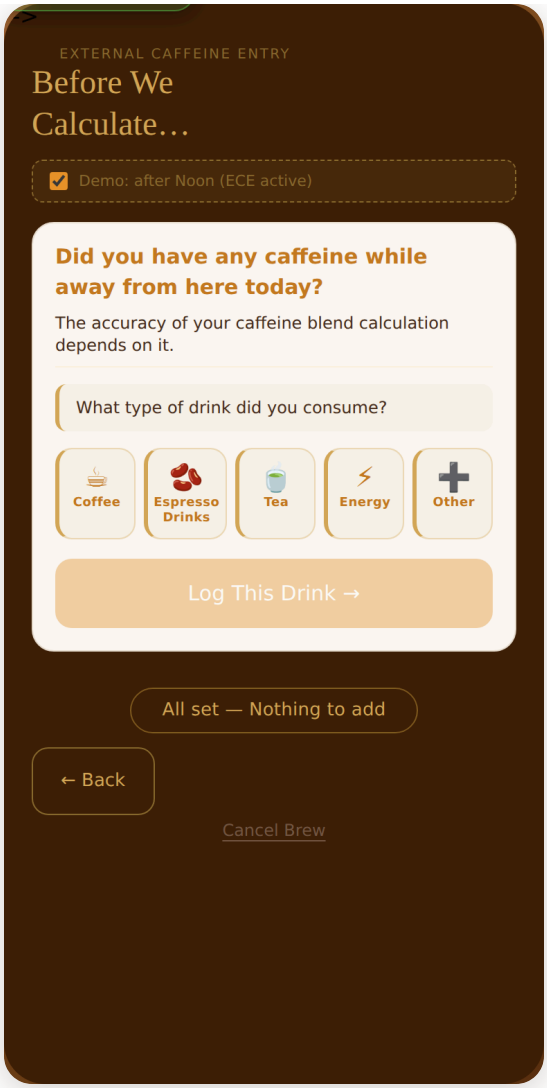


PS1b — Guest Entry

PS1b — Guest Entry. The guest taps *Quick Profile* to begin the personalized guest brew path.



PS1c — Guest Quick Profile

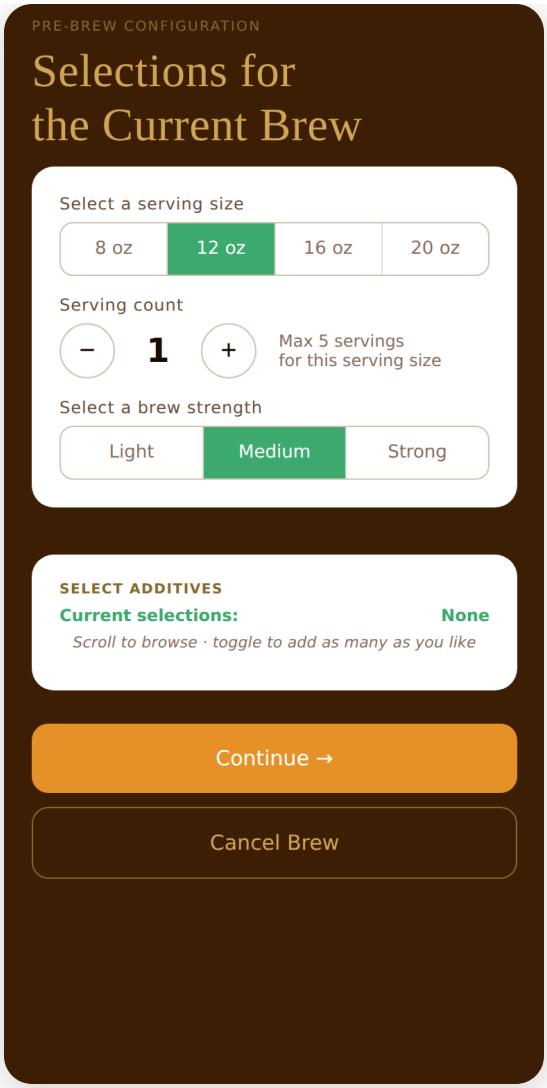


ECE — External Caffeine Entry

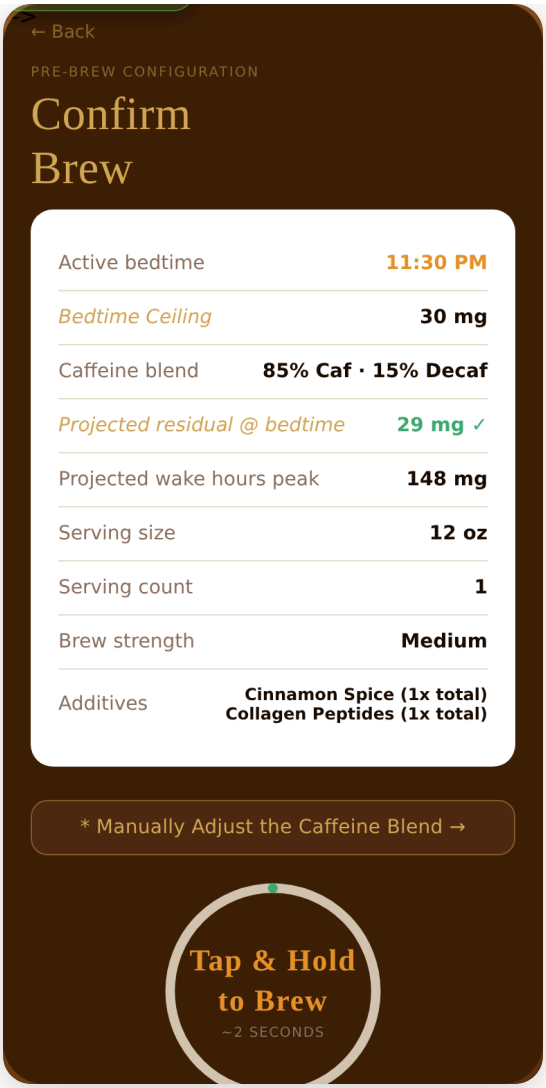
PS1c — Guest Quick Profile. Name, initials, and caffeine sensitivity level (Higher / Moderate / Lower) are entered along with the planned bedtime, establishing the Guest Profile. On confirming, the guest is asked whether they are likely to brew again — if so, the Guest Profile is saved as a Returning Guest Profile for 72 hours; if not, it is used for this brew only and then discarded. As a Guest-only shortcut, the selected sensitivity tier also sets a default Caffeine Decay Rate — Higher = 7.0 hrs, Moderate = 6.0 hrs, Lower = 5.0 hrs — mirroring the same proportional logic as the sensitivity-based ceiling defaults — higher sensitivity pairs a lower ceiling default with a longer assumed half-life (7.0 hr), both the more conservative setting, so the two move in the same protective direction — rather

ECE — External Caffeine Entry. This screen checks for unknown caffeine consumption before recommending the caffeine blend. Any unknown caffeine consumption that is captured here is factored into the blend recommendation.

than the CBI-derived personalized rate a Resident User Profile establishes. This tier-to-decay-rate link applies only to the Guest quick-profile default; for Resident Users, sensitivity (ceilings) and Caffeine Decay Rate remain independent traits, each derived from its own separate **Caffeine Behavior Instrument** question.



PS2a — Brew-Time Selections



PS6-A — Recommendation & Brew

PS2a — Brew-Time Selections. Drink size, number of servings, brew strength, and additive selections are selected for this guest brew.

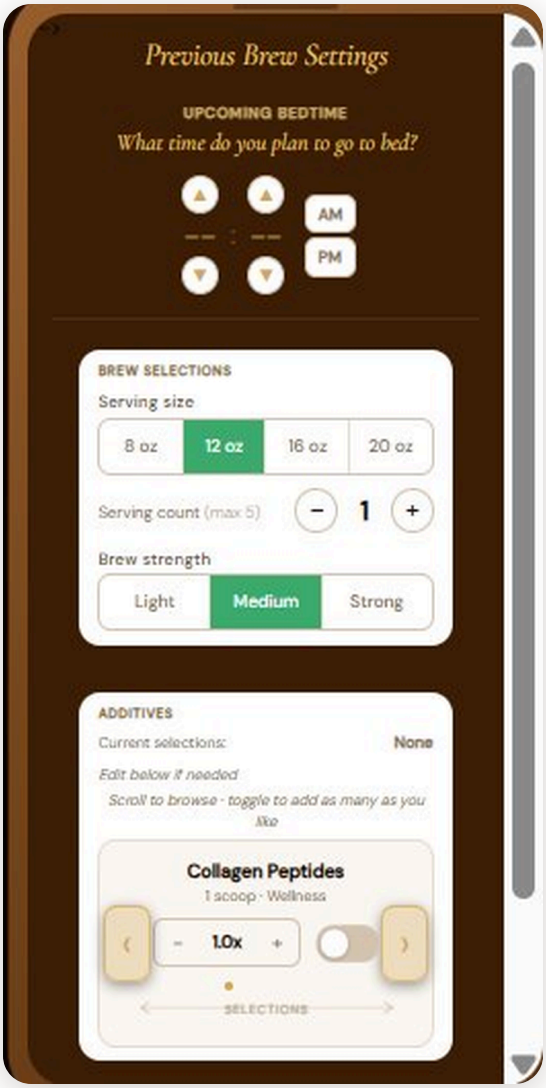
PS6-A — Recommendation & Brew. A personalized caffeine blend recommendation is generated based on the guest's selected sensitivity level and planned bedtime. *Hold to Brew* confirms and initiates the actual brew cycle. If the guest wishes to change the caffeine content of the brew from the recommendation, *Adjust Manually* is tapped to open the Adjust Caffeine Blend slider.

Guest Brew Paths — Returning Guest Path

A guest whose Guest Profile was saved as a Returning Guest Profile within the past 72 hours has a profile card on PS1b. Tapping it loads their saved settings directly for confirmation or adjustment.



PS1b — Guest Entry



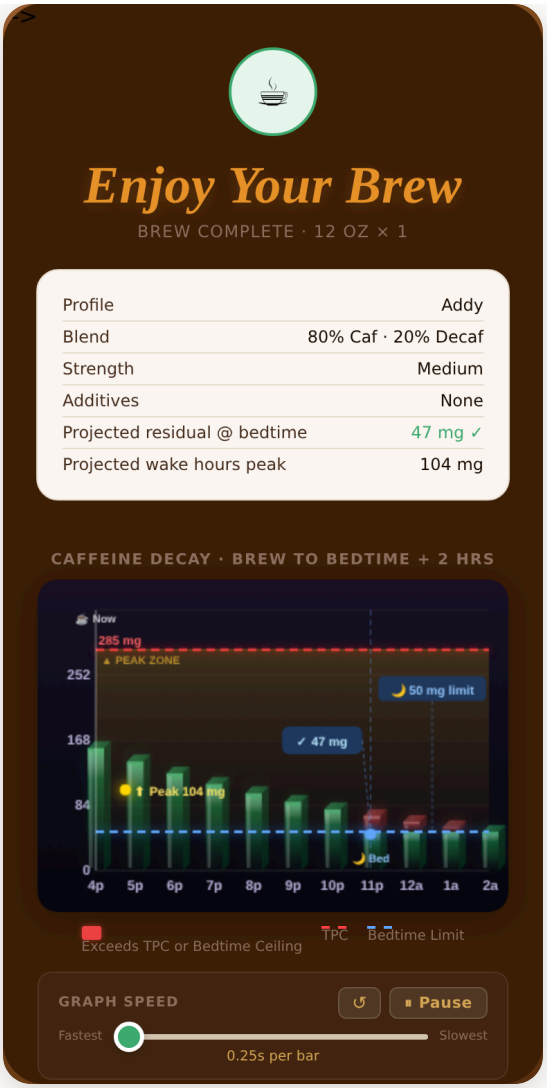
PS2rg — Returning Guest Review

PS1b — Guest Entry. The returning guest's card appears with a direct ▶ Brew route. Tapping it loads saved settings without re-entering the Quick Profile flow.

PS2rg — Returning Guest Review. Pre-populated with the guest's last brew settings — size, strength, and additives. The guest confirms or adjusts before continuing.



PS6-C1 — Guest Brew Confirmation



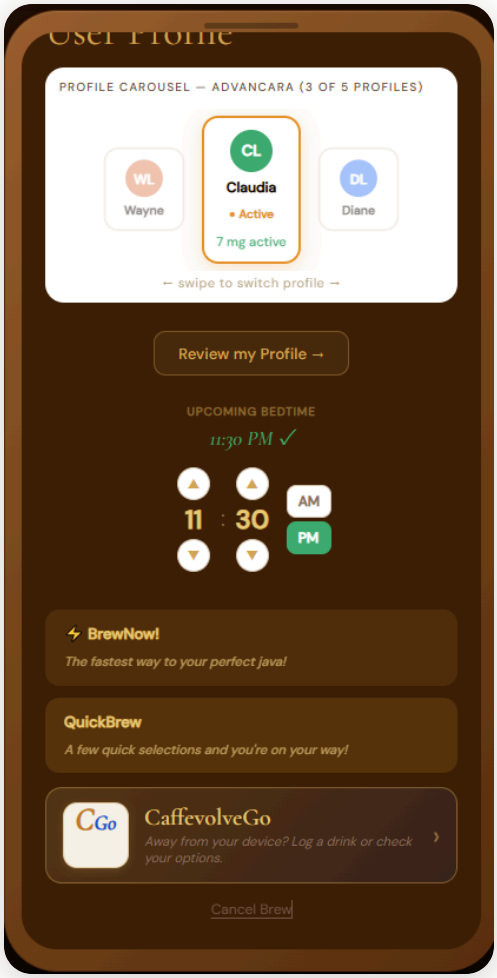
PS8 — Enjoy Your Brew

PS6-C1 — Guest Brew Confirmation. The blend recommendation, brew parameters, and selected additives are displayed for final review before *Hold to Brew* is held to initiate the brew cycle. The recommended caffeine blend can be changed by tapping *Adjust Manually*.

PS8 — Enjoy Your Brew. The brew complete screen. A session summary card confirms the profile, blend, strength, additives, projected residual caffeine at bedtime, and projected wake-hours peak for the brew just completed. A Caffeine Decay chart animates the projected caffeine curve from brew time through bedtime plus two hours, with reference lines marking the Wake Hours Ceiling and Bedtime Ceiling. The screen auto-advances to the home screen after a short countdown; the user may also tap *All set — Thank you!* to return immediately.

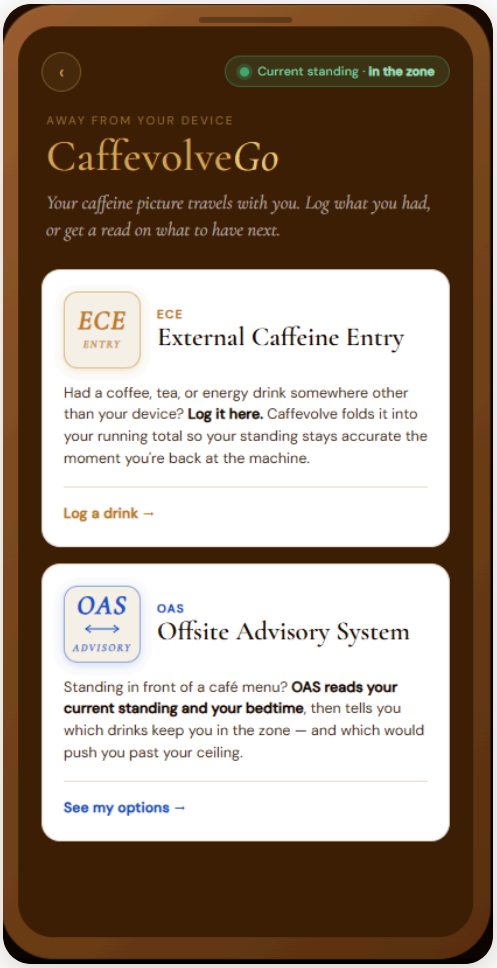
8.3 CaffevolveGo

CaffevolveGo is reached from the profile screen (PS1a). Once a bedtime is set for the session, the user taps *Away* (CaffevolveGo) to open the hub, where the platform's two away-from-home functions are presented as a single branded choice — ***External Caffeine Entry (ECE)*** to log a drink already consumed, and the ***Offsite Advisory System (OAS)*** to weigh what to consume next. Both subsections that follow begin here.



PS1a — Profile Selection

PS1a — Profile Selection. The user selects their profile, confirms the upcoming bedtime, and taps *Away* (CaffevolveGo) at the bottom of the screen to open the hub.



CaffevolveGo — Hub

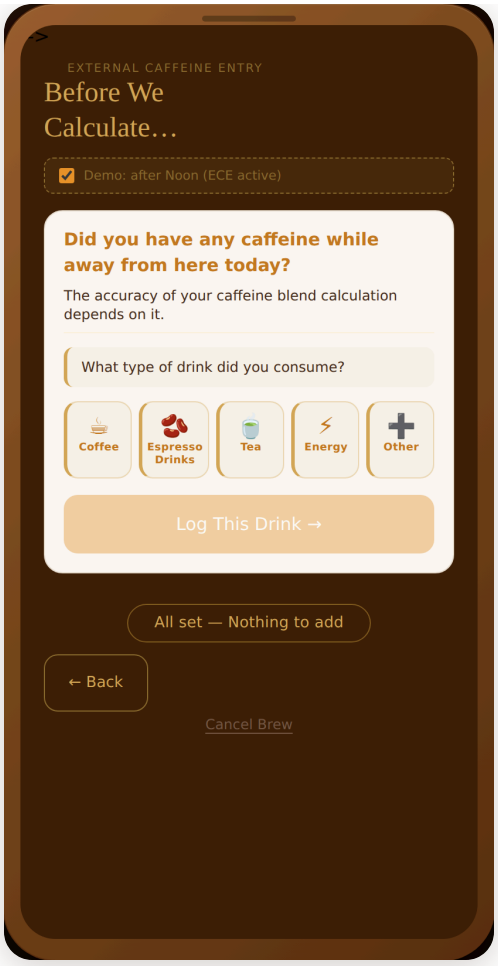
CaffevolveGo — Hub. The hub presents both functions side by side: the user taps *Log a drink* to record an external caffeine event (ECE), or *See my options* to open the Offsite Advisory System (OAS). The two subsections that follow pick up from this screen.

External Caffeine Entry (ECE)

The External Caffeine Entry screens are used to record caffeine activity recorded away from the device — logging a past event to update the residual model and the Offsite Advisory System (OAS).

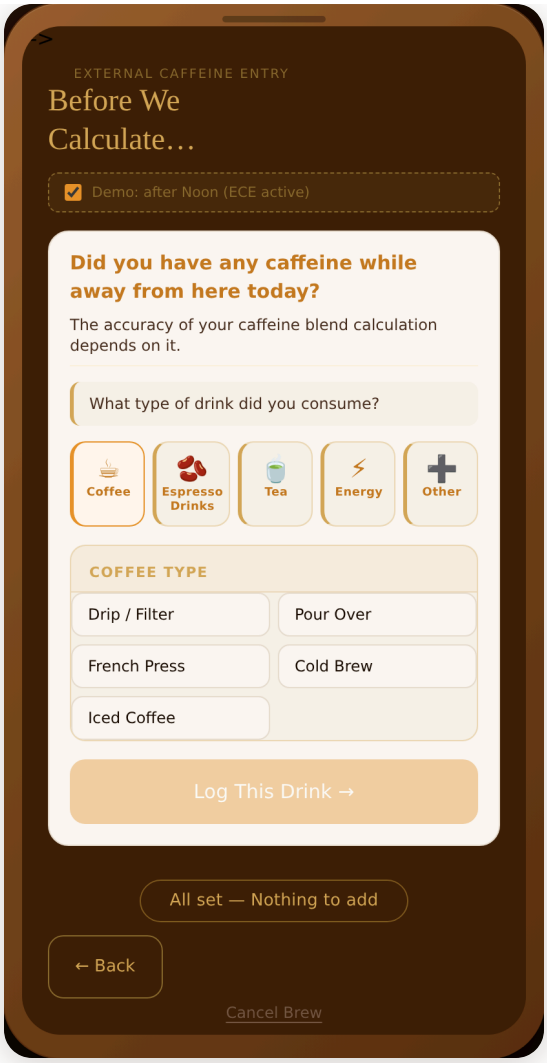
ECE operates in two modes. **ECEr (Reactive)** is user-initiated at any time — the user opens the ECE option from the CaffevolveGo app to log an offsite caffeine event on demand. **ECEp (Proactive)** is system-initiated: it fires automatically before a brew or OAS session when the user's last declared bedtime has passed and no device brew or ECE entry has been recorded since that bedtime. Two ECEp windows are defined per wake cycle, each 6 consecutive hours long — Window 1 opens on the first brew or ECE entry after the last declared bedtime has passed, and Window 2 opens 6 hours after Window 1. ECEp fires once per window; answering or skipping suppresses it for the remainder of that window, so it never fires twice for the same gap. Because the daily log is shared in real time between the device and companion app, any caffeine already logged via OAS is present when ECEp opens — the user is asked only to account for what those channels haven't already captured. For guests, the ECEp flag is keyed to the guest session and clears when the session ends.

Log a Previous Event

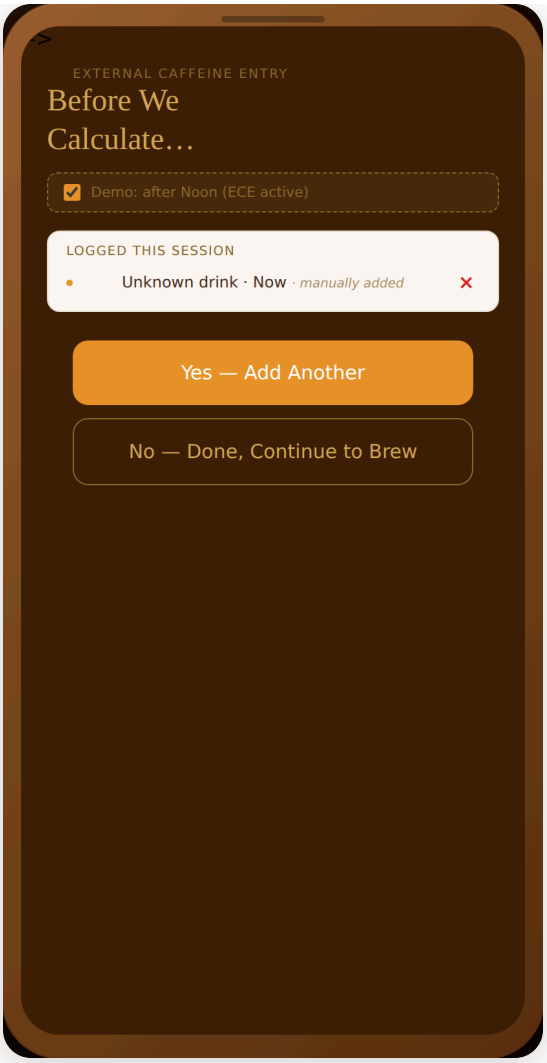


ECE — External Caffeine Entry

ECE — External Caffeine Entry. The user enters drink type, size, servings and approximate time of the external event. The residual model is updated immediately and all blend recommendations in the session reflect the logged intake. Tapping *All set — Nothing to add* skips the entry and returns to the brew selection screen.



ECE — Drink Selection

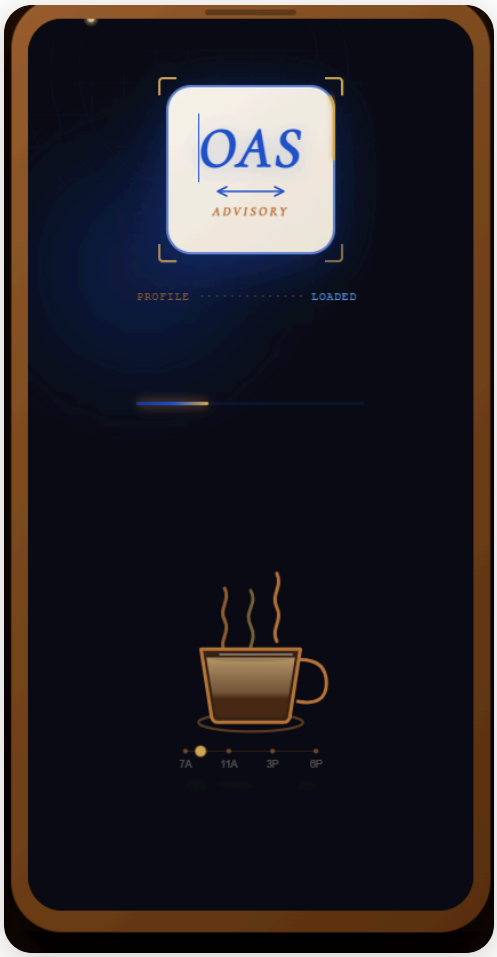


ECE — Session Log

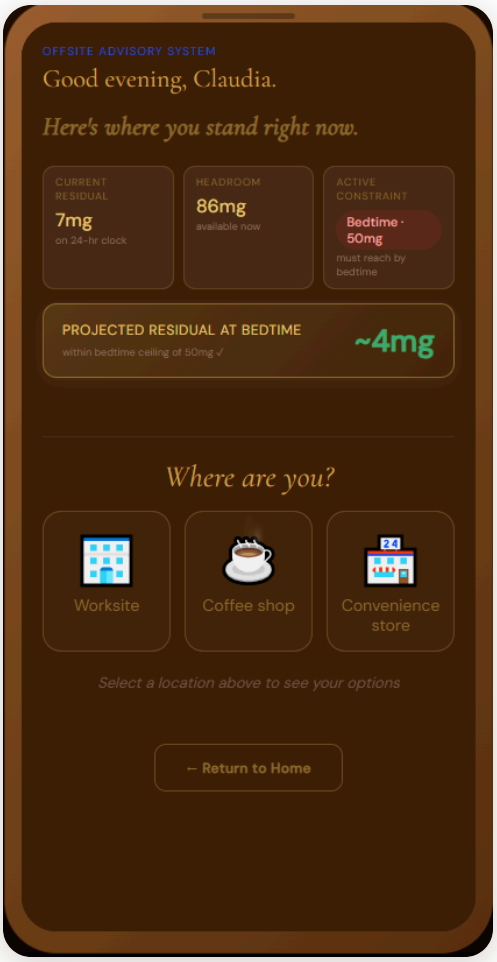
ECE — Drink Selection. After selecting the drink category, the user chooses the specific drink type, sets the number of servings, and selects when the drink was consumed — now, or an earlier time window. Tapping *Log This Drink →* records the entry.

ECE — Session Log. Logged entries appear in the session list with the option to add another drink or continue to brew. Tapping *No — Done, Continue to Brew* closes the ECE session and returns to the brew flow with an updated residual model.

Offsite Advisory System (OAS)



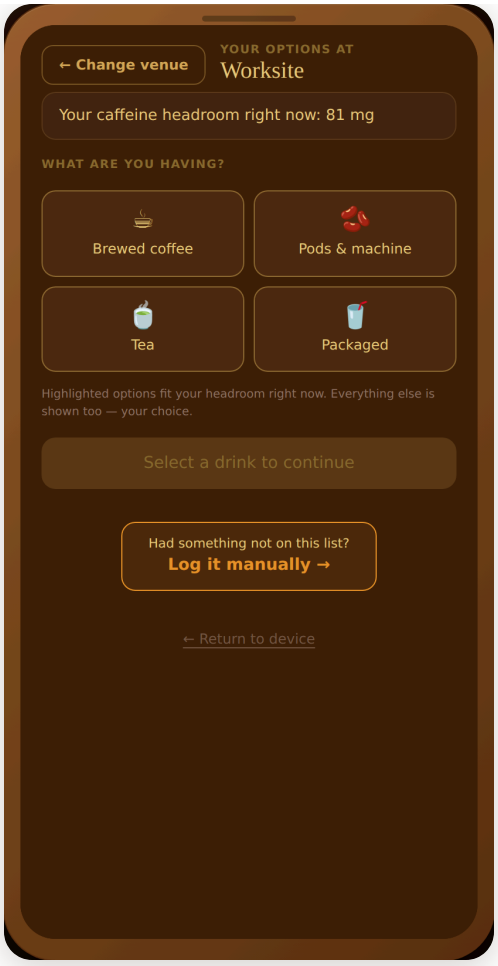
OAS0 — Advisory Launch



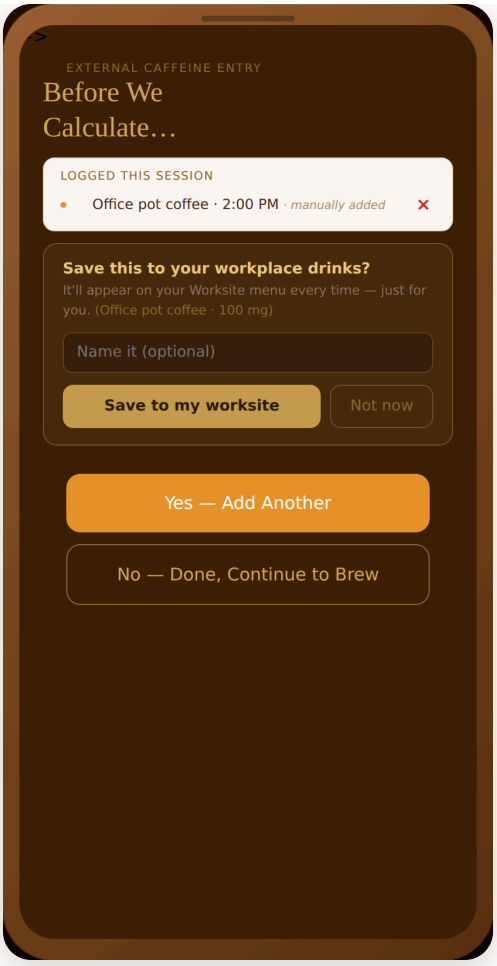
OAS1a — Location Type

OAS0 — Advisory Launch. A brief branded interstitial confirms the profile is loaded and the current standing is being read before the advisory opens.

OAS1a — Location Type. The OAS opening screen presents the user's current caffeine metrics — residual level, available headroom, and active constraint — alongside a venue selector. The user taps their current location type to load the appropriate drink menu. The venue selector leads with **Worksite** — the workplace setting — followed by the other common venues such as café and convenience store.



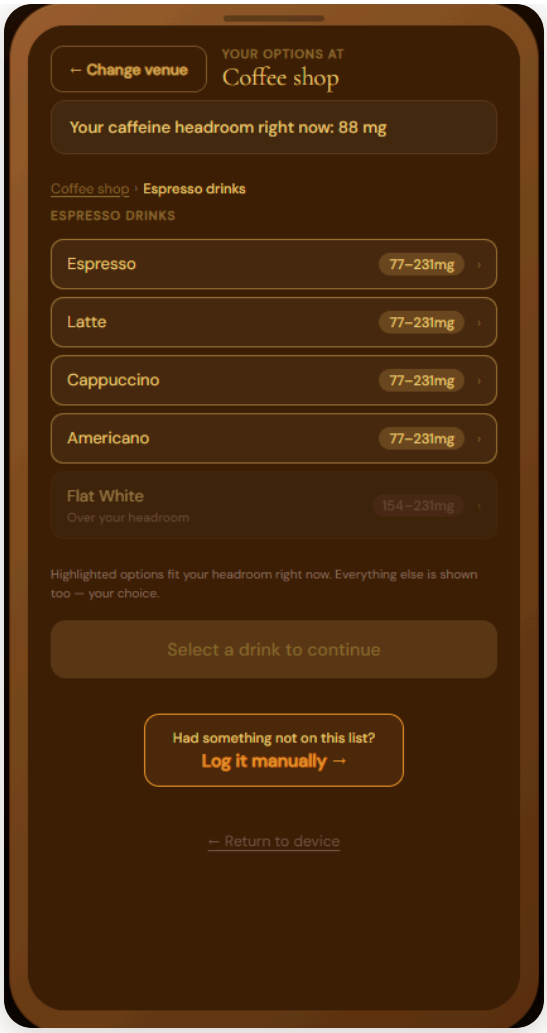
OAS1b — Worksite Venue



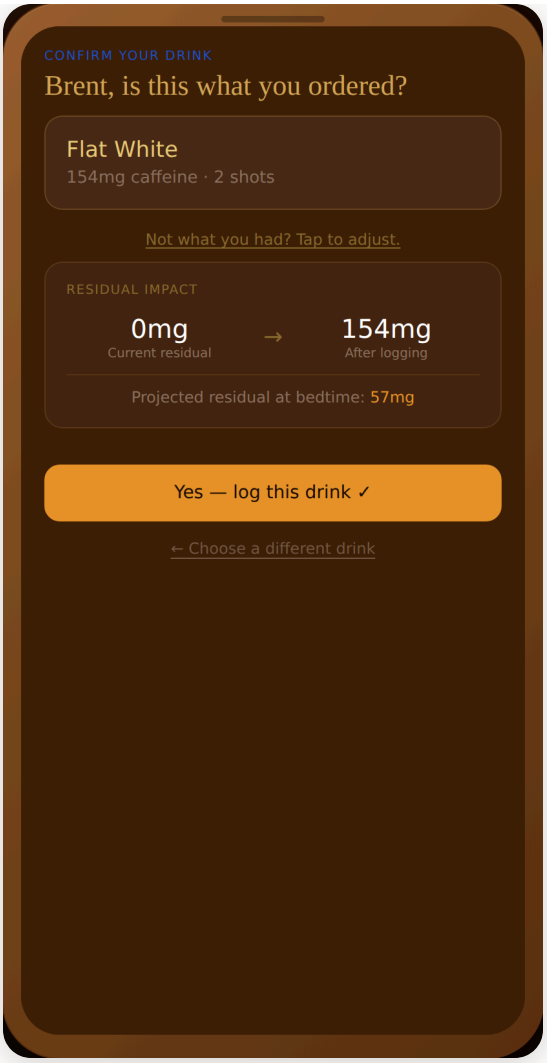
ECE — Save to Workplace Drinks

Worksite venue. Worksite is the first venue in the OAS selector and covers the drinks a user is most likely to encounter at their workplace, organized into categories — brewed coffee, pods and single-serve machines, tea, and packaged or canned drinks. Because the strength of a shared office pot is not something the user controls, serving volume is the primary lever at this venue: each listed drink is offered on a pour ladder — a full cup, two-thirds, or one-third — so the user can size the drink to the headroom available. The menu presents the full set of options with those that fit the user's current headroom highlighted; the highlighting is guidance, not a gate, consistent with the platform's Advisory Philosophy.

Saved workplace options. When a user logs a workplace drink that is not on the standard list — using the same Alternative Drink Choice path available elsewhere in the OAS — the platform offers to save that drink to a personal *My workplace drinks* group, with an optional rename. On later visits to the Worksite venue, saved drinks appear alongside the standard options and are highlighted and ranked against the user's current headroom in the same way. A user manages their saved workplace drinks — renaming or removing them — from the Profile Summary (SU13).



OAS1b — Drink Selection



OAS2 — Confirm & Log

OAS1b — Drink Selection. The drink menu populates for the selected venue and presents the full set of likely options, with those that fit the user's current headroom highlighted and ranked first; options that exceed it appear below, de-emphasized rather than hidden. The highlighting is guidance, not a gate — consistent with the platform's Advisory Philosophy — and a **Log a different drink** path is always available for anything not listed, recorded via the Alternative Drink Choice path. Every selection is written to the shared daily log and reflected immediately in the residual model.

OAS2 — Confirm & Log. The selected drink is displayed with its caffeine content and projected impact — current residual before and after logging, and projected bedtime residual. The user taps *Yes — log this drink* to record the event, and return to the Sleep screen.

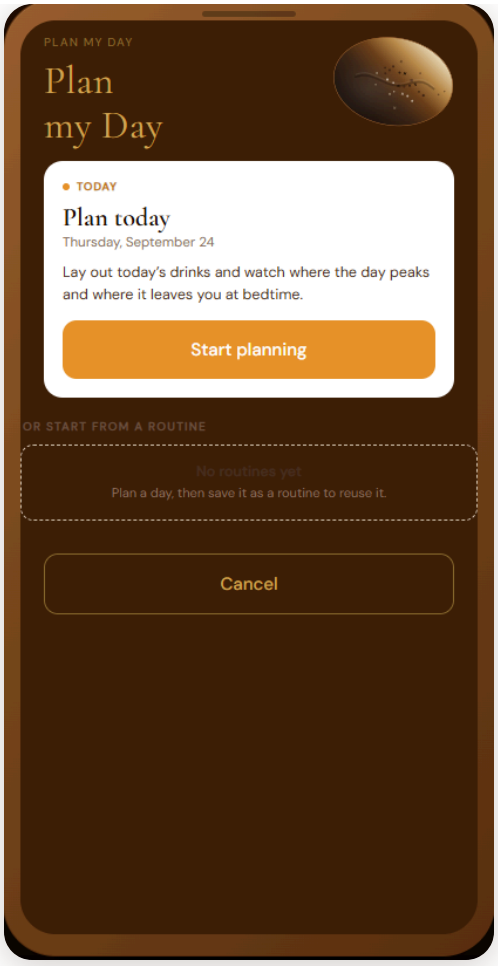
8.4 Plan my Day

The Planner is covered in Section 5; the screens below show it in use. In the walkthrough, the user lays out a typical day and tunes each drink until the projected day sits inside their Caffeine Comfort Zone.

Plan my Day — Planner

Reached by tapping the Planner on the profile screen, the user initiates the process by creating a plan for the day. The plan should start by emulating a typical day — a work day, for example — and the user lays out what, when, and where drinks are usually consumed on this type of day. As the plan is being built, the user can see how each drink contributes to the caffeine level in their system as the day progresses — and most importantly, whether the plan is likely to compromise their personal Caffeine Comfort Zone boundaries, which could impact their sleep or productivity. If the plan does create a compromise, each of the drinks can be adjusted — depending on where they are being consumed — at home with a size or bean blend adjustment, or away from home with a size or strength change, to get the caffeine level in their system back to a comfortable range. Once the user is content with the plan, it can be saved and assigned to different days of the week, allowing it to be in play and provide guidance on those specific days.

Diane builds her workday from a blank plan. She adds her four usual drinks: a 16 oz French press at home at 7:30, then three cups at her office, which has a Caffevolve device: a 12 oz pour-over at 9:45, already eased to 80% caffeine, a drip at 11, and another drip at 2:15. She protects the 11 AM drip, the one cup she won't give up. Counting what is still in her system from last night, the day peaks at 404 mg, inside her 490 mg limit, but leaves 97 mg in her system at her 11:30 PM bedtime, over her 75 mg ceiling. She taps Rebalance for me. Keeping her 11 AM cup at full strength, the Planner eases the other three in a gentle taper: the French press to 65% caffeine, the pour-over to 60% and the 2:15 drip to 65%. Her bedtime level falls to 74 mg and her peak to 335 mg. Had she wanted her French press untouched, she could have protected it as well, and Rebalance would have taken the pour-over to 50% and the 2:15 drip to 55% instead. She then saves the day as a routine called Workday, set to repeat Monday through Friday. It stays her plan for today, and the next time she opens the Planner, Workday is there to start from.



PMD1 — Plan my Day

Plan my Day. Opened from the Planner on the profile screen. With no routines saved yet, Diane starts today's plan from scratch.



PMD2 — A Fresh Day

A fresh day. An empty Thursday. What is still in Diane's system from last night is already counted, so the tiles start from her real state rather than from zero.



PMD3 — Add a Drink



PMD4 — The Day Laid Out

Add a drink. Each drink is set by time, place, type, and size. For a coffee made on a Caffevolve device, the bean blend dial sets its caffeine anywhere from full strength to decaf.

The day laid out. Diane's four drinks, with her 11 AM drip starred as the one she protects. The day peaks at 404 mg, within her 490 mg limit, but bedtime lands at 97 mg — over her 75 mg ceiling, and flagged in red. Because the day is over a ceiling, *Rebalance for me* appears.



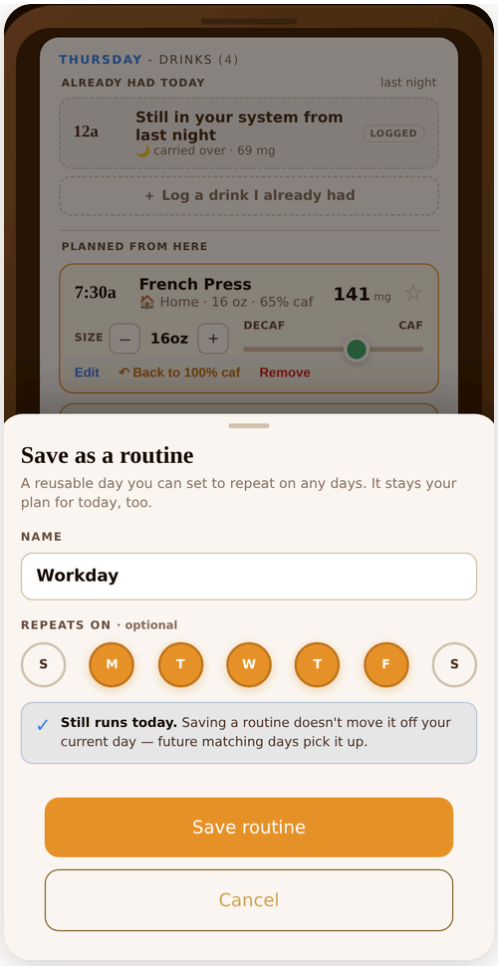
PMD5 — Rebalanced



PMD6 — The Day, Rebalanced

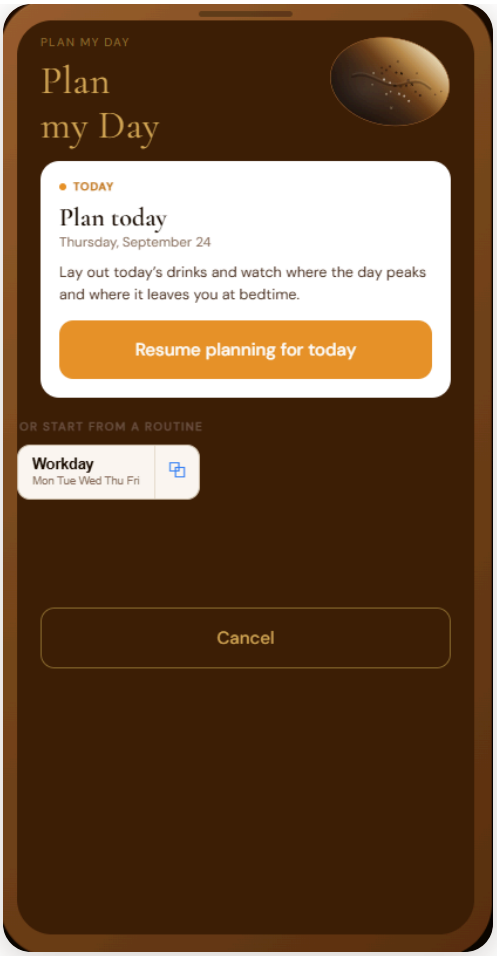
Rebalanced. One tap adjusts the three unprotected cups in a gentle taper — the French press to 65%, the pour-over to 60%, the 2:15 drip to 65% — while the protected 11 AM cup stays at full strength. Each changed cup is outlined and carries a back arrow that returns it to its earlier setting.

The day, rebalanced. Bedtime falls to 74 mg, just under her 75 mg ceiling — amber, for little margin — and the peak eases to 335 mg. The projection now steps down toward evening.



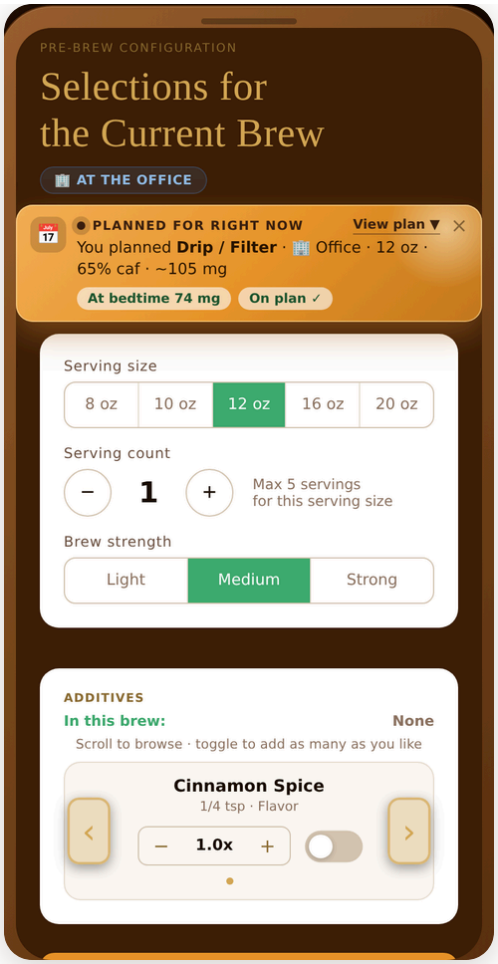
PMD7 — Save as Routine

Save as routine. Diane names the day *Workday* and sets it to repeat Monday through Friday. It remains her plan for today.



PMD8 — Routine Ready

Routine ready. The next time she opens the Planner, Workday is there as a starting point, and today's plan can be resumed.



PMD9 — Planned for Right Now

Planned for right now. At 2:15, when Diane starts a brew at the office, Plan Check shows the drink her plan holds for now — a 12 oz drip at 65% caffeine, about 105 mg — and re-solves the rest of her day against what she has actually had: 74 mg at bedtime, on plan. *View plan* opens the whole day, and × dismisses the notice. Had an unplanned cup pushed the day over, the notice would offer Rebalance (5.4).

A saved plan then works quietly in the background. As a planned day progresses and the user goes to make or buy a drink, Plan Check surfaces in-path — on a Caffevolve device at home or at an equipped office, or in the CaffevolveGo app when away — showing the drink the plan holds for that time and where the rest of the day now lands (Sections 5.4 and 5.5). The user can then make an informed decision. However, as in all other situations, the platform only advises; the user still decides what to brew or buy.

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